ID	Problem Title	Optimised Pattern
1	Two Sum	Hash Map (One Pass)
2	Add Two Numbers	Digit-by-Digit Math w/ Carry
3	Longest Substring Without Repeating Characters	Sliding Window (Hash Set)
4	Median of Two Sorted Arrays	Binary Search on Partitions
5	Longest Palindromic Substring	Expand Around Center
6	Zigzag Conversion	Math / Index Hopping
7	Reverse Integer	Digit Extraction (Pop & Push)
8	String to Integer (atoi)	Deterministic One-Pass (DFA)
9	Palindrome Number	Revert Half the Number
10	Regular Expression Matching	Dynamic Programming (2D Table)
11	Container With Most Water	Two Pointers (Greedy Squeeze)
12	Integer to Roman	Greedy Subtraction (Coin Change)
13	Roman to Integer	Compare Current vs Next
14	Longest Common Prefix	Horizontal Shrinking
15	3Sum	Sort + Anchor & Squeeze
16	3Sum Closest	Sort + Anchor & Squeeze
17	Letter Combinations of a Phone Number	Backtracking (DFS)
18	4Sum	Sort + Two Anchors + Squeeze
19	Remove Nth Node From End of List	Fast & Slow Pointers (One Pass)
20	Valid Parentheses	Stack (LIFO)
21	Merge Two Sorted Lists	Two Pointers + Dummy Node
22	Generate Parentheses	Backtracking (Constrained DFS)
23	Merge k Sorted Lists	Min-Heap (Priority Queue)
24	Swap Nodes in Pairs	Pointer Manipulation + Dummy Node
25	Reverse Nodes in k-Group	In-Place Pointer Reversal
26	Remove Duplicates from Sorted Array	Two Pointers (Reader/Writer)
27	Remove Element	Two Pointers (In-Place Write)
28	Find the Index of the First Occurrence in a String	KMP Algorithm (LPS Array)
29	Divide Two Integers	Bit Manipulation (Exponential Subtraction)
30	Substring with Concatenation of All Words	Sliding Window + Hash Maps
31	Next Permutation	Find Dip, Swap, Reverse
32	Longest Valid Parentheses	Two Counters (Left/Right Passes)
33	Search in Rotated Sorted Array	Modified Binary Search
34	Find First and Last Position of Element in Sorted Array	Double Binary Search
35	Search Insert Position	Standard Binary Search
36	Valid Sudoku	Single Pass (Hash Sets)
37	Sudoku Solver	Backtracking (DFS)
38	Count and Say	Two Pointers + String Builder
39	Combination Sum	Backtracking (Include/Exclude)
40	Combination Sum II	Sort + Backtracking (Skip Dupes)
41	First Missing Positive	Cyclic Sort (Index as Map)
42	Trapping Rain Water	Two Pointers (Inward Squeeze)
43	Multiply Strings	Array Math (Grade School Multiplication)
44	Wildcard Matching	Two Pointers + Greedy Backtrack
45	Jump Game II	Greedy (BFS-Style Window)
46	Permutations	Backtracking (In-Place Swap)
47	Permutations II	Sort + Backtracking (Used Array / Skip)
48	Rotate Image	Transpose + Reverse Rows
49	Group Anagrams	Hash Map (Sorted Key / Char Count)
50	Pow(x, n)	Binary Exponentiation
51	N-Queens	Backtracking (Row-by-Row + Bitsets)
52	N-Queens II	Backtracking (Counter Only)
53	Maximum Subarray	Kadane’s Algorithm (Greedy/DP)
54	Spiral Matrix	Boundary Shrinking
55	Jump Game	Greedy (Max Reach)
56	Merge Intervals	Sorting + Linear Scan
57	Insert Interval	Three-Phase Linear Scan
58	Length of Last Word	Backward Pointer Scan
59	Spiral Matrix II	Boundary Contraction

60	Permutation Sequence	Factorial Number System (Math)
61	Rotate List	The Circular Link (Ring)
62	Unique Paths	Dynamic Programming (1D/2D)
63	Unique Paths II	DP with Obstacle Guard
64	Minimum Path Sum	Dynamic Programming (In-Place)
65	Valid Number	Deterministic Finite Automaton (DFA)
66	Plus One	Reverse Linear Scan
67	Add Binary	Two-Pointer Bitwise Addition
68	Text Justification	Greedy Word Packing + Round Robin Spacing
69	Sqrt(x)	Binary Search (Range [0, x])
70	Climbing Stairs	Iterative DP (Fibonacci)
71	Simplify Path	Stack (Push/Pop on Chunks)
72	Edit Distance	2D Dynamic Programming
73	Set Matrix Zeroes	In-Place State Marking
74	Search a 2D Matrix	Binary Search (Flattened Index)
75	Sort Colors	Dutch National Flag (3-Pointer)
76	Minimum Window Substring	Sliding Window (Two Pointers + Map)
77	Combinations	Backtracking (DFS + Lexicographic)
78	Subsets	Cascading / Bitmasking
79	Word Search	Backtracking (DFS + In-Place Mark)
80	Remove Duplicates from Sorted Array II	Two Pointers (Slow/Fast with Offset)
81	Search in Rotated Sorted Array II	Binary Search (Shrinking Bound)
82	Remove Duplicates from Sorted List II	Sentinel Node + Two Pointers
83	Remove Duplicates from Sorted List	Single Pass (In-Place)
84	Largest Rectangle in Histogram	Monotonic Stack
85	Maximal Rectangle	DP + Largest Histogram (Q84)
86	Partition List	Two Dummy Nodes (Merging Streams)
87	Scramble String	3D Dynamic Programming / Memoization
88	Merge Sorted Array	Three Pointers (Backward Fill)
89	Gray Code	Iterative Mirroring / Bitwise Formula
90	Subsets II	Sort + Backtracking (Sibling Skip)
91	Decode Ways	1D Dynamic Programming (Fibonacci-like)
92	Reverse Linked List II	In-Place "Leapfrog" Reversal
93	Restore IP Addresses	Backtracking with Pruning
94	Binary Tree Inorder Traversal	Iterative with Stack (or Morris)
95	Unique Binary Search Trees II	Recursion with Memoization
96	Unique Binary Search Trees	Dynamic Programming (Catalan Number)
97	Interleaving String	2D Dynamic Programming
98	Validate Binary Search Tree	Recursion with Range (Min/Max)
99	Recover Binary Search Tree	Morris Inorder Traversal
100	Same Tree	Recursive DFS (Structural Comparison)
101	Symmetric Tree	Two-pointer recursive DFS (or BFS with queue).
102	Binary Tree Level Order Traversal	BFS using a Queue.
103	Binary Tree Zigzag Level Order Traversal	BFS with a Double-ended Queue (Deque) and a boolean direction flag.
104	Maximum Depth of Binary Tree	Post-order Traversal (Bottom-Up DFS). 1 + max(left, right)
105	Construct Binary Tree from Preorder and Inorder Traversal	Hash Map for O(1) inorder index lookup + Recursive DFS.
106	Construct Binary Tree from Inorder and Postorder Traversal	Hash Map for O(1) inorder index lookup + Recursive DFS (Process postorder backwards: Root, Right, Left).
107	Binary Tree Level Order Traversal II	BFS with a Queue + Prepending to a Deque/Linked List for the result (or standard array reversal, which is practically optimal in m
108	Convert Sorted Array to Binary Search Tree	Divide and Conquer / DFS. Find mid, make root, recurse left half, recurse right half.
109	Convert Sorted List to Binary Search Tree	Inorder Simulation (DFS). Calculate list length. Build left subtree, assign current list node as root, advance list pointer, build right
110	Balanced Binary Tree	Bottom-Up DFS. Return the height of the subtree if it is balanced. If it is ever unbalanced, return -1 immediately to short-circuit.
111	Minimum Depth of Binary Tree	BFS (Level Order Traversal) with an early exit on the first leaf node encountered.
112	Path Sum	DFS (Pre-order) with target subtraction (passing `target - node.val` down the recursion).
113	Path Sum II	DFS with Backtracking. Maintain a single shared path list: `push` node, recurse, `pop` node.
114	Flatten Binary Tree to Linked List	Reverse Post-order Traversal (Right, Left, Root) maintaining a global `prev` pointer.
115	Distinct Subsequences	2D Dynamic Programming (Bottom-Up Tabulation).
116	Populating Next Right Pointers in Each Node	Level-by-Level Linked List Traversal (using existing `next` pointers to traverse horizontally).
117	Populating Next Right Pointers in Each Node II	Linked List Traversal with a Dummy Head (tracking the start of the next level).
118	Pascal's Triangle	Dynamic Programming (Tabulation). Building the current row iteratively from the directly previous row.
119	Pascal's Triangle II	1D Dynamic Programming (Rolling Array) calculated strictly from right-to-left.
120	Triangle	Bottom-Up Dynamic Programming (In-place or rolling 1D array).

121	Best Time to Buy and Sell Stock	One-pass Greedy / Kadane’s-like Sliding Window.
122	Best Time to Buy and Sell Stock II	Greedy Algorithm (Summing Positive Slopes).
123	Best Time to Buy and Sell Stock III	Dynamic Programming (4-state variables) or Prefix/Suffix Profit Arrays.
124	Binary Tree Maximum Path Sum	Bottom-Up DFS (Post-order Traversal) with a Global Max tracker.
125	Valid Palindrome	Two-Pointer Technique (Meeting in the center).
126	Word Ladder II	BFS (to find shortest distances) + DFS/Backtracking (to reconstruct paths).
127	Word Ladder	BFS with Character Mutation (a to z).
128	Longest Consecutive Sequence	Hash Set (Intelligent Starting Points).
129	Sum Root to Leaf Numbers	DFS with Arithmetic Accumulation.
130	Surrounded Regions	Border-to-Interior DFS (Inverse Flood Fill).
131	Palindrome Partitioning	Backtracking with DP/Memoization (pre-checking palindromes).
132	Palindrome Partitioning II	1D Dynamic Programming with 2D Palindrome Pre-processing.
133	Clone Graph	DFS/BFS with a Hash Map (Mapping Original Nodes to Cloned Nodes).
134	Gas Station	Greedy One-Pass (The Total Net Gain Rule).
135	Candy	Two-Pass Greedy (Left-to-Right and Right-to-Left).
136	Single Number	Bit Manipulation (XOR).
137	Single Number II	Bit Manipulation (Modulo 3 Summing).
138	Copy List with Random Pointer	Interweaving Nodes (In-place).
139	Word Break	1D Dynamic Programming (Tabulation).
140	Word Break II	Backtracking + Memoization (Top-Down DP).
141	Linked List Cycle	Floyd’s Cycle-Finding Algorithm (Tortoise and Hare).
142	Linked List Cycle II	Two-Pointer Meeting Point + Distance Logic.
143	Reorder List	Find Middle + Reverse Second Half + Interweave.
144	Binary Tree Preorder Traversal	Iterative Traversal with a Stack.
145	Binary Tree Postorder Traversal	Iterative Traversal using Two Stacks (or One Stack + Reverse).
146	LRU Cache	Hash Map + Doubly Linked List (DLL).
147	Insertion Sort List	Insertion Sort with Dummy Head and Predecessor Pointers.
148	Sort List	Merge Sort (Bottom-Up Iterative for O(1) space, or Top-Down Recursive).
149	Max Points on a Line	Hash Map for Slopes (per starting point).
150	Evaluate Reverse Polish Notation	Stack-based Evaluation.
151	Reverse Words in a String	Two-Pointer In-place Reverse (or String Builder).
152	Maximum Product Subarray	Dynamic Programming (Kadane's variation with Min/Max tracking).
153	Find Minimum in Rotated Sorted Array	Binary Search (O(log N)).
154	Find Minimum in Rotated Sorted Array II	Binary Search with Duplicate Handling (O(log N) average, O(N) worst).
155	Min Stack	Two Stacks (Data Stack + Min Stack) or One Stack with Pairs.
156	Binary Tree Upside Down	Iterative Pointer Manipulation (Bottom-Up Logic).
157	Read N Characters Given Read4	Buffered Chunk Processing with Bound Checking.
158	Read N Characters Given Read4 II	Global State Buffer (Persistent Pointer/Counter).
159	Longest Substring with At Most Two Distinct Characters	Sliding Window with Hash Map (Frequency Counter).
160	Intersection of Two Linked Lists	Two-Pointer Sync (Head Switching).
161	One Edit Distance	One-Pass String Comparison.
162	Find Peak Element	Binary Search on Slope.
163	Missing Ranges	Linear Scan of Adjacent Elements.
164	Maximum Gap	Bucket Sort (Pigeonhole Principle).
165	Compare Version Numbers	Two-Pointer Integer Parsing.
166	Fraction to Recurring Decimal	Long Division + Hash Map (Remainder Tracking).
167	Two Sum II - Input Array Is Sorted	Two Pointers (Left and Right).
168	Excel Sheet Column Title	Base-26 Conversion (with Offset Correction).
169	Majority Element	Boyer-Moore Voting Algorithm.
170	Two Sum III - Data structure design	Hash Map (Value rightarrow Frequency Count).
171	Excel Sheet Column Number	Polynomial Rolling Sum (Base-26 conversion).
172	Factorial Trailing Zeroes	Count Factors of 5 (Legendre's Formula).
173	Binary Search Tree Iterator	Controlled Stack-based DFS (Lazy Loading).
174	Dungeon Game	Bottom-Up Dynamic Programming (Reverse Tabulation).
175	Combine Two Tables	SQL Left Outer Join.
176	Second Highest Salary	Subquery with LIMIT 1 OFFSET 1 wrapped in IFNULL .
177	Nth Highest Salary	Function with Local Variable Offset (N-1) and LIMIT .
178	Rank Scores	Window Function: DENSE_RANK() OVER(ORDER BY Score DESC) .
179	Largest Number	Custom Greedy Sorting (Comparator: S1+S2 vs S2+S1).
180	Consecutive Numbers	Window Functions: LEAD() or LAG() .
181	Employees Earning More Than Their Managers	Self-Join on ManagerId = Id .

182	Duplicate Emails	GROUP BY with HAVING COUNT(*) > 1 .
183	Customers Who Never Order	LEFT JOIN where Orders.Id IS NULL (or NOT IN with subquery).
184	Department Highest Salary	Join with a GROUP BY subquery or Window Function.
185	Department Top Three Salaries	Window Function: DENSE_RANK() .
186	Reverse Words in a String II	Double In-place Reversal (Reverse Whole + Reverse Words).
187	Repeated DNA Sequences	Sliding Window + Hash Set (or Rolling Hash).
188	Best Time to Buy and Sell Stock IV	Dynamic Programming (DP Table with State).
189	Rotate Array	The Reversal Algorithm.
190	Reverse Bits	Bit Manipulation (Extract and Shift).
191	Number of 1 Bits	Brian Kernighan’s Algorithm (n & (n-1)).
192	Word Frequency	Unix Pipeline (cat | tr | sort | uniq -c | sort -nr).
193	Valid Phone Numbers	Regex Pattern Matching (grep -P or awk).
194	Transpose File	AWK Column Accumulation.
195	Tenth Line	sed or tail + head .
196	Delete Duplicate Emails	Self-Join Delete.
197	Rising Temperature	Self-Join with DATEDIFF .
198	House Robber	1D Dynamic Programming (Iterative Tabulation).
199	Binary Tree Right Side View	Recursive DFS (Root arrow Right arrow Left).
200	Number of Islands	DFS with In-place Sink (or BFS/Union-Find).
201	Bitwise AND of Numbers Range	Bit Manipulation (Common Prefix Match)
202	Happy Number	Floyd’s Cycle-Finding Algorithm (Fast & Slow Pointers)
203	Remove Linked List Elements	Dummy Node + Linear Traversal
204	Count Primes	Sieve of Eratosthenes
205	Isomorphic Strings	Two Hash Maps / Two Frequency Arrays
206	Reverse Linked List	Iterative In-Place Reversal (Three Pointers: prev, curr, next)
207	Course Schedule	Topological Sort (Kahn's Algorithm / BFS)
208	Implement Trie (Prefix Tree)	Trie (Prefix Tree) Node Structure
209	Minimum Size Subarray Sum	Sliding Window (Two Pointers)
210	Course Schedule II	Topological Sort (DFS with 3 States or Kahn's BFS)
211	Design Add and Search Words Data Structure	Trie (Prefix Tree) + DFS Backtracking for '.'
212	Word Search II	Trie + Backtracking DFS on Grid
213	House Robber II	Dynamic Programming (Two Passes)
214	Shortest Palindrome	KMP Partial Match Table (LPS Array)
215	Kth Largest Element in an Array	Quickselect (Hoare's Partition) OR Min-Heap
216	Combination Sum III	Backtracking with Early Pruning
217	Contains Duplicate	Hash Set (Seen Filter)
218	The Skyline Problem	Sweep Line Algorithm + Max Heap / TreeMap
219	Contains Duplicate II	Sliding Window + Hash Set
220	Contains Duplicate III	Bucket Sort / Sliding Window
221	Maximal Square	Dynamic Programming (2D Table or 1D Optimized)
222	Count Complete Tree Nodes	Binary Search on Leaf Nodes + Tree Height Properties
223	Rectangle Area	Inclusion-Exclusion Principle (Area1 + Area2 - Overlap)
224	Basic Calculator	Stack-Based Single Pass with Sign Tracking
225	Implement Stack using Queues	Single Queue Rotation
226	Invert Binary Tree	Recursive or Iterative In-Place Swap (DFS)
227	Basic Calculator II	Single-Pass Stack (Precedence Handling)
228	Summary Ranges	Two-Pointer / Sliding Boundary
229	Majority Element II	Boyer-Moore Voting Algorithm (Extended)
230	Kth Smallest Element in a BST	Iterative In-order Traversal with Early Stop
231	Power of Two	Bit Manipulation: (n > 0) and (n & (n-1) == 0)
232	Implement Queue using Stacks	Two Stacks (Amortized O(1) - "Lazy Transfer")
233	Number of Digit One	Digit DP / Mathematical Counting (Digit-by-Digit)
234	Palindrome Linked List	Slow/Fast Pointers + In-Place Reversal of Second Half
235	Lowest Common Ancestor of a BST	BST Property Traversal (Value Comparison)
236	Lowest Common Ancestor of a Binary Tree	Recursive Backtracking (Post-Order Traversal)
237	Delete Node in a Linked List	Value Overwriting / "Node Identity Theft"
238	Product of Array Except Self	Two-Pass Prefix/Suffix Accumulation
239	Sliding Window Maximum	Monotonic Deque (Double-Ended Queue)
240	Search a 2D Matrix II	Staircase Search (Top-Right Start)
241	Different Ways to Add Parentheses	Divide and Conquer + Memoization (Top-Down DP)

242	Valid Anagram	Frequency Counter (Fixed-size Array)
243	Shortest Word Distance	Two-Pointer Index Tracking (One Pass)
244	Shortest Word Distance II	Hash Map Pre-processing + Two Pointers
245	Shortest Word Distance III	One-Pass with Toggle/Previous Tracking
246	Strobogrammatic Number	Two Pointers + Mirror Map
247	Strobogrammatic Number II	Recursive Backtracking (Inside-Out Construction)
248	Strobogrammatic Number III	Recursive Generation with Range Pruning
249	Group Shifted Strings	Hash Map with Canonical Form Key
250	Count Univalue Subtrees	Post-Order DFS (Bottom-Up Flagging)
251	Flatten 2D Vector	Two-Pointer Iterators (Row & Col)
252	Meeting Rooms	Sorting by Start Time
253	Meeting Rooms II	Min-Heap (Priority Queue) of End Times
254	Factor Combinations	Backtracking with Path Tracking
255	Verify Preorder Sequence in BST	Monotonic Stack (Simulation of DFS)
256	Paint House	Dynamic Programming (In-place or 1D)
257	Binary Tree Paths	DFS with Backtracking / String Builder
258	Add Digits	Digital Root (Mathematical Modulo 9)
259	3Sum Smaller	Sort + Two Pointers (Left/Right)
260	Single Number III	Bit Manipulation (XOR + Rightmost Set Bit)
261	Graph Valid Tree	Union Find with Path Compression
262	Trips and Users (Algo Equivalent)	Hash Join / Hash Map Filtering
263	Ugly Number	Iterative Division (Prime Squeezing)
264	Ugly Number II	Three Pointers (Dynamic Programming)
265	Paint House II	DP with Minimum Tracking (Two-Minima Optimization)
266	Palindrome Permutation	Frequency Counting (Odd Count Limit)
267	Palindrome Permutation II	Backtracking (Half-String Construction)
268	Missing Number	Gauss' Sum Formula OR XOR Gates
269	Alien Dictionary	Topological Sort (Kahn's or DFS)
270	Closest BST Value	Binary Search (Iterative Descent)
271	Encode and Decode Strings	Chunked Transfer (Length + Delimiter)
272	Closest BST Value II	Two Stacks (Predecessor & Successor)
273	Integer to English Words	Recursive Triplet Grouping (Base 1000)
274	H-Index	Bucket Sort (Counting Citations)
275	H-Index II	Binary Search on H-Boundary
276	Paint Fence	Dynamic Programming (Two-State: Same/Different)
277	Find the Celebrity	Greedy Elimination (Two Pointers)
278	First Bad Version	Binary Search (Leftmost Boundary)
279	Perfect Squares	Dynamic Programming (1D) or BFS
280	Wiggle Sort	One-Pass Greedy Swapping
281	Zigzag Iterator	Queue of Iterators
282	Expression Add Operators	Backtracking with Evaluation-on-the-fly
283	Move Zeroes	Two Pointers (Read & Write)
284	Peeking Iterator	Look-Ahead (Caching)
285	Inorder Successor in BST	BST Property Navigation (Succ = smallest of right subtree)
286	Walls and Gates	Multi-Source BFS
287	Find the Duplicate Number	Floyd’s Cycle Finding (Tortoise and Hare)
288	Unique Word Abbreviation	Hash Map Pre-processing
289	Game of Life	In-Place State Encoding (Bit Manipulation)
290	Word Pattern	Two Hash Maps / One Map + Indexing
291	Word Pattern II	Backtracking with Bi-directional Mapping
292	Nim Game	Mathematical Modulo (Game Theory)
293	Flip Game	String Iteration / Simulation
294	Flip Game II	Backtracking + Memoization (Top-Down DP)
295	Find Median from Data Stream	Two Heaps (Min-Heap & Max-Heap)
296	Best Meeting Point	Median Property (Independent Dimensions)
297	Serialize and Deserialize Binary Tree	Pre-Order Traversal (DFS) with Sentinel Markers
298	Binary Tree Longest Consecutive Sequence	Single-Pass Bottom-Up / Top-Down DFS
299	Bulls and Cows	Single-Pass Frequency Array
300	Longest Increasing Subsequence	Patience Sorting (Binary Search + Tails Array)
301	Remove Invalid Parentheses	Breadth-First Search (BFS) / Level-order traversal with a Visited HashSet.
302	Smallest Rectangle Enclosing Black Pixels	Binary Search (on 1D projections of the 2D grid)

303	Range Sum Query - Immutable	Prefix Sum Array
304	Range Sum Query 2D - Immutable	2D Prefix Sum (Inclusion-Exclusion Principle)
305	Number of Islands II	Disjoint Set Union (Union-Find)
306	Additive Number	Backtracking with Pruning / Iterative Prefix Checking
307	Range Sum Query - Mutable	Segment Tree or Binary Indexed Tree (Fenwick Tree)
308	Range Sum Query 2D - Mutable	2D Binary Indexed Tree (Fenwick Tree)
309	Best Time to Buy and Sell Stock with Cooldown	Dynamic Programming (State Machine)
310	Minimum Height Trees	Topological Sort / BFS from Leaves (Peeling the Onion)
311	Sparse Matrix Multiplication	Zero-Skipping Sparse Traversal / Compressed Sparse Row (CSR)
312	Burst Balloons	Dynamic Programming (Divide and Conquer with Memoization)
313	Super Ugly Number	K-way Merge (Dynamic Programming with Multiple Pointers)
314	Binary Tree Vertical Order Traversal	Breadth-First Search (BFS) with Column Tracking (Hash Map)
315	Count of Smaller Numbers After Self	Merge Sort (Divide & Conquer) tracking original indices
316	Remove Duplicate Letters	Monotonic Stack + Greedy + Frequency Map
317	Shortest Distance from All Buildings	BFS from Buildings (Multi-Source Distance Accumulation)
318	Maximum Product of Word Lengths	Bitmasking (Bitwise AND Comparison)
319	Bulb Switcher	Mathematical Pattern (Square Root)
320	Generalized Abbreviation	Backtracking (Keep or Abbreviate)
321	Create Maximum Number	Monotonic Stack + Lexicographical Merge
322	Coin Change	Dynamic Programming (Bottom-Up)
323	Number of Connected Components	Disjoint Set Union (DSU)
324	Wiggle Sort II	Quickselect + Virtual Indexing (3-way Partition)
325	Max Size Subarray Sum Equals k	Prefix Sum + Hash Map (Two-Sum Variation)
326	Power of Three	Mathematical Constant Division (Base-10 Log or Max Int)
327	Count of Range Sum	Modified Merge Sort (Divide & Conquer)
328	Odd Even Linked List	Two-Pointer In-Place Re-linking
329	Longest Increasing Path in a Matrix	DFS + Memoization (Topological Sort Pattern)
330	Patching Array	Greedy Range Expansion
331	Verify Preorder Serialization of a Binary Tree	Graph Degree Logic (Out-degree vs In-degree) / Slot Counting
332	Reconstruct Itinerary	Hierholzer's Algorithm (Eulerian Path via DFS)
333	Largest BST Subtree	Post-order Traversal (Bottom-Up Information Passing)
334	Increasing Triplet Subsequence	Greedy Candidate Maintenance (Two-Variable Sweep)
335	Self Crossing	Pattern Categorization (6-Segment Window)
336	Palindrome Pairs	Trie (Prefix/Suffix) or Hashing with Word Splitting
337	House Robber III	Tree Dynamic Programming (Post-order Traversal)
338	Counting Bits	Dynamic Programming (Most Significant Bit / Least Significant Bit)
339	Nested List Weight Sum	Depth-First Search (DFS) or Breadth-First Search (BFS)
340	Longest Substring with At Most K Distinct Characters	Sliding Window + Hash Map (Two Pointers)
341	Flatten Nested List Iterator	Lazy Iterator with Stack (Depth-First Search)
342	Power of Four	Bit Manipulation + Masking
343	Integer Break	Dynamic Programming or Mathematical Optimization (Rule of 3s)
344	Reverse String	Two Pointers (In-place Swap)
345	Reverse Vowels of a String	Two Pointers (Vowel-Selective Swap)
346	Moving Average from Data Stream	Double-Ended Queue (Deque) + Running Sum
347	Top K Frequent Elements	Bucket Sort or Min-Heap
348	Design Tic-Tac-Toe	Row/Column/Diagonal Sum Counters
349	Intersection of Two Arrays	Two HashSets (Unique Intersection)
350	Intersection of Two Arrays II	Frequency Map or Sort + Two Pointers
351	Android Unlock Patterns	Backtracking with Pre-calculated Jump Table
352	Data Stream as Disjoint Intervals	Ordered Map (TreeMap) / Balanced BST
353	Design Snake Game	Deque + HashSet (Constant-Time Movement and Collision)
354	Russian Doll Envelopes	Sort + Longest Increasing Subsequence (LIS) via Binary Search
355	Design Twitter	Max-Heap / K-way Merge (Pull Model)
356	Line Reflection	Hash Set + Extremity Midpoint
357	Count Numbers with Unique Digits	Combinatorics (Permutations)
358	Rearrange String k Distance Apart	Max-Heap + Cooling Queue (Greedy)
359	Logger Rate Limiter	Hash Map with Expiration Timestamps
360	Sort Transformed Array	Two Pointers (Parabolic Extremity)
361	Bomb Enemy	Dynamic Programming (Row & Column Pre-computation)
362	Design Hit Counter	Circular Buffer (Fixed-size Array)
363	Max Sum of Rectangle No Larger Than K	2D Prefix Sum + 1D TreeSet Range Search

364	Nested List Weight Sum II	Single-Pass Level Accumulation
365	Water and Jug Problem	Number Theory (GCD / Bézout's Identity)
366	Find Leaves of Binary Tree	DFS with Height-Based Grouping
367	Valid Perfect Square	Binary Search (Range [1, num])
368	Largest Divisible Subset	Dynamic Programming (Sort + Longest Path in DAG)
369	Plus One Linked List	Two-Pointer (Sentinel + Last Non-9)
370	Range Addition	Difference Array (Prefix Sum Optimization)
371	Sum of Two Integers	Bitwise XOR (Sum) & AND-Shift (Carry) Loop
372	Super Pow	Modular Exponentiation + Recursive Property
373	Find K Pairs with Smallest Sums	Min-Heap (Priority Queue) with Dijkstra-like Expansion
374	Guess Number Higher or Lower	Binary Search
375	Guess Number Higher or Lower II	Interval Dynamic Programming (Minimax)
376	Wiggle Subsequence	Greedy (One-Pass Peak-Valley Counting)
377	Combination Sum IV	Dynamic Programming (Bottom-Up 1D)
378	Kth Smallest Element in a Sorted Matrix	Binary Search on Value Range
379	Design Phone Directory	Queue + HashSet (Available Pool)
380	Insert Delete GetRandom O(1)	HashMap + Dynamic Array (Swap with Last)
381	Insert Delete GetRandom O(1) - Duplicates allowed	HashMap of Sets + Dynamic Array
382	Linked List Random Node	Reservoir Sampling
383	Ransom Note	Frequency Array (Counting Sort Pattern)
384	Shuffle an Array	Fisher-Yates Shuffle Algorithm
385	Mini Parser	Stack-Based Iterative Parsing
386	Lexicographical Numbers	DFS (Pre-order Traversal of a 10-ary Tree)
387	First Unique Character in a String	Frequency Array (Hash Map Pattern)
388	Longest Absolute File Path	Stack / Array Depth Tracking
389	Find the Difference	Bit Manipulation (XOR) or Character Summation
390	Elimination Game	Recursive Pattern / Head-Pointer Tracking
391	Perfect Rectangle	Corner Count Set + Area Summation
392	Is Subsequence	Two Pointers (Greedy Matching)
393	UTF-8 Validation	Bit Manipulation (Bitmasking)
394	Decode String	Two Stacks (Count Stack & String Stack)
395	Longest Substring with At Least K Repeating Characters	Divide and Conquer (Split on Invalid Characters)
396	Rotate Function	Mathematical DP (Linear Recurrence)
397	Integer Replacement	Greedy (Bit Manipulation)
398	Random Pick Index	Reservoir Sampling (Memory Efficient)
399	Evaluate Division	Graph + BFS/DFS or Disjoint Set Union (DSU) with Weights
400	Nth Digit	Mathematical Interval Binary Search
401	Binary Watch	Backtracking Iterate through the 10 LEDs, choosing to turn them on/off until `turnedOn` is met.
402	Remove K Digits	Monotonic Stack Keep digits in increasing order. Pop if current digit < top of stack and K > 0.
403	Frog Jump	DP (Hash Map of Sets) Map each stone to a Set of jump lengths that reached it.
404	Sum of Left Leaves	DFS/BFS Traversal (Same as Brute force, as you must visit every node to check).
405	Convert a Number to Hexadecimal	Bit Manipulation Group every 4 bits using bitwise AND (num & 15), then right shift by 4 (num >>> 4).
406	Queue Reconstruction by Height	Sort + Insert Sort descending by height (h), then ascending by (k). Insert people into a list at index k.
407	Trapping Rain Water II	Min-Heap + BFS Push perimeter to min-heap. Pop lowest boundary, process neighbors, add to heap. "Shrinking perimeter".
408	Valid Word Abbreviation	Two Pointers Pointer `i` traverses the word, `j` traverses abbreviation. Parse numbers to skip `i` forward.
409	Longest Palindrome	Hash Map / Frequency Array Count char frequencies. Add all even counts. Add (odd-1) for odd counts. Add 1 if any odd exists.
410	Split Array Largest Sum	Binary Search on Answer Binary search the "max sum" (Low: max(arr), High: sum(arr)). Greedily count chunks needed for each g
411	Minimum Unique Word Abbreviation	Bitmask + Backtracking Find character differences as bitmasks. Use DFS to find the shortest abbreviation mask that covers all diff
412	Fizz Buzz	String Concatenation (Better for scale/many conditions). Append "Fizz" if mod 3, append "Buzz" if mod 5. Output if empty.
413	Arithmetic Slices	1D DP / Two Pointers If arr[i]-arr[i-1] == arr[i-1]-arr[i-2], current slices = prev slices + 1. Add to total sum.
414	Third Maximum Number	Top-3 Variables Keep `first`, `second`, `third` initialized to -Infinity. Update them sequentially as you iterate O(N).
415	Add Strings	Two Pointers (Right to Left) Traverse both strings backwards. Add digits plus carry. Append to StringBuilder and reverse.
416	Partition Equal Subset Sum	0/1 Knapsack DP (1D) Use a boolean array `dp` where `dp[i]` is true if sum `i` is possible.
417	Pacific Atlantic Water Flow	Multi-source BFS/DFS Start from the Pacific edges and "flow uphill." Do the same for Atlantic. Find the intersection.
418	Sentence Screen Fitting	DP / Pre-calculated Jumps Join sentence with spaces. At each screen column start, pre-calculate which word the next row will star
419	Battleships in a Board	Single Pass (Head Counting) Iterate through the grid. Only count an 'X' if it doesn't have an 'X' above it or to its left.
420	Strong Password Checker	Greedy + Case Analysis Categorize by length: <6 (Insert), 6-20 (Replace), >20 (Delete then Replace). Priority on deleting from r
421	Max XOR of Two Numbers	Bitwise Trie Insert all numbers into a Binary Trie. For each number, traverse the trie trying to pick the opposite bit.
422	Valid Word Square	Direct Indexing Compare `words[i].charAt(j)` with `words[j].charAt(i)` with boundary checks.
423	Reconstruct Digits from English	Unique Char Counting Observe unique identifiers: 'z' is only in zero, 'w' in two, 'x' in six. Subtract counts in a specific order.

424	Longest Repeating Char Replacement	Sliding Window (Expand/Contract) Use two pointers. Keep track of `maxFreq` in current window. If `windowSize - maxFreq > k`
425	Word Squares	Trie-Based Backtracking Use a Trie to instantly find all words that start with a specific prefix (the prefix being the current column's
426	Convert BST to Sorted DLL	In-Place In-order Maintain a `first` and `last` pointer. Connect `last.right = current` and `current.left = last` during traversal.
427	Construct Quad Tree	Divide & Conquer Recursively check if a quadrant is "leaf". If all 4 children are leaves and same value, merge.
428	Serialize/Deserialize N-ary Tree	Pre-order with Child Count Serialize as `Value, ChildCount`. E.g., `1,3,2,0,3,0,4,0` (Node 1 has 3 children...).
429	N-ary Tree Level Order Traversal	BFS (Queue) Use a Queue. Process all nodes in the current queue (one level) before moving to the next.
430	Flatten Multilevel DLL	Iterative Stack / Pointer Weaving When you hit a `child`, "insert" the child branch between `current` and `current.next`.
431	Encode N-ary Tree to Binary Tree	LCRS (Recursive Pointer Re-assignment) Use a recursive DFS to transform pointers without creating new nodes.
432	All O`one Data Structure	Hash Map + Doubly Linked List of Buckets Map strings to nodes. Nodes are buckets of strings with the same count. Buckets are lir
433	Minimum Genetic Mutation	BFS (Shortest Path) Use a Queue. For each mutation, check if it's in the "bank". Use a visited set to avoid cycles.
434	Number of Segments in a String	Single Pass Count Iterate through characters. Count a segment whenever you see a non-space character that follows a space (or
435	Non-overlapping Intervals	Greedy (Sort by End Time) Sort intervals by their end times. Always pick the one that ends earliest to leave maximum room for o
436	Find Right Interval	Binary Search + Sort Store all `start` values in a sorted list with original indices. For each `end`, binary search the list.
437	Path Sum III	Prefix Sum + DFS Use a Hash Map to store "Prefix Sum -> Frequency". If (CurrentSum - Target) exists in the map, a path is foun
438	Find All Anagrams in a String	Sliding Window + Frequency Map Use an array of size 26. As the window moves, increment the "entering" char and decrement th
439	Ternary Expression Parser	Stack (Right-to-Left) Scan the string backwards. If you hit a `?`, pop the next two values from the stack and push the result base
440	K-th Smallest in Lexicographical Order	Denary Tree Traversal Visualize numbers as a 10-ary tree (1, 10, 100...). Calculate "steps" between prefixes to skip whole subtree
441	Arranging Coins	Binary Search / Math Search for k such that frac{k(k+1)}{2} le N or use the quadratic formula directly.
442	Find All Duplicates in an Array	In-place Marking For each number x, go to index |x|-1 and flip the value to negative. If already negative, it's a duplicate.
443	String Compression	Two Pointers (Read/Write) Use `read` and `write` pointers. `read` finds the end of a group; `write` updates the array in-place.
444	Sequence Reconstruction	Topological Sort (Kahn's) Build a graph of dependencies. Check if at every step there is exactly 1 node with an in-degree of 0.
445	Add Two Numbers II	Two Stacks Push all nodes of both lists onto two stacks. Pop from both to add from right-to-left.
446	Arithmetic Slices II - Subsequence	DP with Map Matrix Use an array of HashMaps `dp[i]`, where `dp[i][diff]` stores count of slices ending at i with difference `diff`.
447	Number of Boomerangs	Hash Map of Distances For each point i, calculate distances to all other points j. Store counts in a Map `dist: count`. Result is sum
448	Find All Numbers Disappeared in an Array	In-place Negation For each value x, treat |x|-1 as an index and make the value at that index negative. Positives remaining are the
449	Serialize and Deserialize BST	Pre-order + Bounds Use Pre-order without null markers. Rebuild using BST property (current value must be within [min, max]).
450	Delete Node in a BST	Recursive Re-linking Find node. If it's a leaf, remove. If 1 child, replace with child. If 2 children, replace with In-order Successor (
451	Sort Characters By Frequency	Bucket Sort Use an array where the index is the frequency. Put characters into the bucket at `frequency[char]` index.
452	Minimum Number of Arrows to Burst Balloons	Greedy (Sort by End) Sort balloons by their end coordinates. Shoot one arrow at the end of the first balloon; skip all that overlap.
453	Minimum Moves to Equal Array Elements	Mathematical Reduction Incrementing n-1 elements is mathematically equivalent to decrementing 1 element. Sum up the distance
454	4Sum II	Hash Map (Meet-in-the-Middle) Calculate all A+B and store in a Map `sum: count`. Then iterate C+D and check for -(C+D) in the
455	Assign Cookies	Two Pointers + Greedy Sort both arrays. Give the smallest cookie that satisfies the child with the smallest greed factor.
456	132 Pattern	Monotonic Stack (Right-to-Left) Keep a stack of potential "2" values (middle magnitude) and track the largest "3" value found so f
457	Circular Array Loop	Two Pointers (Slow/Fast) Use Floyd's Cycle-Finding algorithm. A fast pointer moves 2 steps, slow moves 1. Check for direction cha
458	Poor Pigs	Information Theory (N-Dimensional Hypercube) Each pig represents a dimension. The number of states a pig can have is (Tests +
459	Repeated Substring Pattern	String Doubling Trick Concatenate S+S. Remove the first and last characters. If S is still a substring of the result, it has a repeate
460	LFU Cache	Hash Map + Linked List of DLLs Map keys to nodes. Have a second "Frequency Map" that points to a Doubly Linked List of all keys
461	Hamming Distance	Bitwise XOR + Bit Counting XOR the two numbers (x oplus y). Count the set bits using Brian Kernighan’s algorithm.
462	Min Moves to Equal Elements II	Median Target (Greedy) The point that minimizes the sum of absolute differences is the median. Sort and find the middle.
463	Island Perimeter	Single Pass (Shared Edge) Count land cells x 4. Subtract 2 for every shared edge (horizontal or vertical neighbor).
464	Can I Win	Recursion + Bitmask Memoization Use an integer bitmask to represent which numbers are used. Cache result in a Map `mask: wi
465	Optimal Account Balancing	Backtracking on Net Balances Flatten everyone into their net balance. Try to settle one person's balance with any other person of
466	Count The Repetitions	Cycle Detection (Pigeonhole) Record the `index` in s_2 after each s_1. If the same index repeats, a cycle is found. Calculate rema
467	Unique Substrings in Wraparound String	DP (Max Length per Char) Track the longest "consecutive" sequence ending at each character ('a' through 'z'). The sum of these 2
468	Validate IP Address	Regex or Modular Parsing Use pre-defined Regex patterns for IPv4 and IPv6 to handle boundaries and valid character ranges in or
469	Convex Polygon	Cross Product (Orientation) For every three consecutive points, calculate the cross product. All cross products must have the same
470	Implement Rand10() Using Rand7()	Rejection Sampling (7x7 Matrix) Create a range of 1-49 using `7 * (rand7() - 1) + rand7()`. If the result is > 40, reject it. Other
471	Encode String with Shortest Length	Interval DP `dp[i][j]` stores the shortest encoded string for `s[i...j]`. Check for patterns using `(s+s).indexOf(s, 1)`.
472	Concatenated Words	Trie + DFS/Memoization Sort words by length. Insert into a Trie. For each word, use DFS to see if it can be formed by existing Trie
473	Matchsticks to Square	DFS with Pruning / Bitmask DP Sort matchsticks descending. Try placing in a side; if it exceeds `target`, backtrack immediately.
474	Ones and Zeroes	2D Knapsack DP `dp[i][j]` is max strings using `i` zeros and `j` ones. Update backwards to save space.
475	Heaters	Binary Search (Sorted Heaters) Sort heaters. For each house, binary search for the two closest heaters (one left, one right).
476	Number Complement	Bitwise Mask XOR Find the highest set bit. Create a mask of all 1s of that length. Result is `num ^ mask`.
477	Total Hamming Distance	Bit-by-Bit Counting For each bit position (0-31), count how many numbers have a '1' (k) and how many have a '0' (n-k). Distance
478	Generate Random Point in a Circle	Inverse Transform (Polar) Generate theta in [0, 2n] and r = R √{rand()}. The square root is crucial for uniform density.
479	Largest Palindrome Product	Greedy Palindrome Generation Start from the largest possible palindrome (e.g., 999999) and work downwards. For each, check if
480	Sliding Window Median	Two Heaps (Balance) / TreeSet Use a Max-Heap for the left half and a Min-Heap for the right half. Use "Lazy Removal" with a Hash
481	Magical String	Two Pointers (Generator/Consumer) Use an `int[]` or `char[]` and two pointers: `head` (reading how many) and `tail` (writing
482	License Key Formatting	Reverse Traversal Start from the end of the string. Collect K characters, add a dash, and repeat. Reverse the result at the end.
483	Smallest Good Base	Binary Search on Length The number of '1's can only be between 2 and 64. For each fixed length m, binary search for base k.
484	Find Permutation	Stack-Based Reversal Use a stack to push 1, 2, 3 dots If you see 'I', pop everything to the result. If you see 'D', keep pushing.

485	Max Consecutive Ones	Single Pass Counter Iterate through the array. Increment `count` if 1; if 0, update `max` and reset `count`.
486	Predict the Winner	DP (1D/2D) `dp[i][j]` stores max score diff from subarray i...j. `dp[i][j] = max(nums[i]-dp[i+1][j], nums[j]-dp[i][j-1])`.
487	Max Consecutive Ones II	Sliding Window (Expandable) Maintain a window with at most one '0'. If zeros > 1, shrink left pointer.
488	Zuma Game	DFS + Memoization + Pruning Use string compression (remove 3+ matches) and a Map to store "State: MinBalls". Skip redundant
489	Robot Room Cleaner	Backtracking with Coordinate Set Maintain a `visited` Set of `(row, col)`. Always try to clean, move, recurse, then return and fac
490	The Maze	BFS/DFS with Sliding Logic From a cell, move in one direction using a `while` loop until hitting a wall. Treat the *stop point* as a
491	Non-decreasing Subsequences	DFS with Local Set Use backtracking. At each recursive level, use a `HashSet` to skip duplicate numbers to avoid redundant subs
492	Construct the Rectangle	Start at Square Root Start W at lfloor√{N}rfloor and decrement until N % W == 0. This ensures L and W are as close as possible.
493	Reverse Pairs	Merge Sort (Divide & Conquer) During the merge step, both halves are sorted. Use two pointers to count reverse pairs before mer
494	Target Sum	DP (Subset Sum) Transform to finding a subset with sum (S + target)/2. Solve using 1D Knapsack DP.
495	Teemo Attacking	One-Pass Greedy Iterate through attacks. If the gap between attacks < duration, add the gap. Else, add the full duration.
496	Next Greater Element I	Monotonic Stack + HashMap Pre-process nums2 using a stack to find all "Next Greater" pairs. Store in a Map `num: next_greater
497	Random Point in Non-overlapping Rectangles	Prefix Sum + Binary Search Calculate areas of all rectangles. Create a prefix sum array. Randomly pick a "total area" value and bi
498	Diagonal Traverse	Simulation (Direction Flag) Use a direction flag (Up/Down). Calculate new i, j based on boundaries (e.g., if at last column, move d
499	The Maze III	Dijkstra with String Priority Use a Priority Queue. Store `(distance, path_string, x, y)`. Only update if distance is smaller OR dista
500	Keyboard Row	Pre-mapped Char-to-Row Create a Map or Array of size 26 where `row[char]` is 1, 2, or 3. Check if all chars in a word return the
501	Find Mode in Binary Search Tree	Inorder Traversal (Morris Traversal) with State Tracking variables.
502	IPO	Two Heaps (Min-Heap for Capital, Max-Heap for Profit) / Greedy.
503	Next Greater Element II	Monotonic Decreasing Stack (Looping 2N times).
504	Base 7	Math / Simulation (Repeated Division and Modulo).
505	The Maze	BFS (Shortest Path) using a Queue and a Distance 2D Array / Dijkstra.
506	Relative Ranks	Sorting with Index Tracking / Max-Heap.
507	Perfect Number	Math / Pair Divisors up to Square Root of N.
508	Most Frequent Subtree Sum	Post-order Traversal (Bottom-Up DFS) + Hash Map.
509	Fibonacci Number	Iterative Bottom-Up Dynamic Programming (State variables).
510	Inorder Successor in BST II	BST Traversal leveraging Parent Pointers.
511	Game Play Analysis I (SQL)	SQL `GROUP BY` with Aggregate Function `MIN()`.
512	Game Play Analysis II (SQL)	Window Function `ROW_NUMBER() OVER(PARTITION BY player_id ORDER BY event_date)`.
513	Find Bottom Left Tree Value	BFS (Level Order Traversal) but strictly Right-to-Left.
514	Freedom Trail	Dynamic Programming (Top-Down Memoization or 2D Bottom-Up).
515	Find Largest Value in Each Tree Row	BFS (Level Order Traversal) tracking max per level.
516	Longest Palindromic Subsequence	2D Dynamic Programming (Bottom-Up).
517	Super Washing Machines	Math / Prefix Sum (Tracking flow and bottlenecks).
518	Coin Change II	1D Dynamic Programming (Unbounded Knapsack).
519	Random Flip Matrix	Virtual 1D Array + Hash Map mapping (Fisher-Yates Shuffle).
520	Detect Capital	Single-Pass Character Counting.
521	Longest Uncommon Subsequence I	Brain Teaser / Math (Length Comparison).
522	Longest Uncommon Subsequence II	Custom Sort + Two Pointers (Subsequence Check).
523	Continuous Subarray Sum	Prefix Sum + Modulo Hash Map.
524	Longest Word in Dictionary through Deleting	Sort Dictionary + Two Pointers.
525	Contiguous Array	Prefix Sum (Treat 0 as -1) + Hash Map.
526	Beautiful Arrangement	Backtracking with Pruning + Visited Array (or Bitmask).
527	Word Abbreviation	Group by Length/Last Char + Trie for Longest Common Prefix (LCP).
528	Random Pick with Weight	Prefix Sum Array + Binary Search.
529	Minesweeper	DFS / BFS from the click point with strict boundary/number checks.
530	Minimum Absolute Difference in BST	In-Order Traversal (natively sorted) + Track `prev` Node.
531	Lonely Pixel I	Row/Column Tally Arrays.
532	K-diff Pairs in an Array	Frequency Hash Map or Sorted Two-Pointers.
533	Lonely Pixel II	Hash Map for Row Patterns.
534	Game Play Analysis III (SQL)	Window Function: `SUM(games_played) OVER(PARTITION BY player_id ORDER BY event_date)`.
535	Encode and Decode TinyURL	Hash Map with Random Base62 Encoding.
536	Construct Binary Tree from String	Stack-based Iterative Construction.
537	Complex Number Multiplication	Math (FOIL Method) + String Split.
538	Convert BST to Greater Tree	Reverse In-Order Traversal (Right → Root → Left).
539	Minimum Time Difference	Sorting or Bucket Sort (Boolean Array).
540	Single Element in a Sorted Array	Binary Search on Even Indices.
541	Reverse String II	Two-Pointer In-Place Swap (Jumping by 2k).
542	01 Matrix	Multi-Source BFS or 2-Pass DP.
543	Diameter of Binary Tree	Bottom-Up DFS (Post-order Traversal).
544	Output Contest Matches	In-place Array Transformation.
545	Boundary of Binary Tree	Pre-order / In-order / Post-order Coordination.

546	Remove Boxes	3D Dynamic Programming (Top-Down with Memoization).
547	Number of Provinces	DFS / BFS or Union-Find (Disjoint Set Union).
548	Split Array with Equal Sum	Prefix Sum + Hash Set for Mid-Point Optimization.
549	Binary Tree Longest Consecutive Sequence II	Bottom-Up DFS (returning pairs).
550	Game Play Analysis IV (SQL)	Subquery / CTE with Date Arithmetic.
551	Student Attendance Record I	Single Pass State Tracking.
552	Student Attendance Record II	Multi-Dimensional Dynamic Programming (State Compression).
553	Optimal Division	Math / Greedy Observation.
554	Brick Wall	Hash Map of Edge Frequencies.
555	Split Concatenated Strings	Greedy Pre-reversal + Optimized Split Scan.
556	Next Greater Element III	Next Permutation Algorithm (Linear Scan).
557	Reverse Words in a String III	Two-Pointer In-Place Swap.
558	Logical OR of Two Quadtrees	Recursive Tree Merging (Divide & Conquer).
559	Maximum Depth of N-ary Tree	Simple DFS Recursion.
560	Subarray Sum Equals K	Prefix Sum + Hash Map.
561	Array Partition I	Sorting + Greedy Pairing.
562	Longest Line of Consecutive One in Matrix	3D Dynamic Programming (Spatial State).
563	Binary Tree Tilt	Post-order Traversal (Bottom-Up Summing).
564	Find the Closest Palindrome	Prefix Manipulation + Candidate Testing.
565	Array Nesting	Visited Set / In-Place Marking.
566	Reshape the Matrix	Direct Index Mapping (Modulo Arithmetic).
567	Permutation in String	Sliding Window + Frequency Map (Fixed Size).
568	Maximum Vacation Days	2D Dynamic Programming (Bottom-Up).
569	Median Employee Salary (SQL)	Row Numbering / Window Functions.
570	Managers with at Least 5 Direct Reports (SQL)	Group By + Having Clause.
571	Find Median Given Frequency of Numbers (SQL)	Running Total (Cumulative Frequency) Analysis.
572	Subtree of Another Tree	Merkle Hashing or String Serialization (KMP).
573	Squirrel Simulation	Greedy Distance Calculation (Total Distance - Max Saving).
574	Winning Candidate (SQL)	Join + Group By + Limit.
575	Distribute Candies	Set Size vs. Half-Length Comparison.
576	Out of Boundary Paths	3D Dynamic Programming (Memoization).
577	Employee Bonus (SQL)	Left Join with Null/Value Filtering.
578	Get Highest Answer Rate Question (SQL)	Group By + Aggregate Ratio + Order By.
579	Find Cumulative Salary of an Employee (SQL)	Window Function (ROWS BETWEEN).
580	Count Student Number in Departments (SQL)	Left Join + Group By.
581	Shortest Unsorted Continuous Subarray	Two-Pass Monotonic Logic (Min/Max Tracking).
582	Kill Process	Hash Map Adjacency List + BFS/DFS.
583	Delete Operation for Two Strings	2D Dynamic Programming (LCS).
584	Find Customer Referee (SQL)	Filtered Select with Null Handling.
585	Investments in 2016 (SQL)	Window Functions or Subqueries with Group By.
586	Customer Placing the Largest Number of Orders (SQL)	Group By + Order By Count + Limit 1.
587	Erect the Fence	Monotone Chain Algorithm (Sorting + Cross Product).
588	Design In-Memory File System	Trie-based Directory Structure.
589	N-ary Tree Preorder Traversal	Recursive DFS or Iterative Stack.
590	N-ary Tree Postorder Traversal	Recursive DFS or Iterative "Reverse Preorder".
591	Tag Validator	Stack-based Parsing + State Machine.
592	Fraction Addition and Subtraction	Common Denominator (LCM) + Final GCD.
593	Valid Square	Distance Sorting / Hash Set of Lengths.
594	Longest Harmonious Subsequence	Frequency Hash Map.
595	Big Countries (SQL)	Simple WHERE clause with OR.
596	Classes More Than 5 Students (SQL)	Group By + Having Clause.
597	Friend Requests I: Overall Acceptance Rate (SQL)	Distinct Counting + Null Handling.
598	Range Addition II	Minimum Boundary Tracking.
599	Minimum Index Sum of Two Lists	Hash Map for Index Lookup.
600	Non-negative Integers without Consecutive Ones	Digit DP / Fibonacci Pattern.
601	Human Traffic of Stadium (SQL)	Window Functions (id - ROW_NUMBER() grouping).
602	Friend Requests II: Who Has the Most Friends (SQL)	UNION ALL followed by GROUP BY and ORDER BY LIMIT 1.
603	Consecutive Available Seats (SQL)	Self Join with absolute difference = 1 OR Window Functions (LAG/LEAD).
604	Design Compressed String Iterator	Lazy Evaluation / Two Pointers / State Machine.
605	Can Place Flowers	Greedy Algorithm / Single Pass Array Scan.
606	Construct String from Binary Tree	Preorder DFS with Conditional Appending.

607	Sales Person (SQL)	Subquery / NOT IN clause.
608	Tree Node (SQL)	CASE WHEN with IN clause (Subquery).
609	Find Duplicate File in System	Hash Map grouping (Content -> List of Paths).
610	Triangle Judgement (SQL)	Simple conditional logic: IF or CASE WHEN.
611	Valid Triangle Number	Sort + Two Pointers.
612	Shortest Distance in a Plane (SQL)	Cross Join with Duplicate Filtering (Upper Triangular Check).
613	Shortest Distance in a Line (SQL)	Sorting / Window Functions (LEAD/LAG).
614	Second Degree Follower (SQL)	Inner Join or Group By with IN clause.
615	Average Salary: Departments VS Company (SQL)	Window Functions (AVG OVER PARTITION).
616	Add Bold Tag in String	Boolean Mask + Interval Merging.
617	Merge Two Binary Trees	Recursive In-Place Merge.
618	Students Report By Geography (SQL)	Window Function (ROW_NUMBER) + Pivot (CASE WHEN).
619	Biggest Single Number (SQL)	Subquery Grouping with HAVING count = 1.
620	Not Boring Movies (SQL)	Direct SQL WHERE Clause + ORDER BY.
621	Task Scheduler	Greedy (Math-based Calculation).
622	Design Circular Queue	Two-Pointer Array with Modulo Arithmetic.
623	Add One Row to Tree	Recursive DFS with Depth Tracking.
624	Maximum Distance in Arrays	Greedy Min/Max Tracking.
625	Minimum Factorization	Greedy Factorization (Divisors 9 down to 2).
626	Exchange Seats (SQL)	Bitwise Logic / CASE statement with ID manipulation.
627	Swap Sex of Employees	SQL UPDATE + CASE WHEN
628	Maximum Product of Three Numbers	Sort OR Linear Scan for Top 3 and Bottom 2.
629	K Inverse Pairs Array	DP with Sliding Window Sum.
630	Course Schedule III	Greedy with Max-Priority Queue.
631	Design Excel Sum Formula	Directed Acyclic Graph (DAG) with Topological Updates.
632	Smallest Range Covering Elements from K Lists	Sliding Window with Min-Heap.
633	Sum of Square Numbers	Two Pointers or Fermat's Theorem.
634	Find the Derangement of An Array	Dynamic Programming (Subproblem relation).
635	Log Storage System	Ordered Map / TreeMap with Granularity Masking.
636	Exclusive Time of Functions	Stack-based Interval Calculation.
637	Average of Levels in Binary Tree	BFS (Level Order Traversal).
638	Shopping Offers	Backtracking with Memoization (DFS + Map).
639	Decode Ways II	Iterative Dynamic Programming (1D).
640	Solve the Equation	Linear Expression Reduction (Parsing).
641	Design Circular Deque	Circular Buffer / Fixed-size Array with Head/Tail Pointers.
642	Design Search Autocomplete System	Trie (Prefix Tree) + Min-Heap/Caching.
643	Maximum Average Subarray I	Sliding Window (Fixed Size).
644	Maximum Average Subarray II	Binary Search on Answer + Sliding Window.
645	Set Mismatch	Index-as-Hash / Negation Marker.
646	Maximum Length of Pair Chain	Greedy (Sort by End Time).
647	Palindromic Substrings	Expand Around Center.
648	Replace Words	Trie (Prefix Tree).
649	Dota2 Senate	Two Queues (R_index, D_index).
650	2 Keys Keyboard	Prime Factorization (Greedy/DP).
651	4 Keys Keyboard	Dynamic Programming (1D Array).
652	Find Duplicate Subtrees	Post-order Traversal + Serialization + Hash Map.
653	Two Sum IV - Input is a BST	DFS + Hash Set.
654	Maximum Binary Tree	Monotonic Stack.
655	Print Binary Tree	Recursive Matrix Filling (Height-based).
656	Coin Path	Dynamic Programming (Right-to-Left).
657	Robot Return to Origin	Net Displacement Counter.
658	Find K Closest Elements	Binary Search + Two Pointers.
659	Split Array into Consecutive Subsequences	Greedy with Frequency & Vacancy Maps.
660	Remove 9	Base-9 Conversion.
661	Image Smoother	Single-Pass Iteration / In-Place Bit Manipulation.
662	Maximum Width of Binary Tree	BFS / DFS with Positional Indexing.
663	Equal Tree Partition	Post-order Subtree Sums + Hash Set.
664	Strange Printer	Dynamic Programming (Range DP).
665	Non-decreasing Array	Greedy Local Correction.
666	Path Sum IV	Hash Map Traversal / Parent-Child Mapping.

667	Beautiful Arrangement II	Greedy Constructive Construction.
668	Kth Smallest Number in Multiplication Table	Binary Search on Answer.
669	Trim a Binary Search Tree	Recursive Pointer Re-assignment.
670	Maximum Swap	Greedy / Last-Position Tracking.
671	Second Minimum Node In a Binary Tree	Recursive Branch Pruning.
672	Bulb Switcher II	Mathematical Pattern Reduction.
673	Number of Longest Increasing Subsequence	Dynamic Programming (Two 1D Arrays).
674	Longest Continuous Increasing Subsequence	Sliding Window / Greedy.
675	Cut Off Trees for Golf Event	BFS / Dijkstra on Sorted Checkpoints.
676	Implement Magic Dictionary	Trie with DFS / Generalized Neighbors.
677	Map Sum Pairs	Trie with Weight Accumulation.
678	Valid Parenthesis String	Greedy (Min/Max Balance Tracking).
679	24 Game	Backtracking / Divide and Conquer.
680	Valid Palindrome II	Two Pointers with One-Time Divergence.
681	Next Closest Time	Simulation / Minute-by-Minute Increment.
682	Baseball Game	Stack Data Structure.
683	K Empty Slots	Sliding Window / Monotonic Queue.
684	Redundant Connection	Union-Find (Disjoint Set Union).
685	Redundant Connection II	DSU + In-degree Tracking.
686	Repeated String Match	Upper Bound Estimation + String Search.
687	Longest Univalue Path	Post-order DFS (Bottom-Up).
688	Knight Probability in Chessboard	Dynamic Programming (2D/3D).
689	Maximum Sum of 3 Non-Overlapping Subarrays	DP with Sliding Window / Prefix-Suffix Max.
690	Employee Importance	Hash Map + DFS/BFS.
691	Stickers to Spell Word	DP with Bitmask / Memoization.
692	Top K Frequent Words	Hash Map + Min-Heap (size K).
693	Binary Number with Alternating Bits	Bit Manipulation (XOR Shift).
694	Number of Distinct Islands	DFS + Path Serialization.
695	Max Area of Island	DFS with In-Place Sinking (Visited marking).
696	Count Binary Substrings	Group Consecutive Counts.
697	Degree of an Array	Hash Map (First/Last/Count).
698	Partition to K Equal Sum Subsets	Backtracking with Pruning + State Compression.
699	Falling Squares	Coordinate Compression + Segment Tree / Sparse Map.
700	Search in a Binary Search Tree	BST Property Navigation (Binary Search).
701	Insert into a Binary Search Tree	BST Traversal (Iterative)
702	Search in a Sorted Array of Unknown Size	Exponential Expansion + Binary Search
703	Kth Largest Element in a Stream	Min-Heap (Size K)
704	Binary Search	Two-Pointer Binary Search
705	Design HashSet	Chaining (Array of Linked Lists / BSTs)
706	Design HashMap	Array of Linked Lists (Chaining)
707	Design Linked List	Doubly Linked List w/ Dummy Nodes
708	Insert into a Sorted Circular Linked List	Two-Pointer Ring Traversal
709	To Lower Case	ASCII Bit Manipulation / Offset
710	Random Pick with Blacklist	Hash Map Remapping (Virtual Array)
711	Number of Distinct Islands II	Canonical Hashing (DFS + Sort)
712	Minimum ASCII Delete Sum for Two Strings	Dynamic Programming (LCS Variant)
713	Subarray Product Less Than K	Sliding Window
714	Best Time to Buy and Sell Stock with Transaction Fee	State Machine DP / Greedy
715	Range Module	Interval Merging / Segment Tree (TreeMap)
716	Max Stack	Doubly Linked List + TreeMap (or Lazy Deletion PQ)
717	1-bit and 2-bit Characters	Greedy Pointer / Linear Jump
718	Maximum Length of Repeated Subarray	Dynamic Programming (2D)
719	Find K-th Smallest Pair Distance	Binary Search on Distance + Sliding Window
720	Longest Word in Dictionary	Trie (Prefix Tree) + DFS/BFS
721	Accounts Merge	Disjoint Set Union (DSU) / Union-Find
722	Remove Comments	State Machine / Deterministic Parsing
723	Candy Crush	In-place State Flagging & Two-Pointer Drop
724	Find Pivot Index	Prefix Sum / Total Sum Tracking
725	Split Linked List in Parts	Length Counting + Mathematical Chunking
726	Number of Atoms	Recursion / Stack with HashMaps
727	Minimum Window Subsequence	Dynamic Programming (2D) or Two-Pointer Slide

728	Self Dividing Numbers	Modulo Digit Extraction
729	My Calendar I	Balanced BST / TreeMap (Ordered Map)
730	Count Different Palindromic Subsequences	Dynamic Programming (2D Range DP)
731	My Calendar II	Sweep Line Algorithm (TreeMap)
732	My Calendar III	Segment Tree with Lazy Propagation
733	Flood Fill	Depth-First Search (DFS) / BFS
734	Sentence Similarity	Hash Set of Pair Strings
735	Asteroid Collision	Monotonic Stack
736	Parse Lisp Expression	Recursive Descent Parser + Scoped Symbol Table
737	Sentence Similarity II	Disjoint Set Union (DSU)
738	Monotone Increasing Digits	Greedy Digit Manipulation
739	Daily Temperatures	Monotonic Decreasing Stack
740	Delete and Earn	Dynamic Programming (House Robber Variation)
741	Cherry Pickup	3D Dynamic Programming (Simultaneous Path)
742	Closest Leaf in a Binary Tree	Graph Conversion + BFS
743	Network Delay Time	Dijkstra’s Algorithm (Min-Priority Queue)
744	Find Smallest Letter Greater Than Target	Binary Search (Left-most target)
745	Prefix and Suffix Search	Trie of Combined Strings (suffix + { + prefix)
746	Min Cost Climbing Stairs	Dynamic Programming (Space Optimized)
747	Largest Number At Least Twice of Others	Single Pass Max Tracking
748	Shortest Completing Word	Prime Product Hashing or Frequency Array
749	Contain Virus	Greedy Simulation with BFS/DFS
750	Number Of Corner Rectangles	Row-Pair Comparison (DP-lite)
751	IP to CIDR	Greedy Bit Manipulation (Trailing Zeros)
752	Open the Lock	Breadth-First Search (BFS) / Bidirectional BFS
753	Cracking the Safe	Hierholzer’s Algorithm (Eulerian Path)
754	Reach a Number	Mathematical Greedy + Parity Check
755	Pour Water	Two-Pointer Greedy Simulation
756	Pyramid Transition Matrix	Backtracking with Memoization (Bitmasking)
757	Set Intersection Size At Least Two	Greedy (Interval Sorting)
758	Bold Words in String	Boolean Array Masking
759	Employee Free Time	Min-Heap (K-Way Merge)
760	Find Anagram Mappings	Hash Map Indexing
761	Special Binary String	Recursion + String Sorting (Divide & Conquer)
762	Prime Number of Set Bits in Binary Representation	Bit Manipulation (bitCount) + Constant Set Check
763	Partition Labels	Greedy (Last Occurrence Map)
764	Largest Plus Sign	Dynamic Programming (4-Directional Prefix)
765	Couples Holding Hands	Greedy / Cycle Decomposition (DSU)
766	Toeplitz Matrix	One-Pass Neighbor Comparison
767	Reorganize String	Max-Heap / Greedy with Interleaving
768	Max Chunks To Make Sorted II	Prefix Max & Suffix Min
769	Max Chunks To Make Sorted	Running Maximum Comparison
770	Basic Calculator IV	Polynomial Class / Map of Term-to-Coefficient
771	Jewels and Stones	Hash Set Lookup
772	Basic Calculator III	Recursive Descent with Operator Precedence
773	Sliding Puzzle	Breadth-First Search (BFS)
774	Minimize Max Distance to Gas Station	Binary Search on Answer (Range)
775	Global and Local Inversions	Running Maximum Property
776	Split BST	Recursive Post-Order Split
777	Swap Adjacent in LR String	Two-Pointer Position Invariant
778	Swim in Rising Water	Modified Dijkstra / Min-Heap BFS
779	K-th Symbol in Grammar	Binary Tree Symmetry / Recursion
780	Reaching Points	Modulo Backtracking (Reverse Engineering)
781	Rabbits in Forest	Greedy Grouping (HashMap)
782	Transform to Chessboard	Invariance & Hamming Distance
783	Minimum Distance Between BST Nodes	In-Order Traversal (Sorted Property)
784	Letter Case Permutation	Backtracking / DFS
785	Is Graph Bipartite?	Graph Coloring (BFS/DFS)
786	K-th Smallest Prime Fraction	Binary Search on Value or Min-Heap Merge
787	Cheapest Flights Within K Stops	Bellman-Ford / BFS (Level-based)
788	Rotated Digits	Digit Dynamic Programming

789	Escape The Ghosts	Manhattan Distance Comparison
790	Domino and Tromino Tiling	Dynamic Programming (Matrix Exponentiation)
791	Custom Sort String	Frequency Bucket Counting
792	Number of Matching Subsequences	Bucket Sort of Iterators / Waiting Lists
793	Preimage Size of Factorial Zeroes Function	Binary Search on Answer Range
794	Valid Tic-Tac-Toe State	Win-Condition Invariance Check
795	Number of Subarrays with Bounded Maximum	Two-Pass Inclusion/Exclusion (At-Most K)
796	Rotate String	String Concatenation (Double Check)
797	All Paths From Source to Target	Backtracking (DFS for DAG)
798	Smallest Rotation with Highest Score	Difference Array / Sweep Line
799	Champagne Tower	Dynamic Programming (Pascal's Triangle Variation)
800	Similar RGB Color	Greedy Component Rounding
801	Minimum Swaps To Make Sequences Increasing	Dynamic Programming (State Machine): 2 states per index (Swapped vs Not Swapped).
802	Find Eventual Safe States	Graph Coloring (3-Color DFS) OR Kahn's Algorithm (Topological Sort on reversed edges).
803	Bricks Falling When Hit	Reverse Time Union-Find (Disjoint Set).
804	Unique Morse Code Words	Hash Set.
805	Split Array With Same Average	Meet-in-the-Middle + Dynamic Programming / Math Pruning.
806	Number of Lines To Write String	Single-Pass Greedy Accumulation / Simulation.
807	Max Increase to Keep City Skyline	Precomputation (2D Grid Greedy).
808	Soup Servings	2D Dynamic Programming with Memoization + Math Pruning (Probability limit).
809	Expressive Words	Two Pointers / Run-Length Encoding (RLE) grouping.
810	Chalkboard XOR Game	Mathematical Proof / Bit Manipulation (XOR properties).
811	Subdomain Visit Count	String Parsing + Hash Map (Frequency Counter).
812	Largest Triangle Area	Convex Hull + 3 Pointers (Shoelace Formula).
813	Largest Sum of Averages	2D Dynamic Programming + Prefix Sums.
814	Binary Tree Pruning	Post-Order Traversal (Bottom-Up DFS).
815	Bus Routes	BFS on Busses/Routes (Bipartite-like Graph).
816	Ambiguous Coordinates	String Parsing with Strict Validation Pruning.
817	Linked List Components	Hash Set Lookup + Single Pass Traversal.
818	Race Car	Dynamic Programming / BFS with Memoization & Pruning.
819	Most Common Word	Regex/String Parsing + Hash Set + Hash Map.
820	Short Encoding of Words	Suffix Trie (Inverted Prefix Tree).
821	Shortest Distance to a Character	Two-Pass Min-Sweep (Left-to-Right + Right-to-Left).
822	Card Flipping Game	Hash Set (Impossible Value Filtering).
823	Binary Trees With Factors	DP with Hash Map + Sorting.
824	Goat Latin	StringBuilder / String Join with Index-based Suffix.
825	Friends Of Appropriate Ages	Frequency Map + Binary Search or Sliding Window.
826	Most Profit Assigning Work	Sorting + Two Pointers (Greedy).
827	Making A Large Island	Component Labeling (Island ID) + Hash Map.
828	Count Unique Characters of All Substrings	Contribution of Each Character (Index tracking).
829	Consecutive Numbers Sum	Algebraic Factorization (N = kx + frac{k(k-1)}{2}).
830	Positions of Large Groups	Two Pointers (Start and End of Run).
831	Masking Personal Information	Standardization via Pre-filtering + Template Mapping.
832	Flipping an Image	Two Pointers In-Place (Symmetry Swap).
833	Find And Replace in String	Descending Order Sorting or Single-Pass Mapping.
834	Sum of Distances in Tree	Re-rooting DP (Post-order + Pre-order DFS).
835	Image Overlap	List of Coordinates + Delta Frequency Map.
836	Rectangle Overlap	1D Projection (Interval Overlap).
837	New 21 Game	Sliding Window DP.
838	Push Dominoes	Two-Pass Force Calculation or Two-Pointer Segmenting.
839	Similar String Groups	Union-Find with Optimized Similarity Check.
840	Magic Squares In Grid	Pruning + Constant-Time Symmetry Check.
841	Keys and Rooms	Graph Traversal (BFS or DFS) with Visited Set.
842	Split Array into Fibonacci Sequence	Backtracking with Fibonacci Pruning & Integer Overflow check.
843	Guess the Word	Minimax / Information Theory (Heuristic Guessing).
844	Backspace String Compare	Two Pointers (Reverse Scanning).
845	Longest Mountain in Array	Two-Pass Precomputation OR One-Pass State Machine.
846	Hand of Straights	TreeMap / Frequency Map + Smallest Element Processing.
847	Shortest Path Visiting All Nodes	BFS on (CurrentNode, VisitedMask) State Space.
848	Shifting Letters	Suffix Sum of Shifts.
849	Maximize Distance to Closest Person	Two-Pass Distance Arrays OR Two Pointers (Gap Tracking).

850	Rectangle Area II	Line Sweep + Segment Tree (or Coordinate Compression).
851	Loud and Rich	Topological Sort (Kahn’s) or Memoized DFS.
852	Peak Index in a Mountain Array	Binary Search on Slope Direction.
853	Car Fleet	Sorting + Monotonic Arrival Times.
854	K-Similar Strings	BFS with State Pruning (Shortest Path in String Graph).
855	Exam Room	Max-Heap (Priority Queue) of Intervals.
856	Score of Parentheses	Stack or "Core Value" Summation.
857	Minimum Cost to Hire K Workers	Sorting (by Ratio) + Max-Heap (for Quality).
858	Mirror Reflection	Mathematical Least Common Multiple (LCM).
859	Buddy Strings	Case-Based Comparison (Counter + Mismatch Pointer).
860	Lemonade Change	Greedy Billing (Counter-based).
861	Score After Flipping Matrix	Greedy Property (Row Head + Column Max).
862	Shortest Subarray with Sum at Least K	Monotonic Deque + Prefix Sums.
863	All Nodes Distance K in Binary Tree	Graph Conversion (Parent Pointers) + BFS.
864	Shortest Path to Get All Keys	BFS on State (x, y, bitmask_of_keys).
865	Smallest Subtree with all the Deepest Nodes	Post-Order DFS (Depth-Pair Return).
866	Prime Palindrome	Palindrome Generation + Primality Test.
867	Transpose Matrix	New Grid Mapping.
868	Binary Gap	Bit Manipulation (Right Shift) + Distance Tracker.
869	Reordered Power of 2	Digit Frequency Counting (Anagram Check).
870	Advantage Shuffle	Greedy Strategy (Tian Ji's Horse Racing).
871	Minimum Number of Refueling Stops	Greedy with Max-Heap (Priority Queue).
872	Leaf-Similar Trees	DFS (Leaf-only Extraction).
873	Length of Longest Fibonacci Subsequence	2D Dynamic Programming + Hash Set.
874	Walking Robot Simulation	Hash Set (Coordinate Encoding).
875	Koko Eating Bananas	Binary Search on Answer (Range [1, max(piles)]).
876	Middle of the Linked List	Tortoise and Hare (Slow & Fast Pointers).
877	Stone Game	Mathematical Observation (Parity) OR 2D DP.
878	Nth Magical Number	Binary Search on Answer + Inclusion-Exclusion Principle.
879	Profitable Schemes	3D Dynamic Programming (Knapsack variant).
880	Decoded String at Index	Reverse Work-Backwards with Modulo.
881	Boats to Save People	Sorting + Two Pointers (Greedy).
882	Reachable Nodes In Subdivided Graph	Dijkstra’s Algorithm + Edge Coverage Math.
883	Projection Area of 3D Shapes	Row Max, Column Max, and Non-Zero Count.
884	Uncommon Words from Two Sentences	Unified Hash Map (Frequency Counting).
885	Spiral Matrix III	Simulation with Step Increment (Greedy Walking).
886	Possible Bipartition	Graph Coloring (Bipartite Graph Check).
887	Super Egg Drop	2D DP with Binary Search OR "Moves to Floors" Inverse DP.
888	Fair Candy Swap	Hash Set (Target Difference math).
889	Construct Binary Tree from Preorder and Postorder Traversal	Recursive Range Splitting (Divide and Conquer).
890	Find and Replace Pattern	Dual-Map Bijection or Pattern Normalization.
891	Sum of Subsequence Widths	Sorting + Contribution of Each Element.
892	Surface Area of 3D Shapes	Geometry Math (Column-based Calculation).
893	Groups of Special-Equivalent Strings	Character Frequency Signatures (Even/Odd split).
894	All Possible Full Binary Trees	Recursion with Memoization (Divide and Conquer).
895	Maximum Frequency Stack	Frequency Map + Map of Stacks (Grouped by Freq).
896	Monotonic Array	Single Pass Boolean Flag Scan.
897	Increasing Order Search Tree	In-order Traversal with In-place Re-linking.
898	Bitwise ORs of Subarrays	DP with Pruning (Capped Set size).
899	Orderly Queue	Math Observation (Bubblesort Property).
900	RLE Iterator	Pointer Simulation on Compressed Pairs.
901	Online Stock Spanner	Monotonic Stack (Pair of Price and Span).
902	Numbers At Most N Given Digit Set	Digit DP (Combinatorial Counting).
903	Valid Permutations for DI Sequence	2D Dynamic Programming.
904	Fruit Into Baskets	Sliding Window (Two Pointers) with Hash Map.
905	Sort Array By Parity	Two Pointers (In-place Swap)
906	Super Palindromes	Palindrome Construction (Math/Generator)
907	Sum of Subarray Minimums	Monotonic Stack (Increasing)
908	Smallest Range I	Math / Greedy / Min-Max
909	Snakes and Ladders	Breadth-First Search (BFS)
910	Smallest Range II	Sort + Greedy Linear Scan

911	Online Election	Precomputation + Binary Search
912	Sort an Array	Merge Sort / Quick Sort / Heap Sort (Divide & Conquer)
913	Cat and Mouse	Minimax / Topological Sort on States
914	X of a Kind in a Deck of Cards	Greatest Common Divisor (GCD)
915	Partition Array into Disjoint Intervals	One-Pass Running Maximum
916	Word Subsets	Frequency Map Compression (Maximum Requirements)
917	Reverse Only Letters	Two Pointers (In-place)
918	Maximum Sum Circular Subarray	Kadane's (Min/Max Complement)
919	Complete Binary Tree Inserter	Queue-based Level Order State
920	Number of Music Playlists	Dynamic Programming (2D)
921	Minimum Add to Make Parentheses Valid	Balance Counter (Greedy)
922	Sort Array By Parity II	Two Pointers (In-place)
923	3Sum With Multiplicity	Counting + Combinatorics (Three Cases)
924	Minimize Malware Spread	Union-Find (Connected Components)
925	Long Pressed Name	Two Pointers (Sequential Match)
926	Flip String to Monotone Increasing	Dynamic Programming / Prefix Sums
927	Three Equal Parts	Counting 1s + Suffix Matching
928	Minimize Malware Spread II	Graph Connectivity (Unique Source Tracking)
929	Unique Email Addresses	String Parsing + HashSet
930	Binary Subarrays With Sum	Prefix Sum + Hashmap (or Sliding Window)
931	Minimum Falling Path Sum	Dynamic Programming (In-place)
932	Beautiful Array	Divide and Conquer (Odd/Even Separation)
933	Number of Recent Calls	Sliding Window (Queue)
934	Shortest Bridge	Multi-Source BFS
935	Knight Dialer	Dynamic Programming (State Compression)
936	Stamping The Sequence	Greedy Reverse Matching (Work Backwards)
937	Reorder Data in Log Files	Custom Comparator / Partitioning
938	Range Sum of BST	Recursive Pruning (BST Property)
939	Minimum Area Rectangle	Diagonal Point Matching + Hashset
940	Distinct Subsequences II	Dynamic Programming (Ending-character Tally)
941	Valid Mountain Array	Two-Pointer "Climb"
942	DI String Match	Greedy Two-Pointer Placement
943	Find the Shortest Superstring	Traveling Salesperson (TSP) with Bitmask DP
944	Delete Columns to Make Sorted	Greedy Column-wise Scan
945	Minimum Increment to Make Array Unique	Sort + Tracking Next Available
946	Validate Stack Sequences	Simulation with Real Stack
947	Most Stones Removed with Same Row or Column	Union-Find (Connected Components)
948	Bag of Tokens	Sort + Two Pointers (Greedy)
949	Largest Time for Given Digits	Permutation Validation (Fixed Size)
950	Reveal Cards In Increasing Order	Simulation with Queue (Work Backwards/Index Mapping)
951	Flip Equivalent Binary Trees	Recursive Canonical Comparison
952	Largest Component Size by Common Factor	Union-Find + Prime Factorization
953	Verifying an Alien Dictionary	Direct Mapping + Pairwise Check
954	Array of Doubled Pairs	Frequency Map + Sorted Greedy
955	Delete Columns to Make Sorted II	Greedy with Sortedness Tracking
956	Tallest Billboard	DP (Difference as State)
957	Prison Cells After N Days	Cycle Detection (Hash Map)
958	Check Completeness of a Binary Tree	Level-Order BFS (Null Detection)
959	Regions Cut By Slashes	Grid Expansion + DFS/BFS (or Union-Find)
960	Delete Columns to Make Sorted III	Dynamic Programming (Longest Common Subsequence Variation)
961	N-Repeated Element in Size 2N Array	Sliding Window / Randomized Search
962	Maximum Width Ramp	Monotonic Stack (Decreasing)
963	Minimum Area Rectangle II	Center Point + Diagonal Property
964	Least Operators to Express Number	Dynamic Programming (Base-X Representation)
965	Univalued Binary Tree	Recursive Logical AND
966	Vowel Spellchecker	Precomputed HashMaps (Priority Tiering)
967	Numbers With Same Consecutive Differences	BFS / DFS Digit Construction
968	Binary Tree Cameras	Greedy / Post-Order DP (State Signaling)
969	Pancake Sorting	Greedy "Largest to Back"
970	Powerful Integers	Double Logarithmic Loop (Logarithmic Bounds)

971	Flip Binary Tree To Match Preorder Traversal	Greedy Preorder Simulation
972	Equal Rational Numbers	Fractional Conversion (Math/String)
973	K Closest Points to Origin	Max-Heap (Size K) or QuickSelect
974	Subarray Sums Divisible by K	Prefix Sums + Frequency Map (Remainder Counting)
975	Odd-Even Jump	Monotonic Stack + DP (Suffix Jump Mapping)
976	Largest Perimeter Triangle	Sort + Greedy (Adjacent Triplet)
977	Squares of a Sorted Array	Two Pointers (Inward Meeting)
978	Longest Turbulent Subarray	Sliding Window / DP
979	Distribute Coins in Binary Tree	Post-Order Traversal (Excess/Deficit Flow)
980	Unique Paths III	Backtracking with State (Hamiltonian Path)
981	Time Based Key-Value Store	HashMap + Sorted List (Binary Search)
982	Triplets with Bitwise AND Equal To Zero	Precomputed Pairwise Frequency (Hashmap)
983	Minimum Cost For Tickets	DP (1D Calendar / Days-based)
984	String Without AAA or BBB	Greedy (Heavy-Letter Priority)
985	Sum of Even Numbers After Queries	Global Sum Maintenance (Delta Tracking)
986	Interval List Intersections	Two Pointers (Parallel Scan)
987	Vertical Order Traversal of a Binary Tree	TreeMap of Sorted Lists (Coordinate Grouping)
988	Smallest String Starting From Leaf	DFS with Path Tracking (Bottom-Up Logic)
989	Add to Array-Form of Integer	Schoolbook Addition (Right-to-Left)
990	Satisfiability of Equality Equations	Union-Find (Consistency Check)
991	Broken Calculator	Greedy Reverse Engineering (Work Backwards)
992	Subarrays with K Different Integers	Sliding Window (Exact = AtMost(K) - AtMost(K-1))
993	Cousins in Binary Tree	Level-Order BFS (Same Level, Different Parent)
994	Rotting Oranges	Multi-Source BFS
995	Minimum Number of K Consecutive Bit Flips	Greedy with Difference Array / Sliding Window
996	Number of Squareful Arrays	Hamiltonian Path (Backtracking + Pruning)
997	Find the Town Judge	Trust Score Array (In-Degree vs Out-Degree)
998	Maximum Binary Tree II	Right-Spine Insertion
999	Available Captures for Rook	Rook-Centric Scan
1000	Minimum Cost to Merge Stones	DP (Interval 3D)
1001	Grid Illumination	HashMaps for Line Counting
1002	Find Common Characters	Frequency Intersection (Min-Tally)
1003	Check If Word Is Valid After Substitutions	Stack Simulation
1004	Max Consecutive Ones III	Sliding Window (Expand/Contract)
1005	Maximize Sum Of Array After K Negations	Greedy (Sorting).
1006	Clumsy Factorial	Stack (Simulation of BODMAS/PEMDAS).
1007	Min Domino Rotations For Equal Row	Greedy (Target Candidates).
1008	Construct BST from Preorder Traversal	Recursion with Valid Range (Upper/Lower Bounds).
1009	Complement of Base 10 Integer	Bit Manipulation (XOR Masking).
1010	Pairs of Songs With Total Durations Divisible by 60	Hash Map / Frequency Array (Modulo Arithmetic).
1011	Capacity To Ship Packages Within D Days	Binary Search on Answer Range.
1012	Numbers With Repeated Digits	Digit DP / Combinatorics.
1013	Partition Array Into Three Parts With Equal Sum	Greedy Accumulation (Target Sum Matching).
1014	Best Sightseeing Pair	Dynamic Programming / 1D State Tracking.
1015	Smallest Integer Divisible by K	Math (Modulo Arithmetic / Pigeonhole Principle).
1016	Binary String With Substrings Representing 1 To N	Sliding Window / Hash Set.
1017	Convert to Base -2	Math (Modulo with Negative Base).
1018	Binary Prefix Divisible By 5	Math (Running Modulo).
1019	Next Greater Node In Linked List	Monotonic Stack.
1020	Number of Enclaves	Multi-source DFS/BFS from Boundaries.
1021	Remove Outermost Parentheses	State Tracking (Depth Counter).
1022	Sum of Root To Leaf Binary Numbers	Pre-order Traversal (Bitwise Accumulation).
1023	Camelcase Matching	Two Pointers (Subsequence Match with Constraints).
1024	Video Stitching	Greedy (Interval Scheduling / Jump Game logic).
1025	Divisor Game	Brain Teaser / Math (Parity).
1026	Max Difference Between Node and Ancestor	DFS with Range Tracking (Top-Down).
1027	Longest Arithmetic Subsequence	2D Dynamic Programming (HashMap/Array).
1028	Recover a Tree From Preorder Traversal	Iterative Stack (Depth Tracking).
1029	Two City Scheduling	Greedy (Sorting by Opportunity Cost).
1030	All Cells in Distance Order in a Grid	BFS (Level-Order Expansion) or Bucket Sort.
1031	Maximum Sum of Two Non-Overlapping Subarrays	Dynamic Programming / Prefix Sums.

1032	Stream of Characters	Trie (Prefix Tree) with Reversed Search.
1033	Moving Stones Until Consecutive	Greedy / Case Analysis.
1034	Coloring A Border	DFS / BFS with Connectivity Logic.
1035	Uncrossed Lines	Dynamic Programming (Longest Common Subsequence).
1036	Escape a Large Maze	Bounded BFS / Geometric Limitation.
1037	Valid Boomerang	Math (Cross Product / Slope Comparison).
1038	Binary Search Tree to Greater Sum Tree	Reverse In-order Traversal (Right -> Root -> Left).
1039	Minimum Score Triangulation of Polygon	Dynamic Programming (Matrix Chain Multiplication style).
1040	Moving Stones Until Consecutive II	Sliding Window + Mathematical Logic.
1041	Robot Bounded In Circle	Vector Math / State Check.
1042	Flower Planting With No Adjacent	Greedy Coloring.
1043	Partition Array for Maximum Sum	Dynamic Programming (Linear).
1044	Longest Duplicate Substring	Binary Search on Length + Rolling Hash (Rabin-Karp).
1045	Customers Who Bought All Products	Group By + Having Count (Distinct).
1046	Last Stone Weight	Max-Heap (Priority Queue).
1047	Remove All Adjacent Duplicates In String	Stack (LIFO processing).
1048	Longest String Chain	Dynamic Programming + Hashing (Sorting by Length).
1049	Last Stone Weight II	Dynamic Programming (0/1 Knapsack variation).
1050	Actors and Directors Who Cooperated At Least Three Times	Group By + Having.
1051	Height Checker	Sorting Comparison or Counting Sort.
1052	Grumpy Bookstore Owner	Sliding Window (Fixed Size).
1053	Previous Permutation With One Swap	Greedy (Right-to-Left Scan).
1054	Distant Barcodes	Greedy + Frequency Fill (Even/Odd Interleaving).
1055	Shortest Way to Form String	Greedy (Two Pointers / Multi-Pass).
1056	Confusing Number	Math (Digit Extraction and Remainder).
1057	Campus Bikes	Bucket Sort / Greedy with Priority.
1058	Minimize Rounding Error to Meet Target	Greedy (Sorting by Fractional Part).
1059	All Paths from Source Lead to Destination	DFS with State Marking (3-Coloring).
1060	Missing Element in Sorted Array	Binary Search (Range-based).
1061	Lexicographically Smallest Equivalent String	Union-Find (DSU) with Path Compression.
1062	Longest Repeating Substring	Binary Search on Length + Rolling Hash (Rabin-Karp).
1063	Number of Valid Subarrays	Monotonic Stack.
1064	Fixed Point	Binary Search.
1065	Index Pairs of a String	Trie (Prefix Tree).
1066	Campus Bikes II	DP with Bitmasking.
1067	Digit Count in Range	Math / Digit DP.
1068	Product Sales Analysis I	Inner Join.
1069	Product Sales Analysis II	Group By.
1070	Product Sales Analysis III	Subquery with In-Clause or Window Functions.
1071	Greatest Common Divisor of Strings	Math (Euclidean GCD Logic).
1072	Flip Columns For Maximum Number of Equal Rows	Pattern Canonicalization / Hashing.
1073	Adding Two Negabinary Numbers	Bitwise Simulation (Carry Manipulation).
1074	Number of Submatrices That Sum to Target	2D-to-1D Reduction + Subarray Sum Logic.
1075	Project Employees I	Join + Group By + AVG.
1076	Project Employees II	Group By + Having (Subquery Max).
1077	Project Employees III	Window Function (RANK/DENSE_RANK) or Join with Max-Group.
1078	Occurrences After Bigram	Linear Scan / String Splitting.
1079	Letter Tile Possibilities	Backtracking + Frequency Array.
1080	Insufficient Nodes in Root to Leaf Paths	Post-order Traversal (Bottom-Up Pruning).
1081	Smallest Subsequence of Distinct Characters	Monotonic Stack + Greedy (Frequency Map).
1082	Sales Analysis I	Group By + Having (Subquery Max).
1083	Sales Analysis II	Group By + Having (Conditional Sums).
1084	Sales Analysis III	Group By + Having (Min/Max Dates).
1085	Sum of Digits in the Minimum Number	Linear Scan + Digit Math.
1086	High Five	TreeMap + Max-Heap (or Sorting).
1087	Brace Expansion	Backtracking + Sorting Groups.
1088	Confusing Number II	Digit-based DFS / Backtracking.
1089	Duplicate Zeros	Two-Pointer (In-Place).
1090	Largest Values From Labels	Greedy (Sorting).
1091	Shortest Path in Binary Matrix	BFS (Level-Order Traversal).
1092	Shortest Common Supersequence	Dynamic Programming (LCS Variation).

1093	Statistics from a Large Sample	Frequency-Based Math.
1094	Car Pooling	Difference Array / Prefix Sums.
1095	Find in Mountain Array	Binary Search (Triple Pass).
1096	Brace Expansion II	Recursive Descent Parsing / Stack.
1097	Game Play Analysis V	Join + Group By (Install Date Logic).
1098	Unpopular Books	Left Join + Filtering + Having.
1099	Two Sum Less Than K	Two Pointers (Sort + Shrink).
1100	Find K-Length Substrings With No Repeated Characters	Sliding Window + Frequency Array/Set.
1101	The Earliest Moment When Everyone Become Friends	Union-Find (DSU) with Component Counting.
1102	Path With Maximum Minimum Value	Greedy (Priority Queue / BFS) or Binary Search + DFS.
1103	Distribute Candies to People	Simulation or Mathematical Arithmetic Progression.
1104	Path In Zigzag Labelled Binary Tree	Math (Symmetry / Bit Manipulation).
1105	Filling Bookcase Shelves	1D Dynamic Programming
1106	Parsing A Boolean Expression	Stack (Iterative Evaluation)
1107	New Users Daily Count (SQL)	Common Table Expression (CTE) / GROUP BY
1108	Defanging an IP Address	String Builder / Pre-allocated Array
1109	Corporate Flight Bookings	Difference Array / Prefix Sum
1110	Delete Nodes And Return Forest	Post-order DFS Traversal + Hash Set
1111	Maximum Nesting Depth of Two Valid Parentheses Strings	Greedy / Parity based on Depth
1112	Highest Grade For Each Student (SQL)	Window Function (ROW_NUMBER)
1113	Reported Posts (SQL)	COUNT(DISTINCT) with GROUP BY
1114	Print in Order	Concurrency: Mutexes / Semaphores / Promises
1115	Print FooBar Alternately	Concurrency: Two Alternating Semaphores
1116	Print Zero Even Odd	Concurrency: 3 Semaphores
1117	Building H2O	Semaphores + CyclicBarrier / Phasers
1118	Number of Days in a Month	Pre-computed Array + Leap Year Math
1119	Remove Vowels from a String	Two Pointers / StringBuilder / RegEx
1120	Maximum Average Subtree	Post-order DFS (Bottom-Up)
1121	Divide Array Into Increasing Sequences	Greedy / Frequency Counting
1122	Relative Sort Array	Counting Sort (Frequency Map)
1123	Lowest Common Ancestor of Deepest Leaves	Post-order DFS (Depth-Pair Return)
1124	Longest Well-Performing Interval	Prefix Sum + Hash Map / Monotonic Stack
1125	Smallest Sufficient Team	Bitmask DP
1126	Active Businesses	Window Functions / CTEs
1127	User Purchase Platform (SQL)	CROSS JOIN (Template) + LEFT JOIN
1128	Number of Equivalent Domino Pairs	Key Normalization + Frequency Map
1129	Shortest Path with Alternating Colors	BFS with Multi-State Tracking
1130	Minimum Cost Tree From Leaf Values	Monotonic Stack / Greedy
1131	Maximum of Absolute Value Expression	Math (Expression Expansion / 8 Scenarios)
1132	Reported Posts II (SQL)	GROUP BY + COUNT(DISTINCT) + AVG
1133	Largest Unique Number	Frequency Map / Bucket Sort
1134	Armstrong Number	Mathematical Digit Extraction
1135	Connecting Cities With Minimum Cost	Kruskal's Algorithm (Greedy + DSU)
1136	Parallel Courses	Topological Sort (Kahn's Algorithm / BFS)
1137	N-th Tribonacci Number	Dynamic Programming (Iterative Space Optimized)
1138	Alphabet Board Path	Coordinate Math (Manhattan Distance Pathing)
1139	Largest 1-Bordered Square	2D Prefix Sums (Left and Up counts)
1140	Stone Game II	DP with Memoization (Suffix Sums)
1141	User Activity for the Past 30 Days I	GROUP BY + DATEDIFF / BETWEEN
1142	User Activity for the Past 30 Days II	Aggregated Subquery / IFNULL
1143	Longest Common Subsequence	2D Dynamic Programming (Bottom-Up)
1144	Decrease Elements To Make Array Zigzag	Greedy (Two Scenarios: Even vs Odd indices)
1145	Binary Tree Coloring Game	Tree Partitioning / Subtree Counting
1146	Snapshot Array	History Tracking (Binary Search on Versions)
1147	Longest Chunked Palindrome Decomposition	Greedy + Two Pointers (Rolling Hash)
1148	Article Views I (SQL)	SELECT DISTINCT + ORDER BY
1149	Article Views II (SQL)	GROUP BY + HAVING COUNT(DISTINCT)
1150	Check If a Number Is Majority Element in a Sorted Array	Binary Search (Two Bounds)
1151	Minimum Swaps to Group All 1's Together	Fixed-Size Sliding Window
1152	Analyze User Website Visit Pattern	Sorting + Combinatorics + Hash Map
1153	String Transformation Into Another String	Graph Theory (Out-degree & Cycles)

1154	Day of the Year	Prefix Sum Array + Leap Year Logic
1155	Number of Dice Rolls With Target Sum	2D Dynamic Programming (Space Optimized)
1156	Swap For Longest Repeated Character Substring	Grouping + Sliding Window/Case Analysis
1157	Online Majority Element In Subarray	Boyer-Moore Voting + Random Sampling / Segment Tree
1158	Market Analysis I (SQL)	LEFT JOIN + Conditional Aggregate
1159	Market Analysis II (SQL)	Window Function (ROW_NUMBER) + CTE
1160	Find Words That Can Be Formed by Characters	Frequency Array (Hash Buckets)
1161	Maximum Level Sum of a Binary Tree	BFS (Queue-based Level Order)
1162	As Far from Land as Possible	Multi-Source BFS
1163	Last Substring in Lexicographical Order	Two Pointers (Optimal Suffix Comparison)
1164	Product Price at a Given Date (SQL)	UNION ALL + CTE / Subquery
1165	Single-Row Keyboard	Frequency Map / Index Array
1166	Design File System	Trie (Prefix Tree)
1167	Minimum Cost to Connect Sticks	Min-Heap (Priority Queue)
1168	Optimize Water Distribution in a Village	Kruskal's / Prim's with Virtual Root
1169	Invalid Transactions	Hash Map Grouping + Sorting
1170	Compare Strings by Frequency of the Smallest Character	Pre-computation + Binary Search/Suffix Sum
1171	Remove Zero Sum Consecutive Nodes from Linked List	Prefix Sum + Hash Map (One/Two Pass)
1172	Dinner Plate Stacks	Min-Heap + List of Stacks
1173	Immediate Food Delivery I (SQL)	SUM(CASE) / AVG(IF)
1174	Immediate Food Delivery II (SQL)	IN Clause with Multi-Column Key
1175	Prime Arrangements	Sieve of Eratosthenes + Combinatorics
1176	Diet Plan Performance	Sliding Window (Fixed Size)
1177	Can Make Palindrome from Substring	Prefix Sums + Bitmasking
1178	Number of Valid Words for Each Puzzle	Bitmasking + Subset Enumeration
1179	Reformat Department Table	Conditional Aggregation (SUM + CASE)
1180	Count Substrings with Only One Distinct Character	Two Pointers / Math (Sum of N)
1181	Before and After Puzzle	Hash Map (First Word Mapping)
1182	Shortest Distance to Target Color	Dynamic Programming (Pre-computed Distances)
1183	Maximum Number of Ones	Greedy + Coordinate Modulo Counting
1184	Distance Between Bus Stops	Two-Way Summation (Circular Array)
1185	Day of the Week	Zeller’s Congruence / Sum of Days
1186	Maximum Subarray Sum with One Deletion	Dynamic Programming (Bi-directional Kadane's)
1187	Make Array Strictly Increasing	Dynamic Programming (State: Index + Prev Value)
1188	Design Bounded Blocking Queue	Condition Variables / Semaphores
1189	Maximum Number of Balloons	Frequency Map / Bottleneck Analysis
1190	Reverse Substrings Between Each Pair of Parentheses	Wormhole Teleportation (O(N) Pre-matching)
1191	K-Concatenation Maximum Sum	Kadane's + Total Sum Logic
1192	Critical Connections in a Network	Tarjan's / Bridges (DFS Discovery Time)
1193	Monthly Transactions I (SQL)	GROUP BY + DATE_FORMAT
1194	Tournament Winners (SQL)	UNION ALL + Window Functions
1195	Fizz Buzz Multithreaded	Synchronization (Semaphores / Mutex + Conditions)
1196	How Many Apples Can You Put into the Basket	Greedy (Sorting)
1197	Minimum Knight Moves	BFS with Symmetry + Pruning (or Math)
1198	Find Smallest Common Element in All Rows	Frequency Array / Counting (Rows are Sorted)
1199	Minimum Time to Build Blocks	Greedy (Huffman Coding / Min-Heap)
1200	Minimum Absolute Difference	Sorting + Adjacent Comparison
1201	Ugly Number III	Binary Search on Answer Range + Inclusion-Exclusion Principle
1202	Smallest String With Swaps	Disjoint Set Union (DSU) / Connected Components + Sorting
1203	Sort Items by Groups Respecting Dependencies	Double Topological Sort (Kahn's Algorithm)
1204	Last Person to Fit in the Bus (SQL)	SQL Window Function (Cumulative Sum)
1205	Monthly Transactions II (SQL)	UNION ALL + Unified GROUP BY (CTEs)
1206	Design Skiplist	Probabilistic Multi-Level Linked List
1207	Unique Number of Occurrences	Hash Map (Counting) + Hash Set (Uniqueness)
1208	Get Equal Substrings Within Budget	Sliding Window (Variable Size)
1209	Remove All Adjacent Duplicates in String II	Stack with Frequency Pairs
1210	Minimum Moves to Reach Target with Rotations	Breadth-First Search (BFS) Shortest Path on a 3D State Graph
1211	Queries Quality and Percentage (SQL)	Single-Pass Aggregation with CASE/IF statements
1212	Team Scores in Football Tournament (SQL)	UNION ALL for Event Splitting + GROUP BY
1213	Intersection of Three Sorted Arrays	Three Pointers
1214	Two Sum BSTs	Inorder Traversal to Hash Set (or Two Pointers on Arrays)

1215	Stepping Numbers	BFS / DFS (State Space Generation)
1216	Valid Palindrome III	Dynamic Programming (Longest Palindromic Subsequence)
1217	Minimum Cost to Move Chips to The Same Position	Parity Counting (Math/Greedy)
1218	Longest Arithmetic Subsequence of Given Difference	Dynamic Programming + Hash Map
1219	Path with Maximum Gold	DFS with Backtracking
1220	Count Vowels Permutation	Dynamic Programming (State Transitions)
1221	Split a String in Balanced Strings	Greedy / Running Balance Counter
1222	Queens That Can Attack the King	Reverse Search (King outward to Queens)
1223	Dice Roll Simulation	3D Dynamic Programming
1224	Maximum Equal Frequency	Double Hash Maps (Count of Counts)
1225	Report Contiguous Dates (SQL)	Gaps and Islands (Row Number Difference)
1226	The Dining Philosophers	Resource Hierarchy / Semaphores
1227	Airplane Seat Assignment Probability	Mathematical Induction / Constant Observation
1228	Missing Number In Arithmetic Progression	Binary Search (Range-based Gap Detection)
1229	Meeting Scheduler	Two Pointers on Sorted Intervals
1230	Toss Strange Coins	Dynamic Programming (Linear State)
1231	Divide Chocolate	Binary Search on Answer (Range)
1232	Check If It Is a Straight Line	Cross-Multiplication (Slope Equality)
1233	Remove Sub-Folders from the Filesystem	Sort + Prefix Check (or Trie)
1234	Replace the Substring for Balanced String	Sliding Window (Shrinkable)
1235	Maximum Profit in Job Scheduling	DP + Sorting + Binary Search
1236	Web Crawler	BFS/DFS with Hostname Filtering
1237	Find Positive Integer Solution for a Given Equation	Two Pointers on Monotonic Surface
1238	Circular Permutation in Binary Representation	Gray Code Generation + Bitwise XOR
1239	Maximum Length of a Concatenated String with Unique Characters	Bitmasking + Backtracking (Pruned)
1240	Tiling a Rectangle with the Fewest Squares	Backtracking with Pruning & Skyline State
1241	Number of Comments per Post (SQL)	LEFT JOIN + GROUP BY + DISTINCT
1242	Web Crawler Multithreaded	Thread Pool + Thread-Safe Queue/Set
1243	Array Transformation	In-Place Simulation (Two Pointers/Copying)
1244	Design A Leaderboard	Hash Map + Balanced BST (or TreeMap/SortedSet)
1245	Tree Diameter	Double BFS (or DFS with Height Tracking)
1246	Palindrome Removal	Interval Dynamic Programming
1247	Minimum Swaps to Make Strings Equal	Greedy / Parity Counting
1248	Count Number of Nice Subarrays	Sliding Window / Prefix Sum Hashing
1249	Minimum Remove to Make Valid Parentheses	Stack / Two-Pass Filtering
1250	Check If It Is a Good Array	Number Theory (Greatest Common Divisor)
1251	Average Selling Price (SQL)	LEFT JOIN on Interval + GROUP BY
1252	Cells with Odd Values in a Matrix	Parity Counting (Row/Col Arrays)
1253	Reconstruct a 2-Row Binary Matrix	Greedy Strategy
1254	Number of Closed Islands	Flood Fill (DFS/BFS) + Edge Trimming
1255	Maximum Score Words Formed by Letters	Backtracking with Frequency State Management
1256	Encode Number	Binary Offset Mapping
1257	Smallest Common Region	Hash Map (Parent Pointer) + Set Intersection
1258	Synonymous Sentences	DSU (Disjoint Set Union) + Backtracking + Sorting
1259	Handshakes That Don't Cross	Dynamic Programming (Catalan Number Variation)
1260	Shift 2D Grid	1D Index Mapping (Modulo Arithmetic)
1261	Find Elements in a Contaminated Binary Tree	Pre-computation with Hash Set
1262	Greatest Sum Divisible by Three	Dynamic Programming (Remainder State)
1263	Minimum Moves to Move a Box to Their Target Location	Nested BFS / State BFS (Box Pos + Person Pos)
1264	Page Recommendations (SQL)	CTE / UNION for Bidirectional Friends + JOIN
1265	Print Immutable Linked List in Reverse	Recursive Backtracking (Implicit Stack)
1266	Minimum Time Visiting All Points	Chebyshev Distance (Max of ΔX, ΔY)
1267	Count Servers that Communicate	Row/Column Frequency Counting
1268	Search Suggestions System	Sort + Binary Search (Two Pointers)
1269	Number of Ways to Stay in the Same Place After Some Steps	Dynamic Programming (State Compression)
1270	All People Report to the Given Manager	Recursive CTE (Common Table Expression)
1271	Hexspeak	Direct String Transformation + Regex
1272	Remove Interval	Linear Boundary Comparison
1273	Delete Tree Nodes	Post-Order Traversal (Bottom-Up)
1274	Number of Ships in a Rectangle	Divide and Conquer (Quadtree Decomposition)

1275	Find Winner on a Tic Tac Toe Game	Row/Column/Diagonal Counters
1276	Number of Burgers with No Waste	Linear Algebra (System of Equations)
1277	Count Square Submatrices with All Ones	2D Dynamic Programming
1278	Palindrome Partitioning III	Interval DP + Partition DP
1279	Traffic Light Controlled Intersection	Synchronization (Mutex/Locking)
1280	Students and Examinations (SQL)	CROSS JOIN (Cartesian Base) + LEFT JOIN
1281	Subtract the Product and Sum of Digits of an Integer	Iterative Modulo (%) and Division (/)
1282	Group the People Given the Group Size They Belong To	Hash Map of Lists (Greedy Accumulation)
1283	Find the Smallest Divisor Given a Threshold	Binary Search on Answer Range
1284	Minimum Number of Flips to Convert Binary Matrix to Zero Matrix	BFS on Matrix State (Bitmasking)
1285	Find the Start and End Number of Continuous Ranges (SQL)	Gaps and Islands (Row Number Difference)
1286	Iterator for Combination	Bitmasking / Backtracking Iterator
1287	Element Appearing More Than 25% In Sorted Array	Binary Search on Candidate Points
1288	Remove Covered Intervals	Sort by Start (Asc) and End (Desc) + Greedy Scan
1289	Minimum Falling Path Sum II	DP with Two Minimums (Optimization)
1290	Convert Binary Number in a Linked List to Integer	Bit Manipulation (Left Shifting)
1291	Sequential Digits	Sliding Window on Digit String "123456789"
1292	Maximum Side Length of a Square with Sum Less than or Equal to Threshold	2D Prefix Sum + Binary Search (or Greedy Side Length)
1293	Shortest Path in a Grid with Obstacles Elimination	BFS on 3D State (row, col, k)
1294	Weather Type in Each Country (SQL)	JOIN + GROUP BY + CASE/IF logic
1295	Find Numbers with Even Number of Digits	Logarithmic Digit Counting / Range Check
1296	Divide Array in Sets of K Consecutive Numbers	Frequency Map + Greedy Sorting
1297	Maximum Number of Occurrences of a Substring	Sliding Window (Observation: minSize only)
1298	Maximum Candies You Can Get from Boxes	BFS with Two-Condition Entry (Keys + Possession)
1299	Replace Elements with Greatest Element on Right Side	Reverse Linear Scan (Right-to-Left)
1300	Sum of Mutated Array Closest to Target	Binary Search on Value Range
1301	Number of Paths with Max Score	2D Dynamic Programming (Suffix-Sum style)
1302	Deepest Leaves Sum	BFS (Level-Order Traversal)
1303	Find the Team Size (SQL)	Window Function (COUNT OVER PARTITION)
1304	Find N Unique Integers Sum up to Zero	Symmetric Construction (Pairing)
1305	All Elements in Two Binary Search Trees	In-order DFS + Two-Pointer Merge
1306	Jump Game III	Graph BFS / DFS with Visited Array
1307	Verbal Arithmetic Puzzle	Backtracking DFS with Column-by-Column Pruning
1308	Running Total for Different Genders (SQL)	Window Functions (SUM OVER)
1309	Decrypt String from Alphabet to Integer	Single-Pass Reverse Scan (Right-to-Left Pointer)
1310	XOR Queries of a Subarray	Prefix XOR Array (Cumulative State Cache)
1311	Get Watched Videos by Your Friends	Level-Order BFS + Frequency Map Sort
1312	Min Insertions to Make String Palindrome	2D DP: Longest Common Subsequence (LCS)
1313	Decompress Run-Length Encoded List	Pre-allocation + Linear Fill
1314	Matrix Block Sum	2D Prefix Sum Array
1315	Sum of Nodes with Even-Valued Grandparent	Top-Down DFS with State Passing
1316	Distinct Echo Substrings	Rolling Hash (Rabin-Karp) + Hash Set
1317	Convert Integer to the Sum of Two No-Zero Integers	Math/Modulo Zero-Check Loop
1318	Minimum Flips to Make a OR b Equal to c	Bitwise Operations & Masking
1319	Number of Operations to Make Network Connected	Union-Find (Disjoint Set)
1320	Minimum Distance to Type a Word	3D Dynamic Programming / Memoization
1321	Restaurant Growth (SQL)	Window Functions (Sliding Window)
1322	Ads Performance (SQL)	Conditional Aggregation (SUM + CASE)
1323	Maximum 69 Number	First-Occurrence String Search / Math
1324	Print Words Vertically	Parallel StringBuilder Appending
1325	Delete Leaves With a Given Value	Bottom-Up Post-Order DFS
1326	Minimum Number of Taps to Open to Water a Garden	Greedy Array Processing (Jump Game II Pattern)
1327	List the Products Ordered in a Period (SQL)	INNER JOIN with GROUP BY and HAVING
1328	Break a Palindrome	Greedy Half-Scan Mutation
1329	Sort the Matrix Diagonally	Map of Min-Heaps (Group by i - j)
1330	Reverse Subarray To Maximize Array Value	Math / Edge-Delta Optimization
1331	Rank Transform of an Array	Sort + De-duplicate + Hash Map
1332	Remove Palindromic Subsequences	Brainteaser / Logical Deduction
1333	Filter Restaurants	Linear Filter Pipeline + Custom Sort
1334	Find the City With Smallest Neighbors	Floyd-Warshall (All-Pairs Shortest Path)
1335	Minimum Difficulty of a Job Schedule	2D Dynamic Programming

1336	Number of Transactions per Visit (SQL)	Recursive CTE (Range Generation) + Left Join
1337	The K Weakest Rows in a Matrix	Binary Search + Max-Heap (Priority Queue)
1338	Reduce Array Size to The Half	Frequency Map + Greedy Sort
1339	Maximum Product of Splitted Binary Tree	Post-Order DFS Sum Tracking
1340	Jump Game V	DP + Memoization or Monotonic Stack Sort
1341	Movie Rating (SQL)	Window Functions / Order By + Limit 1
1342	Number of Steps to Reduce to Zero	Bit Manipulation (Popcount + Bit Length)
1343	Number of Sub-arrays of Size K and Average >= Threshold	Sliding Window (Fixed Size)
1344	Angle Between Hands of a Clock	Mathematical Relative Position
1345	Jump Game IV	BFS with Multi-Value Jump Optimization
1346	Check If N and Its Double Exist	Hash Set (Single Pass)
1347	Minimum Number of Steps to Make Two Strings Anagram	Frequency Array (Character Tally)
1348	Tweet Counts Per Frequency	Hash Map of Sorted Sets (TreeMap)
1349	Maximum Students Taking Exam	Dynamic Programming with Bitmask
1350	Students with Invalid Departments (SQL)	LEFT JOIN + NULL Filtering
1351	Count Negative Numbers in a Sorted Matrix	Staircase Search (Saddleback)
1352	Product of the Last K Numbers	Prefix Product Array with Zero-Reset
1353	Maximum Number of Events That Can Be Attended	Greedy + Min-Heap (Priority Queue)
1354	Construct Target Array with Multiple Sums	Reverse Simulation with Max-Heap
1355	Activity Participants (SQL)	Window Functions (RANK / DENSE_RANK)
1356	Sort Integers by The Number of 1 Bits	Custom Comparator with __builtin_popcount
1357	Apply Discount Every n Orders	Stateful Counter + Hash Map
1358	Number of Substrings Containing All Three Characters	Sliding Window (Two-Pointer / At-Most)
1359	Count All Valid Pickup and Delivery Options	Combinatorics / Math (DP)
1360	Number of Days Between Two Dates	Days Since Epoch (1971) Calculation
1361	Validate Binary Tree Nodes	Graph In-Degree Count + BFS/DFS
1362	Closest Divisors	Square Root Search (Greedy)
1363	Largest Multiple of Three	Greedy Tally + Remainder Math
1364	Number of Trusted Contacts of a Customer (SQL)	Multi-Level LEFT JOIN with Grouping
1365	How Many Numbers Are Smaller Than the Current Number	Bucket Sort / Prefix Sum
1366	Rank Teams by Votes	Multi-Level Sorting (Frequency Matrix)
1367	Linked List in Binary Tree	KMP Pattern Matching on Tree
1368	Minimum Cost to Make at Least One Valid Path in a Grid	0-1 BFS / Dijkstra's Algorithm
1369	Get the Second Most Recent Activity (SQL)	Window Function (ROW_NUMBER + COUNT)
1370	Increasing Decreasing String	Bucket Sort / Counting Tally
1371	Find the Longest Substring Containing Vowels in Even Counts	Bitmask + Prefix XOR (First Occurrence)
1372	Longest ZigZag Path in a Binary Tree	Top-Down DFS with State Passing
1373	Maximum Sum BST in Binary Tree	Post-Order DFS (Bottom-Up State)
1374	Generate a String With Characters...	Greedy Parity Construction
1375	Number of Times Binary String Is Prefix-Aligned	Max-Value Tracking (Greedy)
1376	Time Needed to Inform All Employees	Top-Down DFS (Tree Path Sum)
1377	Frog Position After T Seconds	DFS with Probability Propagation
1378	Replace Employee ID With The Unique Identifier (SQL)	LEFT JOIN (Hash Join)
1379	Find a Corresponding Node of a Binary Tree in a Clone of That Tree	Synchronized DFS / BFS
1380	Lucky Numbers in a Matrix	Precomputed Row-Mins and Col-Maxes
1381	Design a Stack With Increment Operation	Lazy Propagation (Difference Array Logic)
1382	Balance a Binary Search Tree	In-Order Flattening + Midpoint Reconstruction
1383	Maximum Performance of a Team	Greedy + Min-Heap (Bounded Priority Queue)
1384	Total Sales Amount by Year (SQL)	Recursive CTE (Year-Level Pivot)
1385	Find the Distance Value Between Two Arrays	Sorting + Binary Search (Range Check)
1386	Cinema Seat Allocation	Hash Map + Bitmask (Sparse Row Processing)
1387	Sort Integers by The Power Value	Recursion with Memoization (Top-Down DP)
1388	Pizza With 3n Slices	Dynamic Programming (Linear House Robber Variant)
1389	Create Target Array in the Given Order	Self-Balancing BST or Linked List (Ideal)
1390	Four Divisors	Square Root Divisor Hunt
1391	Check if There is a Valid Path in a Grid	Disjoint Set Union (DSU) or Validated BFS
1392	Longest Happy Prefix	KMP Algorithm (Prefix Function)
1393	Capital Gain/Loss (SQL)	Conditional Aggregation (SUM + IF/CASE)
1394	Find Lucky Integer in an Array	Frequency Map (Counting Sort Style)
1395	Count Number of Teams	DP / Middle-Element Counting
1396	Design Underground System	Dual Hash Maps (In-Transit & Completed)

1397	Find All Good Strings	Digit DP + KMP (Automata State)
1398	Customers Who Bought A and B but Not C (SQL)	Group By + Having (Set Membership)
1399	Count Largest Group	Integer Modulo Sum + Frequency Bucket
1400	Construct K Palindrome Strings	Odd Frequency Character Counting
1401	Circle and Rectangle Overlapping	Math / Geometry (Closest Point)
1402	Reducing Dishes	Sorting + Greedy (or 1D DP)
1403	Minimum Subsequence in Non-Increasing Order	Sorting + Greedy Accumulation
1404	Number of Steps to Reduce a Number in Binary Representation to One	Right-to-Left Carry Simulation (Greedy)
1405	Longest Happy String	Max Heap / Priority Queue (Greedy)
1406	Stone Game III	1D Dynamic Programming (Bottom-Up)
1407	Top Travellers (SQL)	LEFT JOIN + GROUP BY + COALESCE
1408	String Matching in an Array	Length Sort + KMP (or Suffix Tree)
1409	Queries on a Permutation With Key	Fenwick Tree (Binary Indexed Tree)
1410	HTML Entity Parser	Single-Pass Two-Pointer / StringBuilder
1411	Number of Ways to Paint N × 3 Grid	State Machine DP / Math
1412	Find the Quiet Students in All Exams (SQL)	Window Functions (RANK / MAX OVER)
1413	Min Value to Get Positive Step by Step Sum	Prefix Sum / Minimum Running Sum
1414	Find Min Fibonacci Numbers Whose Sum Is K	Greedy Algorithm (Coin Change variation)
1415	The k-th Lexicographical String of All Happy Strings	Math / Permutation Pruning (Binary Search style)
1416	Restore The Array	1D Dynamic Programming (Suffix/Prefix Array)
1417	Reformat The String	Two-Pointer / Bucket Splitting
1418	Display Table of Food Orders in a Restaurant	Hash Map of Hash Maps & Tree Sets
1419	Minimum Number of Frogs Croaking	State Machine / Concurrent Counters
1420	Build Array Where You Can Find The Maximum Exactly K Comparisons	3D Dynamic Programming
1421	NPV Queries (SQL)	LEFT JOIN + COALESCE / IFNULL
1422	Maximum Score After Splitting a String	Two-Pass / Prefix Tracking
1423	Maximum Points You Can Obtain from Cards	Sliding Window (Inverse approach)
1424	Diagonal Traverse II	Hash Map Grouping by `Row + Col`
1425	Constrained Subsequence Sum	DP + Monotonic Deque (Sliding Window Maximum)
1426	Counting Elements	Hash Set / Boolean Array
1427	Perform String Shifts	Net Shift Calculation + Slicing
1428	Leftmost Column with at Least a One	Staircase Search (Top-Right Start)
1429	First Unique Number	Linked Hash Map / Queue + Hash Map
1430	Check If a String Is a Valid Sequence from Root to Leaves in a Binary Tree	Pruned DFS (Early Exit)
1431	Kids With the Greatest Number of Candies	Two-Pass Greedy (Max Tracking)
1432	Max Difference You Can Get From Changing an Integer	Greedy Digit Substitution
1433	Check If a String Can Break Another String	Sorting + Frequency Counting
1434	Number of Ways to Wear Different Hats to Each Other	Bitmask Dynamic Programming (Inverted)
1435	Create a Session Bar Chart (SQL)	CASE WHEN + CROSS JOIN (Static Ranges)
1436	Destination City	Set Difference / Hash Map
1437	Check If All 1's Are at Least Length K Places Away	Single-Pass Pointer Distance
1438	Longest Continuous Subarray With Absolute Diff <= Limit	Sliding Window + Monotonic Deques
1439	Find the Kth Smallest Sum of a Matrix With Sorted Rows	Max Heap / Binary Search on Sums
1440	Evaluate Boolean Expression (SQL)	Double JOIN + CASE Statement
1441	Build an Array With Stack Operations	Simulation (Two-Pointer Logic)
1442	Count Triplets That Can Form Two Arrays of Equal XOR	Prefix XOR + Property of XOR
1443	Minimum Time to Collect All Apples in a Tree	Post-Order DFS (Bottom-Up)
1444	Number of Ways of Cutting a Pizza	2D DP + 2D Prefix Sum (Suffix Sum)
1445	Apples & Oranges (SQL)	GROUP BY + SUM(CASE WHEN)
1446	Consecutive Characters	Single-Pass Greedy Counter
1447	Simplified Fractions	Greatest Common Divisor (GCD) Filter
1448	Count Good Nodes in Binary Tree	DFS with Max-So-Far Parameter
1449	Form Largest Integer With Digits That Add up to Target	1D Dynamic Programming (Knapsack Variation)
1450	Number of Students Doing Homework at a Given Time	Single-Pass Range Intersection
1451	Rearrange Words in a Sentence	Stable Sort / Map of Lists
1452	People Whose List of Favorite Companies Is Not a Subset of Another List	Bitmasking / Set Hashing
1453	Maximum Number of Darts Inside of a Circular Dartboard	Angular Sweep / Fixed Circle Centers
1454	Active Users (SQL)	Window Functions (LEAD / DENSE_RANK)
1455	Check If a Word Occurs As a Prefix of Any Word in a Sentence	Tokenization + `startsWith`
1456	Max Number of Vowels in a Substring of Given Length	Sliding Window (Fixed)
1457	Pseudo-Palindromic Paths in a Binary Tree	DFS + Bit Manipulation

1458	Max Dot Product of Two Subsequences	2D Dynamic Programming (LCS Variant)
1459	Rectangles Area (SQL)	Self-Join with Inequality Constraints
1460	Make Two Arrays Equal by Reversing Sub-arrays	Frequency Counting (Hash Map)
1461	Check If a String Contains All Binary Codes of Size K	Sliding Window + Hash Set (or Rolling Hash)
1462	Course Schedule IV	Floyd-Warshall (Transitive Closure)
1463	Cherry Pickup II	3D Dynamic Programming (Top-Down)
1464	Maximum Product of Two Elements in an Array	Two-Pointer / Max Tracking
1465	Maximum Area of a Piece of Cake After Horizontal and Vertical Cuts	Greedy (Independent Max Gaps)
1466	Reorder Routes to Make All Paths Lead to the City Zero	BFS / DFS with Edge Direction Tracking
1467	Probability of a Two Boxes Having The Same Number of Distinct Balls	Dynamic Programming / Backtracking with Memoization
1468	Calculate Salaries (SQL)	Window Function (MAX OVER)
1469	Find All The Lonely Nodes	DFS / BFS (Parental Context)
1470	Shuffle the Array	Single-Pass Two-Pointer (In-place if bits allow)
1471	The k Strongest Values in an Array	Sorting + Two-Pointer
1472	Design Browser History	Two Stacks or Doubly Linked List
1473	Paint House III	3D Dynamic Programming
1474	Delete N Nodes After M Nodes of a Linked List	In-Place Link Manipulation
1475	Final Prices With a Special Discount in a Shop	Monotonic Stack (Increasing)
1476	Subrectangle Queries	Update-History Log (Inverse Search)
1477	Find Two Non-overlapping Sub-arrays Each with Target Sum	Sliding Window + 1D DP (Prefix Min)
1478	Allocate Mailboxes	DP + Median Distance Pre-computation
1479	Sales by Day of the Week (SQL)	CASE WHEN + SUM (Pivoting)
1480	Running Sum of 1d Array	Prefix Sum (In-place)
1481	Least Number of Unique Integers after K Removals	Greedy Frequency Sorting
1482	Minimum Number of Days to Make m Bouquets	Binary Search on Answer Range
1483	Kth Ancestor of a Tree Node	Binary Lifting (Sparse Table)
1484	Group Sold Products By The Date (SQL)	GROUP BY + GROUP_CONCAT / STRING_AGG
1485	Clone Binary Tree With Random Pointer	DFS / BFS with Node Mapping (Memoization)
1486	XOR Operation in an Array	Direct Accumulation (Math optional)
1487	Making File Names Unique	Hash Map with Suffix Tracking
1488	Avoid Flood in The City	Greedy + Greedy with Binary Search (TreeSet)
1489	Find Critical and Pseudo-Critical Edges in MST	Kruskal’s + Edge Exclusion/Inclusion
1490	Clone N-ary Tree	DFS Recursion (Post-Order)
1491	Average Salary Excluding the Minimum and Maximum Salary	One-Pass Min-Max Tracking
1492	The kth Factor of n	Square Root Factor Pairing
1493	Longest Subarray of 1's After Deleting One Element	Sliding Window (Flexible)
1494	Parallel Courses II	Dynamic Programming with Bitmasking
1495	Friendly Movies Streamed Last Month (SQL)	JOIN + Distinct Filtering
1496	Path Crossing	Hash Set of Visited Points
1497	Check If Array Pairs Are Divisible by k	Remainder Frequency Counting
1498	Number of Subsequences That Satisfy the Given Sum Condition	Sorting + Two Pointers
1499	MaxValue of Equation	Monotonic Deque (Sliding Window Max)
1500	Design a File Sharing System	Min-Heap + Map of Sets
1501	Countries You Can Safely Invest In (SQL)	JOIN + GROUP BY + HAVING
1502	Can Make Arithmetic Progression From Sequence	Math / Hash Set (Min/Max finding)
1503	Last Moment Before All Ants Fall Out of a Plank	Brainteaser / Array Math (Ghost Ants)
1504	Count Submatrices With All Ones	Monotonic Stack / Histogram
1505	Minimum Possible Integer After at Most K Adjacent Swaps	Greedy + Fenwick Tree / Segment Tree
1506	Find Root of N-Ary Tree	Math (Summation / XOR)
1507	Reformat Date	String Splitting + Hash Map
1508	Range Sum of Sorted Subarray Sums	Min-Heap / Priority Queue (Multi-way Merge)
1509	Minimum Difference Between Largest and Smallest Value in Three Moves	Sorting + Greedy (Sliding Window)
1510	Stone Game IV	Dynamic Programming (Bottom-Up 1D)
1511	Customer Order Frequency (Premium SQL)	JOIN + Conditional Aggregation
1512	Number of Good Pairs	Hash Map / Combinatorics
1513	Number of Substrings With Only 1s	Math / Contiguous Counting
1514	Path with Maximum Probability	Graph / Dijkstra's (Max-Heap)
1515	Best Position for a Service Centre	Geometry / Gradient Descent (or Weiszfeld's)
1516	Move Sub-Tree of N-Ary Tree	DFS (Parent Tracking) + Pointer Manipulation
1517	Find Users With Valid E-Mails (Database / String)	Regular Expressions (Regex)

1518	Water Bottles	Math Formula
1519	Number of Nodes in the Sub-Tree With the Same Label	Post-Order DFS (Bottom-Up)
1520	Maximum Number of Non-Overlapping Substrings	Interval Expansion + Greedy Search
1521	Find a Value of a Mysterious Function Closest to Target	Bitwise AND Properties / Hash Set
1522	Diameter of N-Ary Tree (Premium)	Post-Order DFS (Tree DP)
1523	Count Odd Numbers in an Interval Range	Math
1524	Number of Sub-arrays With Odd Sum	Prefix Sum + Math (Parity)
1525	Number of Good Ways to Split a String	Prefix & Suffix Arrays (State Tracking)
1526	Minimum Number of Increments on Subarrays to Form a Target Array	Greedy / Array Math
1527	Patients With a Condition (SQL)	SQL LIKE Operator / String Matching
1528	Shuffle String	Direct Array Placement / Cyclic Sort
1529	Minimum Suffix Flips (Bulb Switcher IV)	State Tracking / Greedy
1530	Number of Good Leaf Nodes Pairs	Post-Order DFS (Tree DP)
1531	String Compression II	Dynamic Programming (Top-Down with Memo)
1532	The Most Recent Three Orders (SQL)	Window Function (ROW_NUMBER)
1533	Find the Index of the Large Integer (Interactive)	Binary Search (Ternary Splitting)
1534	Count Good Triplets	Brute Force / Constraint Optimization
1535	Find the Winner of an Array Game	One-Pass Greedy / Simulation
1536	Minimum Swaps to Arrange a Binary Grid	Greedy / Bubble Sort Logic
1537	Get the Maximum Score	Two Pointers / Greedy Prefix Sum
1538	Guess the Majority Element in a Hidden Array (Interactive)	Relative Parity / Math
1539	Kth Missing Positive Number	Binary Search (Range-Based)
1540	Can Convert String in K Moves	Hash Map / Frequency Counting
1541	Minimum Insertions to Balance a Parentheses String	Greedy / Stack Logic (Counter)
1542	Find Longest Awesome Substring	Bitmask + Prefix XOR
1543	Fix Product Name Format (SQL)	String Functions + Aggregation
1544	Make The String Great	Stack (Greedy Elimination)
1545	Find Kth Bit in Nth Binary String	Recursive Symmetry / Divide & Conquer
1546	Maximum Number of Non-Overlapping Subarrays With Sum Equals Target	Prefix Sum + Hash Set + Greedy
1547	Minimum Cost to Cut a Stick	Dynamic Programming (MCM Variant)
1548	The Most Similar Path in a Graph (Premium)	Dynamic Programming (Viterbi-like)
1549	The Most Recent Orders for Each Product (SQL)	Window Function (RANK)
1550	Three Consecutive Odds	Linear Scan / Counter
1551	Minimum Operations to Make Array Equal	Math (Arithmetic Progression)
1552	Magnetic Force Between Two Balls	Binary Search on Answer
1553	Minimum Number of Days to Eat N Oranges	DP + BFS / Optimized Recursion
1554	Strings Differ by One Character	Hashing / Rolling Hash
1555	Bank Account Summary II (SQL)	JOIN + GROUP BY + HAVING
1556	Thousand Separator	Reverse Traversal / String Builder
1557	Minimum Number of Vertices to Reach All Nodes	Graph Theory (In-Degree Zero)
1558	Minimum Numbers of Function Calls to Make Target Array	Greedy / Bit Manipulation
1559	Detect Cycles in 2D Grid	DFS + Backtracking (Parent Tracking)
1560	Most Visited Sector in a Circular Track	Math (Observation)
1561	Maximum Number of Coins You Can Get	Greedy / Sorting
1562	Find Latest Group of Size M	Union-Find or Boundary Mapping
1563	Stone Game V	Dynamic Programming + Prefix Sums
1564	Put Boxes Into the Warehouse I	Greedy / Preprocessing
1565	Unique Orders and Customers Per Month (SQL)	SQL GROUP BY + COUNT(DISTINCT)
1566	Detect Pattern of Length M Repeated K or More Times	Single Pass / Offset Comparison
1567	Maximum Length of Subarray With Positive Product	Dynamic Programming / Greedy State
1568	Minimum Number of Days to Disconnect Island	Graph Theory (Articulation Points / 2-Day Rule)
1569	Number of Ways to Reorder Array to Get Same BST	Divide & Conquer + Combinatorics (Pascal's Triangle)
1570	Dot Product of Two Sparse Vectors	Two Pointers on Compressed Pairs
1571	Warehouse Manager (SQL)	SQL Aggregation / JOIN
1572	Matrix Diagonal Sum	Single Pass / Linear Indexing
1573	Number of Ways to Split a String	Combinatorics / Gap Counting
1574	Shortest Subarray to be Removed to Make Array Sorted	Two Pointers (Prefix/Suffix)
1575	Count All Possible Routes	Dynamic Programming (State: City x Fuel)
1576	Replace All ?'s to Avoid Consecutive Repeating Characters	Greedy / Linear Scan
1577	Number of Ways Where Square of One Number Equals Product of Two Numbers	Hash Map / Frequency Counting
1578	Minimum Time to Make Rope Colorful	Greedy / Two Pointers

1579	Remove Max Number of Edges to Keep Graph Fully Traversable	Union Find (DSU) / Greedy
1580	Put Boxes Into the Warehouse II (Premium)	Greedy / Two Pointers / Preprocessing
1581	Customer Who Visited but Did Not Make Any Transactions (SQL)	LEFT JOIN / WHERE IS NULL
1582	Special Positions in a Binary Matrix	Preprocessing (Row/Col Sums)
1583	Count Unhappy Friends	Rank Mapping (Hash Map / 2D Array)
1584	Min Cost to Connect All Points	Graph (Prim's or Kruskal's MST)
1585	Check If String Is Transformable With Substring Sort Operations	Greedy / Relative Order Tracking
1586	Binary Search Tree Iterator II (Premium)	Lazy Iteration + Stack/List Cache
1587	Bank Account Summary II (SQL)	Hash Aggregation / Index Join
1588	Sum of All Odd Length Subarrays	Combinatorics (Contribution of Each Element)
1589	Maximum Sum Obtained of Any Permutation	Difference Array + Greedy (Sweep Line)
1590	Make Sum Divisible by P	Prefix Sum + Hash Map (Modular Arithmetic)
1591	Strange Printer II	Graph Theory (Topological Sort)
1592	Rearrange Spaces Between Words	String Manipulation / Math
1593	Split a String Into the Max Number of Unique Substrings	Backtracking / Set
1594	Maximum Non Negative Product in a Matrix	Dynamic Programming (Min/Max Tracking)
1595	Minimum Cost to Connect Two Groups of Points	DP + Bitmask
1596	Most Frequently Ordered Products for Each Customer (SQL)	Window Functions (RANK / CTE)
1597	Build Binary Expression Tree From Infix Expression	Shunting-yard Algorithm / Stack
1598	Crawler Log Folder	Stack / Integer Counter
1599	Maximum Profit of Operating a Centennial Wheel	Simulation / Greedy
1600	Throne Inheritance	DFS / Preorder Traversal + Hash Set
1601	Maximum Number of Achievable Transfer Requests	Bitmasking / State Space Search
1602	Find Nearest Right Node in Binary Tree	BFS (Level Order Traversal)
1603	Design Parking System	Array Indexing (Direct Access)
1604	Alert Using Same Key-Card Three or More Times...	Hash Map of Lists + Sorting + Sliding Window (Size 3)
1605	Find Valid Matrix Given Row and Column Sums	Greedy Matrix Allocation
1606	Find Servers That Handled Most Number of Requests	Two Heaps (Priority Queues) / TreeSet
1607	Sellers With No Sales (SQL/DB)	Hash Anti-Join (LEFT JOIN + IS NULL)
1608	Special Array With X Elements...	Counting Sort / Frequency Array
1609	Even Odd Tree	BFS (Level Order Traversal)
1610	Maximum Number of Visible Points	Polar Angle Sorting + Sliding Window
1611	Minimum One Bit Operations to Make Integers Zero	Gray Code Reversal / Bit Manipulation
1612	Check If Two Expression Trees are Equivalent	DFS + Frequency Array (Counting)
1613	Find the Missing IDs (SQL)	SQL Recursive CTE + Anti-Join
1614	Maximum Nesting Depth of the Parentheses	Single Counter Tracking
1615	Maximal Network Rank	Degree Array + Edge Adjacency Matrix
1616	Split Two Strings to Make Palindrome	Two Pointers (Outside-In) + Middle Checking
1617	Count Subtrees With Max Distance Between Cities	Bitmasking Subset Enumeration + DFS / Floyd-Warshall
1618	Maximum Font to Fit a Sentence in a Screen	Binary Search on Monotonic Property
1619	Mean of Array After Removing Some Elements	QuickSelect Partitioning (nth_element)
1620	Coordinate With Maximum Network Quality	Exhaustive Heatmap Search (Brute Force)
1621	Number of Sets of K Non-Overlapping Line Segments	Dynamic Programming / Combinatorics Math
1622	Fancy Sequence	Math (Modular Inverse) + Global Modifiers
1623	All Valid Triplets That Can Represent a Country (SQL)	INNER JOIN with Inequality Filtering
1624	Largest Substring Between Two Equal Characters	Hash Map / Fixed Array (First Occurrence)
1625	Lexicographically Smallest String After Applying Operations	BFS / DFS with Visited State Set
1626	Best Team With No Conflicts	Sorting + DP (Longest Increasing Subsequence Variant)
1627	Graph Connectivity With Threshold	Union-Find (Disjoint Set) with Math Multiples
1628	Design an Expression Tree With Evaluate Function	Object-Oriented Design (Polymorphism) + Postfix Stack Tree
1629	Slowest Key	Single Pass Linear Scan
1630	Arithmetic Subarrays	Hash Set + Min/Max Math (No Sorting)
1631	Path With Minimum Effort	Dijkstra / Binary Search + BFS
1632	Rank Transform of a Matrix	Union-Find + Topological Sort (Indegree)
1633	Percentage of Users Attended a Contest (SQL)	Group By + Scalar Subquery / Cross Join
1634	Add Two Polynomials Represented as Linked Lists	Two Pointers (Merge Sort Style)
1635	Hopper Company Queries I (SQL)	Recursive CTE (Month Generator) + LEFT JOIN + Window Functions
1636	Sort Array by Increasing Frequency	Hash Map + Custom Comparator Sorting
1637	Widest Vertical Area Between Two Points...	X-Coordinate Sorting
1638	Count Substrings That Differ by One Character	DP / Central Expansion (Mismatch Count)
1639	Number of Ways to Form a Target String...	DP + Character Frequency Pre-computation

1640	Check Array Formation Through Concatenation	Hash Map (Piece-Start Mapping)
1641	Count Sorted Vowel Strings	Math (Stars and Bars) / DP
1642	Furthest Building You Can Reach	Greedy + Min-Heap (Priority Queue)
1643	Kth Smallest Instructions	Combinatorial Counting (Pascal's Triangle logic)
1644	Lowest Common Ancestor of a Binary Tree II	DFS (Post-order Traversal) with Existence Flags
1645	Hopper Company Queries II (SQL)	Recursive CTE + LEFT JOIN + SUM/COUNT Window Functions
1646	Get Maximum in Generated Array	Iterative DP / Simulation
1647	Minimum Deletions to Make Character Frequencies Unique	Greedy + Frequency Sorting / Hash Set
1648	Sell Diminishing-Valued Colored Balls	Binary Search on Value Threshold / Greedy Math
1649	Create Sorted Array through Instructions	Fenwick Tree (BIT) / Segment Tree
1650	Lowest Common Ancestor of a Binary Tree III	Two Pointers (Linked List Intersection Style)
1651	Hopper Company Queries III (SQL)	Recursive CTE + Window Function (ROWS BETWEEN)
1652	Defuse the Bomb	Fixed Size Sliding Window
1653	Minimum Deletions to Make String Balanced	Dynamic Programming / Greedy Tally
1654	Minimum Jumps to Reach Home	BFS (Breadth-First Search)
1655	Distribute Repeating Integers	Bitmask DP (Submask Iteration)
1656	Design an Ordered Stream	Pointer-based Simulation
1657	Determine if Two Strings Are Close	Frequency Maps + Character Set Comparison
1658	Minimum Operations to Reduce X to Zero	Sliding Window (Target = TotalSum - x)
1659	Maximize Grid Happiness	DP + Ternary Bitmask (State Compression)
1660	Correct a Binary Tree	BFS (Reverse Level Order) + Hash Set
1661	Average Time of Process per Machine (SQL)	Self-Join / Aggregate CASE WHEN
1662	Check If Two String Arrays are Equivalent	Two Pointers (Nested Indexing)
1663	Smallest String With A Given Numeric Value	Greedy (Fill from Back)
1664	Ways to Make a Fair Array	Prefix Sums (Even/Odd Tracking)
1665	Minimum Initial Energy to Finish Tasks	Greedy (Sort by Difference)
1666	Change the Root of a Binary Tree	Recursive In-place Pointer Reversal
1667	Fix Names in a Table (SQL)	String Concatenation + Substring functions
1668	Maximum Repeating Substring	Dynamic Programming (Suffix-based)
1669	Merge In Between Linked Lists	Two Pointers (Marker Placement)
1670	Design Front Middle Back Queue	Two Deques (Balanced Halves)
1671	Minimum Number of Removals to Make Mountain Array	LIS (Longest Increasing Subsequence) + LDS (Decreasing)
1672	Richest Customer Wealth	Row-wise Iteration (Simple Scan)
1673	Find the Most Competitive Subsequence	Monotonic Stack (Greedy)
1674	Minimum Moves to Make Array Complementary	Difference Array (Line Sweep)
1675	Minimize Deviation in Array	Priority Queue (Max-Heap) + Greedy Shrinking
1676	Lowest Common Ancestor of a Binary Tree IV	DFS (Post-order) + Target Set
1677	Product's Worth Over Invoices (SQL)	LEFT JOIN + GROUP BY + IFNULL/COALESCE
1678	Goal Parser Interpretation	Single-Pass Pointer Scan
1679	Max Number of K-Sum Pairs	Two Pointers (Sorted) or Hash Map (Frequency)
1680	Concatenation of Consecutive Binary Numbers	Bit Manipulation (Shift and Add)
1681	Minimum Incompatibility	Bitmask DP + Submask Iteration
1682	Longest Palindromic Subsequence II	3D Dynamic Programming
1683	Invalid Tweets (SQL)	SQL Length Function (LENGTH/CHAR_LENGTH)
1684	Count the Number of Consistent Strings	Boolean Frequency Array / Hash Set
1685	Sum of Absolute Differences in a Sorted Array	Prefix Sums (Mathematical Simplification)
1686	Stone Game VI	Greedy (Sorting by Combined Value)
1687	Delivering Boxes from Storage to Ports	DP + Monotonic Queue (Sliding Window)
1688	Count of Matches in Tournament	Mathematical Insight (Elimination)
1689	Partitioning Into Minimum Deci-Binary Numbers	Greedy (Maximum Digit)
1690	Stone Game VII	2D Interval Dynamic Programming
1691	Maximum Height by Stacking Cuboids	Dimension Sorting + LIS (Dynamic Programming)
1692	Count Ways to Distribute Candies	DP (Stirling Numbers of Second Kind)
1693	Daily Leads and Partners (SQL)	Group By + Aggregate COUNT(DISTINCT)
1694	Reformat Phone Number	Iterative Cleanup + Block Processing
1695	Maximum Erasure Value	Sliding Window + Hash Set
1696	Jump Game VI	DP + Monotonic Queue (Deque)
1697	Checking Existence of Edge Length Limited Paths	Offline Queries + Union-Find (DSU)
1698	Number of Distinct Substrings in a String	Suffix Automaton (SAM) or Rolling Hash
1699	Number of Calls Between Two Persons (SQL)	SQL LEAST() and GREATEST() Functions

1700	Number of Students Unable to Eat Lunch	Frequency Counting (Greedy)
1701	Average Waiting Time	Greedy / Event Simulation (State tracking).
1702	Maximum Binary String After Change	Greedy / Math (Zero Counting).
1703	Minimum Adjacent Swaps for K Consecutive Ones	Sliding Window over an array of indices + Median greediness.
1704	Determine if String Halves Are Alike	Two Pointers (Left & Right) converging to center, using a Hash Set for vowels.
1705	Maximum Number of Eaten Apples	Min-Heap (Priority Queue) tracking `[rot_day, count]`.
1706	Where Will the Ball Fall	Iterative Grid Simulation.
1707	Maximum XOR With an Element From Array	Offline Queries + Bitwise Trie.
1708	Largest Subarray Length K	Greedy Peak Finding (Single Element Identifier).
1709	Biggest Window Between Visits (SQL)	SQL Window Function: `LEAD()` over Partition.
1710	Maximum Units on a Truck	Greedy + Sorting.
1711	Count Good Meals	Hash Map (Frequency Map) + Two Sum variant for 22 static targets.
1712	Ways to Split Array Into Three Subarrays	Prefix Sums + Binary Search (or Two Pointers).
1713	Minimum Operations to Make a Subsequence	Target Index Mapping + Longest Increasing Subsequence (LIS) via Patience Sorting.
1714	Sum Of Special Evenly-Spaced Elements In Array	Square Root Decomposition (Block separation).
1715	Count Apples and Oranges (SQL)	LEFT JOIN with COALESCE / IFNULL.
1716	Calculate Money in Leetcode Bank	Math (Arithmetic Progression Series).
1717	Maximum Score From Removing Substrings	Greedy + Stack (Two Passes).
1718	Construct the Lexicographically Largest Valid Sequence	Backtracking + Greedy (Place largest available number first).
1719	Number Of Ways To Reconstruct A Tree	Graph Theory / Degree Sorting (Constructive).
1720	Decode XORed Array	Bitwise XOR Properties (`A ^ B = C` implies `A ^ C = B`).
1721	Swapping Nodes in a Linked List	Two Pointers (Fast & Slow Ruler Slide).
1722	Minimize Hamming Distance After Swap	Disjoint Set Union (DSU) / Connected Components + Hash Maps.
1723	Find Minimum Time to Finish All Jobs	Binary Search on Answer + Pruned Backtracking.
1724	Checking Existence of Edge Length Limited Paths II	Kruskal's MST + LCA (Lowest Common Ancestor) with Binary Lifting.
1725	Number Of Rectangles That Can Form The Largest Square	Single Pass Streaming Maximum (King of the Hill).
1726	Tuple with Same Product	Hash Map (Frequency Counting) + Math Combinatorics.
1727	Largest Submatrix With Rearrangements	Dynamic Programming (Histogram Heights) + Greedy Sorting.
1728	Cat and Mouse II	Game Theory (Minimax) + 3D Memoization (Topological Sort from End States).
1729	Find Followers Count (SQL)	SQL Aggregation: `GROUP BY` + `COUNT()`.
1730	Shortest Path to Get Food	BFS (Breadth First Search) for Unweighted Shortest Path.
1731	The Number of Employees Which Report to Each Employee (SQL)	Self-Join on ID = Reports_To + Group By.
1732	Find the Highest Altitude	Prefix Sum (Running Total).
1733	Minimum Number of People to Teach	Greedy + Set Intersection (Filtering Friendships).
1734	Decode XORed Permutation	Bitwise Logic (Total XOR Property).
1735	Count Ways to Make Array With Product	Prime Factorization + Stars and Bars (Combinatorics).
1736	Latest Time by Replacing Hidden Digits	Greedy Positional Logic.
1737	Change Minimum Characters to Satisfy Condition	Prefix Sums + Frequency Maps.
1738	Find Kth Largest XOR Coordinate Value	2D Prefix Sum (DP) + Quickselect/Min-Heap.
1739	Building Boxes	Mathematical Series (Tetrahedral Numbers) + Greedy.
1740	Find Distance in a Binary Tree	Lowest Common Ancestor (LCA) + Node Depth.
1741	Find Total Time Spent by Each Employee (SQL)	SQL Aggregation: `GROUP BY` on multiple columns.
1742	Maximum Number of Balls in a Box	Mathematical Digit Summing + Array Frequency Map.
1743	Restore the Array From Adjacent Pairs	Graph Adjacency List + Degree Finding + DFS/Linear Traversal.
1744	Can You Eat Your Favorite Candy on Your Favorite Day?	Prefix Sums + Range Intersection (Inequality Check).
1745	Palindrome Partitioning IV	2D Dynamic Programming (Palindrome Table).
1746	Maximum Subarray Sum After One Operation	Dynamic Programming (State Machine Kadane's).
1747	Leetflex Banned Accounts (SQL)	SQL Self-Join with Inequality Constraints.
1748	Sum of Unique Elements	Hash Map / Frequency Array.
1749	Maximum Absolute Sum of Any Subarray	Prefix Sum Range (Max - Min).
1750	Minimum Length of String After Deleting Similar Ends	Two Pointers (Inward Convergence).
1751	Maximum Number of Events That Can Be Attended II	DP + Binary Search (Weighted Interval Scheduling).
1752	Check if Array Is Sorted and Rotated	Single Pass Contraction Check.
1753	Maximum Score From Removing Stones	Greedy (Always pick two largest) or Math.
1754	Largest Merge Of Two Strings	Greedy with Suffix Comparison.
1755	Closest Subsequence Sum	Meet-in-the-Middle + Two Pointers.
1756	Design Most Recently Used Queue	Square Root Decomposition or Fenwick Tree.
1757	Recyclable and Low Fat Products (SQL)	SQL Logical Indexing (`AND` condition).
1758	Minimum Changes To Make Alternating Binary String	Greedy Comparison (2 Target States).
1759	Count Number of Homogenous Substrings	Two Pointers (Sliding Window) + Arithmetic Progression.
1760	Minimum Limit of Balls in a Bag	Binary Search on the Answer.

1761	Minimum Degree of a Connected Trio in a Graph	Adjacency Matrix + Degree Precomputation.
1762	Buildings With an Ocean View	Right-to-Left Monotonic Scan.
1763	Longest Nice Substring	Divide and Conquer (Recursive Splitting).
1764	Form Array by Concatenating Subarrays of Another Array	Greedy Pointer Matching.
1765	Map of Highest Peak	Multi-Source BFS.
1766	Tree of Coprimes	DFS + Ancestor Value Tracking (Value-based Map).
1767	Find the Subtasks That Did Not Execute (SQL)	Recursive CTE + LEFT JOIN.
1768	Merge Strings Alternately	Two Pointers (Parallel Pointers).
1769	Minimum Number of Operations to Move All Balls to Each Box	Two-Pass Prefix and Suffix Sum.
1770	Maximum Score from Performing Multiplication Operations	2D Dynamic Programming (Memoization).
1771	Maximize Palindrome Length From Subsequences	2D Dynamic Programming (LPS on word1 + word2).
1772	Sort Features by Popularity	Hash Map (Counter) + Custom Stable Sort.
1773	Count Items Matching a Rule	Index Mapping + Linear Scan.
1774	Closest Dessert Cost	Backtracking with Pruning.
1775	Equal Sum Arrays With Minimum Number of Operations	Greedy + Frequency Counting (Bucket Sort).
1776	Car Fleet II	Monotonic Stack (Right-to-Left) + Collision Time Comparison.
1777	Product's Price for Each Store (SQL)	Pivot Table (CASE WHEN + GROUP BY).
1778	Shortest Path in a Hidden Grid	DFS (to Map) + BFS (to Find Shortest Path).
1779	Find Nearest Point That Has the Same X or Y Coordinate	Single Pass Minimum Tracking.
1780	Check if Number is a Sum of Powers of Three	Base 3 Representation (Ternary Logic).
1781	Sum of Beauty of All Substrings	Nested Loop with Running Frequency Array.
1782	Count Pairs Of Nodes	Two Pointers on Sorted Degrees + Shared Edge Subtraction.
1783	Grand Slam Titles (SQL)	UNION ALL + GROUP BY.
1784	Check if Binary String Has at Most One Segment of Ones	Pattern Searching (Presence of "01").
1785	Minimum Elements to Add to Form a Given Sum	Mathematical Ceiling Division.
1786	Number of Restricted Paths From First to Last Node	Dijkstra (from Node N) + DAG DP (Topological Sort/Memoization).
1787	Make the XOR of All Segments Equal to Zero	DP with Frequency Partitioning (i pmod K).
1788	Maximize the Beauty of the Garden	Prefix Sums + First Occurrence Hash Map.
1789	Primary Department for Each Employee (SQL)	UNION of two filtered sets.
1790	Check if One String Swap Can Make Strings Equal	Mismatch Index Comparison.
1791	Find Center of Star Graph	Logical Comparison of First Two Edges.
1792	Maximum Average Pass Ratio	Max-Heap (Priority Queue) of "Potential Gain".
1793	Maximum Score of a Good Subarray	Two Pointers (Expanding from k).
1794	Count Pairs of Equal Substrings With Minimum Difference	First/Last Occurrence Hash Map.
1795	Rearrange Products Table (SQL)	SQL UNION ALL (Unpivoting).
1796	Second Largest Digit in a String	Single-Pass Scalar Tracking.
1797	Design Authentication Manager	Hash Map (Dictionary) for O(1) Access.
1798	Maximum Number of Consecutive Values You Can Make	Greedy (Sort and Gap-Check).
1799	Maximize Score After N Operations	Bitmask Dynamic Programming + GCD Precomputation.
1800	Maximum Ascending Subarray Sum	Single-Pass Greedy Accumulation (Kadane's Variant).
1801	Number of Orders in the Backlog	Two Priority Queues (Min-Heap for Sell, Max-Heap for Buy).
1802	Maximum Value at a Given Index in a Bounded Array	Binary Search on Answer + Greedy Math (Arithmetic Progression sum).
1803	Count Pairs With XOR in a Range	0-1 Trie (Binary Trie).
1804	Implement Trie II (Prefix Tree)	Standard Trie with wordCount and prefixCount variables at each node.
1805	Number of Different Integers in a String	Two Pointers + Hash Set (skip leading zeros directly).
1806	Minimum Number of Operations to Reinitialize a Permutation	Math/Graph (Cycle Detection). Track only the movement of index 1.
1807	Evaluate the Bracket Pairs of a String	Hash Map for lookups + Single Pass StringBuilder.
1808	Maximize Number of Nice Divisors	Greedy Math (Integer Break logic) + Fast Modular Exponentiation.
1809	Ad-Free Sessions (SQL)	LEFT JOIN with time range condition + IS NULL check.
1810	Minimum Path Cost in a Hidden Grid (Premium)	DFS mapping (to reveal grid) + Dijkstra's Algorithm (Shortest Path).
1811	Find Interview Candidates (SQL)	Window Functions (ROW_NUMBER) to group consecutive sequences + CTEs.
1812	Determine Color of a Chessboard Square	Math (Parity check): Sum of ASCII value of letter and integer value of number.
1813	Sentence Similarity III	Two Pointers / Deque (Match from front, match from back).
1814	Count Nice Pairs in an Array	Math Rearrangement + Hash Map: Check nums[i] - rev(nums[i]) == nums[j] - rev(nums[j]).
1815	Maximum Number of Groups Getting Fresh Donuts	Greedy (modulo matches) + DP with Bitmask / Memoization for remainders.
1816	Truncate Sentence	Single Pointer Iteration (count spaces and substring once).
1817	Finding the Users Active Minutes	Hash Map mapping User ID to a Hash Set of minutes.
1818	Minimum Absolute Sum Difference	Binary Search (bisect) on a sorted copy of nums1 to find the closest value to nums2[i].
1819	Number of Different Subsequences GCDs	Math (Sieve-like approach). Check if a target GCD 'x' can be formed by iterating its multiples.
1820	Maximum Number of Accepted Invitations (Premium)	Bipartite Graph Maximum Matching (Kuhn's Algorithm / DFS with Backtracking).
1821	Find Customers With Positive Revenue this Year (SQL)	WHERE clause filtering directly in SQL engine.

1822	Sign of the Product of an Array	Math (Parity tracking). Count negative numbers and check for zeros.
1823	Find the Winner of the Circular Game	Math (The Josephus Problem) / Bottom-Up DP.
1824	Minimum Sideway Jumps	Dynamic Programming (State Machine) / Greedy.
1825	Finding MK Average	Queue + 3 TreeMaps (Multisets for Bottom K, Middle M-2K, Top K) + Running Sum.
1826	Faulty Sensor (Premium)	Two Pointers / Linear Comparison (Shift checking).
1827	Minimum Operations to Make the Array Increasing	Greedy Math (Single Pass).
1828	Queries on Number of Points Inside a Circle	Math (Distance Squared Comparison).
1829	Maximum XOR for Each Query	Prefix XOR + Bitwise Complement.
1830	Minimum Number of Operations to Make String Sorted	Combinatorics (Cantor Expansion) + Modular Multiplicative Inverse + Precomputed Factorials.
1831	Maximum Transaction Each Day (SQL)	Window Function: RANK() or ALL operator.
1832	Check if the Sentence Is Pangram	Hash Set or Boolean Frequency Array.
1833	Maximum Ice Cream Bars	Greedy Algorithm + Sorting.
1834	Single-Threaded CPU	Min-Priority Queue (Min-Heap) sorted by processing time and original index.
1835	Find XOR Sum of All Pairs Bitwise AND	Math (Distributive Law): (a^b) AND (c^d).
1836	Remove Duplicates From an Unsorted Linked List (Premium)	Hash Map Count + Sentinel Node.
1837	Sum of Digits in Base K	Iterative Modulo and Division.
1838	Frequency of the Most Frequent Element	Sorting + Sliding Window.
1839	Longest Substring Of All Vowels in Order	One-pass Sliding Window / Greedy.
1840	Maximum Building Height	Sorting + Two-pass scan (Left-to-Right & Right-to-Left).
1841	League Statistics (SQL)	UNION ALL within a CTE + Group By.
1842	Next Palindrome Using Same Digits (Premium)	Next Permutation Algorithm on the first half.
1843	Suspicious Bank Accounts (SQL)	Window Function (LEAD/LAG) or PERIOD_DIFF.
1844	Replace All Digits with Characters	In-place modification of Character Array.
1845	Seat Reservation Manager	Min-Priority Queue (Min-Heap).
1846	Maximum Element After Decreasing and Rearranging	Sorting + Greedy Pass (or Counting Sort for O(N)).
1847	Closest Room (Premium)	Offline Queries + Sorting + Balanced BST (TreeSet).
1848	Minimum Distance to the Target Element	Two-way expansion from 'start' index.
1849	Splitting a String Into Descending Consecutive Values	Backtracking with Value Pruning.
1850	Minimum Adjacent Swaps to Reach the Kth Smallest Number	Next Permutation (K times) + Greedy Swap Alignment.
1851	Minimum Interval to Include Each Query	Offline Queries + Sorting + Min-Priority Queue (Heap).
1852	Distinct Numbers in Each Subarray (Premium)	Sliding Window + Frequency Array/Map.
1853	Convert Date Format (SQL)	SQL DATE_FORMAT() function.
1854	Maximum Population Year	Difference Array / Prefix Sum (Line Sweep).
1855	Maximum Distance Between a Pair of Values	Two Pointers (Greedy).
1856	Maximum Subarray Min-Product	Prefix Sum + Monotonic Stack (Next Smaller Element).
1857	Largest Color Value in a Directed Graph	Topological Sort (Kahn's) + Dynamic Programming.
1858	Longest Word With All Prefixes (Premium)	Trie (Prefix Tree) + DFS.
1859	Sorting the Sentence	String Split + Bucket Array mapping.
1860	Incremental Memory Leak	Standard Simulation (Loop) or Math (Square root approach).
1861	Rotating the Box	Two Pointers (Gravity logic) + Standard Matrix Rotation.
1862	Sum of Floored Pairs	Frequency Array + Prefix Sums + Math (Harmonic Series block processing).
1863	Sum of All Subset XOR Totals	Backtracking/Recursion or Bit Manipulation (Math: result is OR(all_elements) x 2^{n-1}).
1864	Minimum Number of Swaps to Make the Binary String Alternate	Greedy Counting (Compare with two possible targets: "0101..." and "1010...").
1865	Finding Pairs With a Certain Sum	Hash Map Frequency Tracking.
1866	Number of Ways to Rearrange Sticks With K Sticks Visible	Dynamic Programming (Stirling Numbers of the First Kind).
1867	Orders With Maximum Quantity Above Average (SQL)	Window Functions or Common Table Expressions (CTEs) with MAX() and AVG().
1868	Product of Two Run-Length Encoded Arrays (Premium)	Two Pointers on compressed segments.
1869	Longer Contiguous Segments of Ones than Zeros	Single Pass Counter (Greedy).
1870	Minimum Speed to Arrive on Time	Binary Search on Answer.
1871	Jump Game VII	Dynamic Programming + Sliding Window (Prefix Sum of Reachable Nodes).
1872	Stone Game VIII	Prefix Sum + Dynamic Programming (Suffix Max).
1873	Calculate Special Bonus (SQL)	`CASE` statement or `IF` function + `MOD()` + `LIKE`.
1874	Minimize Product Sum of Two Arrays (Premium)	Greedy Algorithm (Rearrangement Inequality): Sort one ascending, one descending.
1875	Group Employees of the Same Salary (Premium)	CTE with `COUNT(*) OVER(PARTITION BY salary)` + `DENSE_RANK()`.
1876	Substrings of Size Three with Distinct Characters	Fixed Sliding Window (No Set).
1877	Minimize Maximum Pair Sum in Array	Greedy Strategy + Sorting.
1878	Get Biggest Three Rhombus Sums in a Grid	2D Diagonal Prefix Sums (Prefix Sum on two diagonal directions).
1879	Minimum XOR Sum of Two Arrays	Dynamic Programming + Bitmasking.
1880	Check if Word Equals Summation of Two Words	Direct Math (Positional Notation).
1881	Maximum Value after Insertion	Greedy (First Mismatch).

1882	Process Tasks Using Servers	Two Priority Queues (Available Heap, Busy Heap).
1883	Minimum Skips to Arrive at Meeting on Time	Dynamic Programming (State: [roads][skips]).
1884	Egg Drop With 2 Eggs and N Floors	Math (Triangular Numbers) or DP.
1885	Count Pairs in Two Arrays	Two Pointers on Difference Array.
1886	Determine Whether Matrix Can Be Obtained By Rotation	In-place Matrix Rotation (Transpose + Reverse) or Index Mapping.
1887	Reduction Operations to Make the Array Elements Equal	Sorting + Greedy Counting.
1888	Minimum Number of Flips to Make the Binary String Alternating	Sliding Window on concatenated string (s + s).
1889	Minimum Space Wasted From Packaging	Sorting + Binary Search (Prefix Sums).
1890	The Latest Login in 2020 (SQL)	GROUP BY + MAX() + EXTRACT/LIKE/YEAR().
1891	Cutting Ribbons (Premium)	Binary Search on Answer.
1892	Page Recommendations II (SQL Premium)	Common Table Expression (CTE) + LEFT JOIN filtering.
1893	Check if All the Integers in a Range Are Covered	Difference Array / Prefix Sum (Line Sweep).
1894	Find the Student that Will Replace the Chalk	Prefix Sum + Modulo Arithmetic + Binary Search.
1895	Largest Magic Square	2D Prefix Sums (Rows, Cols, and Diagonals).
1896	Minimum Cost to Change the Final Value of Expression	Stack-based Expression Parsing + Dynamic Programming (bottom-up costs).
1897	Redistribute Characters to Make All Strings Equal	Hash Map / Frequency Count (Modulo Arithmetic).
1898	Maximum Number of Removable Characters	Binary Search on Answer (k) + Two Pointers.
1899	Merge Triplets to Form Target Triplet	Greedy Filtering (One Pass).
1900	The Earliest and Latest Rounds Where Players Compete	Memoized Recursion / Dynamic Programming on (n, firstPlayer, secondPlayer).
1901	Find a Peak Element II	Binary Search on Columns/Rows
1902	Depth of BST Given Insertion Order	TreeMap / Ordered Map (Next/Prev Greater Element)
1903	Largest Odd Number in String	Greedy (Right-to-Left Traversal)
1904	The Number of Full Rounds You Have Played	Math / Modular Arithmetic
1905	Count Sub Islands	Graph / DFS with Simultaneous Verification
1906	Minimum Absolute Difference Queries	Prefix Sums over Frequency Array
1907	Count Salary Categories	Single Pass Bucketing (or SQL UNION/GROUP BY)
1908	Game of Nim	Sprague-Grundy Theorem (Bitwise XOR / Nim-Sum)
1909	Remove One Element to Make the Array Strictly Increasing	Single Pass Greedy Violation Detection
1910	Remove All Occurrences of a Substring	Stack-based Pattern Matching
1911	Maximum Alternating Subsequence Sum	Dynamic Programming / State Machine
1912	Design Movie Rental System	Multiple TreeSets / Priority Queues + HashMaps
1913	Maximum Product Difference Between Two Pairs	Single Pass Greedy Tracking
1914	Cyclically Rotating a Grid	1D Array Extraction & Modular Rotation
1915	Number of Wonderful Substrings	Bitmasking + Prefix XOR + Frequency Hash Map
1916	Count Ways to Build Rooms in an Ant Colony	Tree DP with Combinatorics
1917	Leetcodify Friends Recommendations (SQL)	Self-Join with Anti-Join (NOT IN)
1918	Kth Smallest Subarray Sum	Binary Search on Answer + Sliding Window
1919	Leetcodify Similar Friends (SQL)	Filtered Inner Join Pipeline
1920	Build Array from Permutation	In-Place Encoding (Math / Modulo)
1921	Eliminate Maximum Number of Monsters	Greedy Sorting by "Time-to-Arrival"
1922	Count Good Numbers	Binary Exponentiation (Fast Power)
1923	Longest Common Subpath	Binary Search on Length + Rolling Hash (Rolling Hash)
1924	Erect a Fence II	Welzl's Algorithm (Recursive Smallest Enclosing Circle)
1925	Count Square Sum Triples	Double Loop with Square Root Check
1926	Nearest Exit from Entrance in Maze	Breadth-First Search (BFS)
1927	Sum Game	Greedy Mathematical Balancing
1928	Minimum Cost to Reach Destination in Time	Modified Dijkstra (or 2D DP)
1929	Concatenation of Array	Single-Pass Index Mapping
1930	Unique Length-3 Palindromic Subsequences	First/Last Index Pre-computation
1931	Painting a Grid With Three Different Colors	Bitmask / Profile DP
1932	Merge BSTs to Create Single BST	Hash Map Root Lookup + DFS Validation
1933	Check if String Is Decomposable Into Value-Equal Substrings	Greedy Block Counting (Run-Length)
1934	Confirmation Rate (SQL)	LEFT JOIN with Conditional Aggregation
1935	Maximum Number of Words You Can Type	Hash Set / Boolean Array Lookup
1936	Add Minimum Number of Rungs	Greedy with Modular Arithmetic
1937	Maximum Number of Points with Cost	DP with Left/Right Prefix Max
1938	Maximum Genetic Difference Query	Offline Queries + DFS + Binary Trie
1939	Users That Actively Request Confirmation Messages	Window Function (LAG/LEAD)
1940	Longest Common Subsequence Between Sorted Arrays	Frequency Counting (Since arrays are sorted & distinct)
1941	Check if All Characters Have Equal Number of Occurrences	Hash Map / Frequency Array
1942	The Number of the Smallest Unoccupied Chair	Two Min-Heaps (Priority Queues)

1943	Describe the Painting	Sweep Line / Difference Array
1944	Number of Visible People in a Queue	Monotonic Stack (Decreasing)
1945	Sum of Digits of String After Convert	Direct Integer Accumulation
1946	Largest Number After Mutating Substring	Greedy One-Pass
1947	Max Compatibility Score Sum	DP with Bitmask
1948	Delete Duplicate Folders in System	Trie + Merkle Hashing (Post-order)
1949	Strong Friendship (SQL)	Self-Join on Common Friends
1950	Max of Min Values in All Subarrays	Monotonic Stack + Range Extension
1951	All the Pairs With the Maximum Number of Common Followers (SQL)	Self-Join on Follower Column
1952	Three Divisors	Square of Prime Property
1953	Maximum Number of Weeks for Which You Can Work	Greedy Gap Filling / Pigeonhole Principle
1954	Minimum Garden Perimeter to Collect Enough Apples	Binary Search on Cubic Formula
1955	Count Number of Special Subsequences	Dynamic Programming (3-State Machine)
1956	Minimum Time For K Virus Variants to Spread	Binary Search on Answer + Intersection Checks
1957	Delete Characters to Make Fancy String	Single-Pass Greedy Scan
1958	Check if Move is Legal	8-Directional Raycasting from Move
1959	Minimum Total Space Wasted With K Resizing Operations	DP with Interval Precomputation
1960	Maximum Product of the Length of Two Palindromic Substrings	Manacher's Algorithm + Prefix/Suffix Max
1961	Check If String Is a Prefix of Array	Early Exit Iterator
1962	Remove Stones to Minimize the Total	Max-Priority Queue (Heap)
1963	Minimum Number of Swaps to Make the String Balanced	Greedy "Unbalanced" Counter
1964	Find the Longest Valid Obstacle Course at Each Position	Patience Sorting (LIS Variation)
1965	Employees With Missing Information (SQL)	FULL OUTER JOIN (or UNION of LEFT JOINs)
1966	Binary Searchable Numbers in an Unsorted Array	Prefix Max & Suffix Min Pass
1967	Number of Strings That Appear as Substrings in Word	KMP or Optimized Built-in Search
1968	Array With Elements Not Equal to Average of Neighbors	Sort & Interleave (Wiggle Sort)
1969	Minimum Non-Zero Product of the Array Elements	Greedy Pairing + Modular Exponentiation
1970	Last Day Where You Can Still Cross	Binary Search on Days + BFS/DFS
1971	Find if Path Exists in Graph	Union-Find (Disjoint Set Union)
1972	First and Last Call On the Same Day (SQL)	Window Functions (ROW_NUMBER/RANK)
1973	Count Nodes Equal to Sum of Descendants	Post-order Traversal (Bottom-Up)
1974	Minimum Time to Type Word Using Special Typewriter	Greedy Modular Distance
1975	Maximum Matrix Sum	Greedy Parity Property
1976	Number of Ways to Arrive at Destination	Dijkstra + DP (Counting)
1977	Number of Ways to Separate Numbers	DP + LCP (Suffix) + Prefix Sums
1978	Employees Whose Manager Left the Company (SQL)	Self-Anti-Join / WHERE NOT IN
1979	Find Greatest Common Divisor of Array	Euclidean Algorithm (Recursive/Iterative)
1980	Find Unique Binary String	Cantor's Diagonalization
1981	Minimize the Difference Between Target and Chosen Elements	DP with Bitset Optimization
1982	Find Array Given Subset Sums	Recursive Divide and Conquer (Sign Partitioning)
1983	Widest Pair of Indices With Equal Range Sum	Prefix Sum + Hash Map (Diff Tracking)
1984	Min Diff Between Highest and Lowest of K Scores	Sorting + Fixed-Size Sliding Window
1985	Find the Kth Largest Integer in the Array	Custom Sort (Length then Lexicographical)
1986	Minimum Number of Work Sessions to Finish the Tasks	DP with Bitmask (State: Mask of completed tasks)
1987	Number of Unique Good Subsequences	Dynamic Programming (Two-State: Ends in 0, Ends in 1)
1988	Find Cut-Off Score for Each School (SQL)	LEFT JOIN + Aggregation
1989	Maximum Number of People That Can Be Caught in Tag	Two Pointers (Greedy)
1990	Count the Number of Experiments (SQL)	Cross Join (Master Template) + Left Join
1991	Find the Middle Index in Array	Prefix Sum (Total Sum Balance)
1992	Find All Groups of Farmland	Greedy Boundary Detection (Top-Left to Bottom-Right)
1993	Operations on Tree	Hash Map State Tracking + DFS Pruning
1994	The Number of Good Subsets	DP with Bitmask (Prime Representation)
1995	Count Special Quadruplets	Hash Map (Meet-in-the-Middle logic)
1996	The Number of Weak Characters in the Game	Sorting (Attack Desc, Defense Asc) + 1-Pass Greedy
1997	First Day Where You Have Been in All the Rooms	Dynamic Programming (Prefix Sum Transition)
1998	GCD Sort of an Array	Union-Find + Sieve of Eratosthenes
1999	Smallest Greater Multiple Made of Two Digits	BFS (Digit Permutation Generation)
2000	Reverse Prefix of Word	Two-Pointer In-Place Swap
2001	Number of Pairs of Interchangeable Rectangles	Hash Map (Frequency Counter) with GCD reduction for fractions
2002	Maximum Product of the Length of Two Palindromic Subsequences	Bitmask Enumeration + Precomputed Palindromes
2003	Smallest Missing Genetic Value in Each Subtree	DFS Post-order Traversal + Tracking Node with Value 1

2004	The Number of Seniors and Juniors to Join the Company (SQL)	SQL Window Functions (Cumulative Sum/Running Total) + CTEs
2005	Subtree Removal Game with Fibonacci Tree	Game Theory (Sprague-Grundy/Nim) & Modulo Math Pattern
2006	Count Number of Pairs With Absolute Difference K	Hash Map (Frequency Count) / Single Pass
2007	Find Original Array From Doubled Array	Sorting + Frequency Map (or Queue)
2008	Maximum Earnings From Taxi	Sorting + Dynamic Programming + Binary Search
2009	Minimum Number of Operations to Make Array Continuous	Sort + Deduplicate + Sliding Window (Two Pointers)
2010	The Number of Seniors and Juniors to Join the Company II (SQL)	SQL Window Functions + Common Table Expressions (CTEs)
2011	Final Value of Variable After Performing Operations	Middle Character Extraction
2012	Sum of Beauty in the Array	Prefix Max and Suffix Min Arrays
2013	Detect Squares	Hash Map Point Frequencies + Diagonal Anchoring
2014	Longest Subsequence Repeated k Times	Breadth-First Search (BFS) String Building + Frequency Pruning
2015	Average Height of Buildings in Each Segment	Line Sweep Algorithm + TreeMap (Event Sorting)
2016	Maximum Difference Between Increasing Elements	Single Pass with Running Minimum (Prefix Min)
2017	Grid Game	Prefix Sum & Suffix Sum Arrays (Minimax pattern)
2018	Check if Word Can Be Placed In Crossword	String Splitting (by '#') + Segment Matching
2019	The Score of Students Solving Math Expression	Interval DP (Matrix Chain Multiplication) + HashSet
2020	Number of Accounts That Did Not Stream (SQL)	LEFT JOIN with NULL check / EXCEPT operator
2021	Brightest Position on Street	Line Sweep Algorithm + TreeMap (Difference Array)
2022	Convert 1D Array Into 2D Array	Math/Index Mapping (i / n, i % n)
2023	Number of Pairs of Strings With Concatenation Equal to Target	Hash Map (Frequency) + String Prefix/Suffix Splitting
2024	Maximize the Confusion of an Exam	Sliding Window (Two Pointers) Run Twice
2025	Maximum Number of Ways to Partition an Array	Prefix Sums + Hash Maps for Difference Tracking
2026	Low-Quality Problems (SQL)	Direct SQL Filtering (Math inside WHERE clause)
2027	Minimum Moves to Convert String	Greedy Algorithm (Pointer Jumping)
2028	Find Missing Observations	Math / Greedy Distribution (Base + Remainder)
2029	Stone Game IX	Math (Modulo 3 Categorization) & Case Analysis
2030	Smallest K-Length Subsequence With Occurrences of a Letter	Monotonic Stack with Budget Constraints
2031	Count Subarrays With More Ones Than Zeros	Prefix Sum + Binary Indexed Tree (Fenwick) / Merge Sort
2032	Two Out of Three	Frequency Hash Map or Bitmasking on an Array
2033	Minimum Operations to Make a Uni-Value Grid	Math (Median Finding) + Divisibility Check
2034	Stock Price Fluctuation	Hash Map (Timestamp -> Price) + Two Heaps (or TreeMap)
2035	Partition Array Into Two Arrays to Minimize Sum Difference	Meet in the Middle + Binary Search
2036	Maximum Alternating Subarray Sum	Dynamic Programming (Kadane's Algorithm Variant)
2037	Minimum Number of Moves to Seat Everyone	Sorting + Greedy Matching
2038	Remove Colored Pieces if Both Neighbors are the Same Color	Single Pass Counting (Independent Moves)
2039	The Time When the Network Becomes Idle	Breadth-First Search (BFS) + Mathematical Formula
2040	Kth Smallest Product of Two Sorted Arrays	Binary Search on Answer Range + Two Pointers/Binary Search for Counting
2041	Accepted Candidates From the Interviews (SQL)	SQL JOIN + GROUP BY + HAVING
2042	Check if Numbers Are Ascending in a Sentence	Single Pass Tokenization / On-the-fly Parsing
2043	Simple Bank System	1D Array Direct Indexing (1-based to 0-based mapping)
2044	Count Number of Maximum Bitwise-OR Subsets	Backtracking (DFS) / Pick & Skip State Tracking
2045	Second Minimum Time to Reach Destination	Modified BFS (Dijkstra-like) with Top-2 Tracking + Traffic Light Math
2046	Sort Linked List Already Sorted Using Absolute Values	Head Re-linking (Two-Pointer Shift)
2047	Number of Valid Words in a Sentence	Regex Matching or Structured Single-Pass Scan
2048	Next Greater Numerically Balanced Number	Precomputation / Backtracking with Pruning
2049	Count Nodes With the Highest Score	DFS with Subtree Size Precomputation
2050	Parallel Courses III	Topological Sort (Kahn's) + Dynamic Programming
2051	The Category of Each Member in the Store (SQL)	Left Join + CASE Statement + Aggregate Functions
2052	Minimum Cost to Separate Sentence Into Rows	Dynamic Programming (1D)
2053	Kth Distinct String in an Array	Hash Map (Frequency Counter) + Second Pass
2054	Two Best Non-Overlapping Events	Sorting + Binary Search + Suffix Max
2055	Plates Between Candles	Prefix Sums + Nearest Candle Precomputation
2056	Number of Valid Move Combinations On Chessboard	Backtracking + Recursive State Validation
2057	Smallest Index With Equal Value	Single-Pass Early Exit
2058	Find the Minimum and Maximum Number of Nodes Between Critical Points	Single Pass with Three-Node Window + Edge Point Tracking
2059	Minimum Operations to Convert Number	BFS (Breadth-First Search) + Visited Array (0-1000)
2060	Check if an Original String Exists Given Two Encoded Strings	Dynamic Programming (State: i, j, diff)
2061	Number of Spaces Cleaned by Robot	Simulation + 3D State Tracking (r, c, direction)
2062	Count Vowel Substrings of a String	Sliding Window (Two Pointers) with Sub-problem Logic
2063	Vowels of All Substrings	Math / Contribution Technique
2064	Minimized Maximum of Products Distributed to Any Store	Binary Search on Answer Range

2065	Maximum Path Quality of a Graph	Backtracking (DFS) with Time Pruning
2066	Account Balance After Transactions (SQL)	SQL Common Table Expressions (CTE) + SUM() Aggregate
2067	Number of Equal Count Substrings	Fixed-Window Sliding (Multiple Passes)
2068	Check Whether Two Strings are Almost Equivalent	Dual Frequency Array Comparison
2069	Walking Robot Simulation II	Perimeter Modulo Math
2070	Most Beautiful Item for Each Query	Sorting + Prefix Maximum + Binary Search
2071	Maximum Number of Tasks You Can Assign	Binary Search on Answer + Greedy with Monotonic Queue (Deque)
2072	The Winner University (SQL)	Independent Subqueries / Common Table Expressions (CTE)
2073	Time Needed to Buy Tickets	One-Pass Mathematical Summation
2074	Reverse Nodes in Even Length Groups	In-Place Group Iteration & Re-linking
2075	Decode the Slanted Ciphertext	Direct Index Mapping (One-Pass)
2076	Process Restricted Friend Requests	Disjoint Set Union (DSU) with Restrictions Check
2077	Paths in Maze That Lead to Same Room	Adjacency Set Intersection (Triangle Counting)
2078	Two Furthest Houses With Different Colors	Greedy Extremes Check
2079	Watering Plants	Math-Based Accumulator
2080	Range Frequency Queries	Hash Map of Sorted Indices + Binary Search
2081	Sum of k-Mirror Numbers	Palindrome Generation (Length-based)
2082	The Number of Rich Customers (SQL)	COUNT(DISTINCT customer_id)
2083	Substrings That Begin and End With the Same Letter	Combinatorics / Hash Map
2084	Drop Type 1 Orders for Customers With Type 0 Orders (SQL)	Window Function (MIN) or Grouped Filtering
2085	Count Common Words With One Occurrence	Two Hash Maps (Double Tally)
2086	Minimum Number of Food Buckets to Feed the Hamsters	Greedy / Linear Scan
2087	Minimum Cost Homecoming of a Robot in a Grid	Greedy Coordinate Accumulation (Math)
2088	Count Fertile Pyramids in a Land	2D Dynamic Programming (Bottom-Up)
2089	Find Target Indices After Sorting Array	Counting / Frequency (Linear Scan)
2090	K Radius Subarray Averages	Sliding Window (Fixed Size) / Prefix Sums
2091	Removing Minimum and Maximum From Array	Three-Case Greedy Analysis
2092	Find All People With Secret	Sorted Time Processing + DSU with Reset
2093	Minimum Cost to Reach City With Discounts	Dijkstra with State (City, DiscountsUsed)
2094	Finding 3-Digit Even Numbers	Frequency Counting + Iterative Value Check
2095	Delete the Middle Node of a Linked List	Fast and Slow Pointers
2096	Step-By-Step Directions From a Binary Tree Node to Another	LCA (Lowest Common Ancestor) + Path Comparison
2097	Valid Arrangement of Pairs	Hierholzer’s Algorithm (Eulerian Path)
2098	Subsequence of Size K With the Largest Even Sum	Greedy Selection + Smallest-Odd/Even Swap
2099	Find Subsequence of Length K With the Largest Sum	Sorting by Value + Sorting by Index
2100	Find Good Days to Rob the Bank	Precomputed Prefix/Suffix Counts (DP)
2101	Detonate the Maximum Bombs	Graph Traversal (Directed Graph) + BFS/DFS.
2102	Sequentially Ordinal Rank Tracker	Two Heaps (Min-Heap / Max-Heap) or Balanced BST.
2103	Rings and Rods	Bitmasking + Array.
2104	Sum of Subarray Ranges	Monotonic Stack (O(N)).
2105	Watering Plants II	Two Pointers (Greedy Simulation).
2106	Maximum Fruits Harvested After at Most K Steps	Prefix Sum + Binary Search (or Sliding Window).
2107	Number of Unique Flavors After Sharing K Candies	Sliding Window + Frequency HashMap.
2108	Find First Palindromic String in the Array	Two Pointers (String validation).
2109	Adding Spaces to a String	Two Pointers + Pre-allocated Array/StringBuilder.
2110	Number of Smooth Descent Periods of a Stock	Dynamic Programming / Math (Running Count).
2111	Minimum Operations to Make the Array K-Increasing	Longest Non-Decreasing Subsequence (Patience Sorting / Binary Search).
2112	The Airport With the Most Traffic	Graph Degree Counting (Array-based Frequency Map).
2113	Elements in Array After Removing and Replacing Elements	Math / Modulo Arithmetic.
2114	Maximum Number of Words Found in Sentences	String Iteration / Character Counting.
2115	Find All Possible Recipes from Given Supplies	Topological Sort (Kahn's Algorithm) / BFS.
2116	Check if a Parentheses String Can Be Valid	Greedy (Two-Pass Scanning).
2117	Abbreviating the Product of a Range	Math (Trailing Zeros + Modulo/Logarithms).
2118	Build the Equation (SQL)	SQL String Aggregation (`GROUP_CONCAT` / `STRING_AGG`).
2119	A Number After a Double Reversal	Math / Observation (O(1)).
2120	Execution of All Suffix Instructions Staying in a Grid	O(M^2) Simulation (M = instruction length).
2121	Intervals Between Identical Elements	Prefix and Suffix Sums / Hash Map of Lists.
2122	Recover the Original Array	Sorting + Greedy Candidate Validation.
2123	Minimum Operations to Remove Adjacent Ones in Matrix	Bipartite Graph Matching / Min Vertex Cover.
2124	Check if All A's Appears Before All B's	Linear Scan / Substring Search.

2125	Number of Laser Beams in a Bank	Math (Combinatorics) + Filtering.
2126	Destroying Asteroids	Greedy + Sorting.
2127	Maximum Employees to Be Invited to a Meeting	Graph (Functional Graph / Cycle Detection + Topological Sort).
2128	Remove All Ones With Row and Column Flips	Math / Binary Pattern Recognition.
2129	Capitalize the Title	String Builder / Two Pointers (In-place if allowed).
2130	Maximum Twin Sum of a Linked List	Fast/Slow Pointers + Reverse Linked List.
2131	Longest Palindrome by Concatenating Two Letter Words	Frequency Hash Map + Greedy Matching.
2132	Stamping the Grid	2D Prefix Sum + 2D Difference Array.
2133	Check if Every Row and Column Contains All Numbers	Hash Set Validation (or Boolean Array Tracking).
2134	Minimum Swaps to Group All 1's Together II	Fixed Sliding Window on a Concatenated Array.
2135	Count Words Obtained After Adding a Letter	Bitmasking / Sorting + Hash Set.
2136	Earliest Possible Day of Full Bloom	Greedy + Sorting.
2137	Pour Water Between Buckets to Make Water Levels Equal	Binary Search on Answer.
2138	Divide a String Into Groups of Size k	String Chunking / Substring Math.
2139	Minimum Moves to Reach Target Score	Greedy (Work Backwards from Target).
2140	Solving Questions With Brainpower	1D Dynamic Programming (Right to Left).
2141	Maximum Running Time of N Computers	Binary Search on Answer + Greedy.
2142	The Number of Passengers in Each Bus (SQL)	Sorting + Window Functions (Lead/Lag) or Inequality Joins.
2143	Choose Numbers From Two Arrays in Range	Dynamic Programming (Space Optimized HashMap).
2144	Minimum Cost of Buying Candies With Discount	Greedy (Sorting).
2145	Count the Hidden Sequences	Prefix Sum + Range Math.
2146	K Highest Ranked Items Within a Price Range	BFS + Priority Queue (Min-Heap / Max-Heap).
2147	Number of Ways to Divide a Long Corridor	Combinatorics / Greedy Math.
2148	Count Elements With Strictly Smaller and Greater Elements	Min/Max Tracking.
2149	Rearrange Array Elements by Sign	Two Pointers (One-Pass).
2150	Find All Lonely Numbers in the Array	Frequency Hash Map.
2151	Maximum Good People Based on Statements	Bitmasking (Brute Force is the only way, but Bitmasking is the production-ready implementation).
2152	Minimum Number of Lines to Cover Points	Bitmask DP / State Compression.
2153	The Number of Passengers in Each Bus II (SQL)	Iterative Simulation with Window Functions.
2154	Keep Multiplying Found Values by Two	Hash Set / Sorting + Binary Search.
2155	All Divisions With the Highest Score of a Binary Array	Prefix Sum / Running Count.
2156	Find Substring With Given Hash Value	Rolling Hash (Rabin-Karp) - Reversed.
2157	Groups of Strings	Bitmasking + Disjoint Set Union (DSU).
2158	Amount of New Area Painted Each Day	Merge Intervals / Segment Tree / Jump Pointers (DSU).
2159	Order Two Columns Independently (SQL)	Window Functions (`ROW_NUMBER`).
2160	Minimum Sum of Four Digit Number After Splitting Digits	Greedy (Sorting).
2161	Partition Array According to Given Pivot	Three-Pointer Simulation (Single-Pass Result Array).
2162	Minimum Cost to Set Cooking Time	Greedy (Two-Candidate Check).
2163	Minimum Difference in Sums After Removal of Elements	Two Heaps (Priority Queues) + Prefix/Suffix Sums.
2164	Sort Even and Odd Indices Independently	In-Place Extraction + Sorting.
2165	Smallest Value of the Rearranged Number	Greedy Sorting.
2166	Design Bitset	Lazy Flipping (Flag + Two Arrays).
2167	Minimum Time to Remove All Cars Containing Illegal Goods	Dynamic Programming / Prefix-Suffix Min.
2168	Unique Substrings With Equal Digit Frequency	Rolling Hash (Rolling Frequencies) + Trie.
2169	Count Operations to Obtain Zero	Math (Euclidean-like division).
2170	Minimum Operations to Make the Array Alternating	Frequency Counting + Top 2 Candidates.
2171	Removing Minimum Number of Magic Beans	Sorting + Math (Prefix Sum Logic).
2172	Maximum AND Sum of Array	Bitmask DP (Ternary/Base-3 or Base-2 with double slots).
2173	Longest Winning Streak (SQL)	Window Functions (Gaps and Islands).
2174	Remove All Ones With Row and Column Flips II	Bitmask DP / Backtracking with State Compression.
2175	The Change in Global Rankings (SQL)	Window Functions (`RANK()` or `ROW_NUMBER()`).
2176	Count Equal and Divisible Pairs in an Array	Frequency Hash Map (Value -> List of Indices).
2177	Find Three Consecutive Integers That Sum to a Number	Math (Algebraic Equation).
2178	Maximum Split of Positive Even Integers	Greedy (Smallest Even Strategy).
2179	Count Good Triplets in an Array	Fenwick Tree (BIT) / Index Mapping
2180	Count Integers With Even Digit Sum	Math / Parity Analysis.
2181	Merge Nodes in Between Zeros	In-Place Two Pointers / Link Manipulation.
2182	Construct String With Repeat Limit	Greedy + Frequency Array (Counting Sort).
2183	Count Array Pairs Divisible by K	Number Theory (GCD + Frequency Map).
2184	Number of Ways to Build Sturdy Brick Wall	Bitmask DP + State Compression.
2185	Counting Words With a Given Prefix	String Comparison (In-place `startsWith`).

2186	Minimum Number of Steps to Make Two Strings Anagrams II	Frequency Array (Counting Sort).
2187	Minimum Time to Complete Trips	Binary Search on Answer.
2188	Minimum Time to Finish the Race	Dynamic Programming + Precomputed Best Times.
2189	Number of Ways to Build House of Cards	2D Dynamic Programming / Memoization.
2190	Most Frequent Number Following Key In an Array	Single-Pass Frequency Map.
2191	Sort the Jumbled Numbers	Pre-calculation + Stable Sorting.
2192	All Ancestors of a Node in a DAG	Reverse Graph Traversal or Topological Sort.
2193	Minimum Number of Moves to Make Palindrome	Greedy (Two Pointers).
2194	Cells in a Range on an Excel Sheet	Nested Char Loops.
2195	Append K Integers With Minimal Sum	Sorting + Arithmetic Progression.
2196	Create Binary Tree From Descriptions	Hash Map (Val -> Node) + Set (Child Tracking).
2197	Replace Non-Coprime Numbers in Array	Stack (Greedy Simplification).
2198	Number of Single Divisor Triplets	Frequency Array + Triple Loop (Values 1-100).
2199	Finding the Topic of Each Post (SQL)	SQL Join + String Aggregation.
2200	Find All K-Distant Indices in an Array	Two Pointers / Single-Pass Sliding Window.
2201	Count Artifacts That Can Be Extracted	Hash Set / Direct Indexing
2202	Maximize the Topmost Element After K Moves	Greedy / Math Analysis
2203	Minimum Weighted Subgraph With the Required Paths	3-Way Dijkstra's Algorithm
2204	Distance to a Cycle in Undirected Graph	Kahn's Algorithm (Topological Sort) + Multi-source BFS
2205	The Number of Users That Are Eligible for Discount	Fixed Sliding Window + Frequency Map
2206	Divide Array Into Equal Pairs	Frequency Map / Hash Table
2207	Maximize Number of Subsequences in a String	Greedy / Prefix Counters
2208	Minimum Operations to Halve Array Sum	Max Heap (Priority Queue)
2209	Minimum White Tiles After Covering With Carpets	2D Dynamic Programming
2210	Count Hills and Valleys in an Array	Single Pass + Prev Pointer
2211	Count Collisions on a Road	Two Pointers / String Trimming
2212	Maximum Points in an Archery Competition	Bitmask / Binary Enumeration
2213	Longest Substring of One Repeating Character	Segment Tree
2214	Minimum Health to Beat Game	Greedy Math / Total Sum
2215	Find the Difference of Two Arrays	Hash Sets / Set Difference
2216	Minimum Deletions to Make Array Beautiful	Greedy / Virtual Deletion
2217	Find Palindrome With Fixed Length	Math / Half-String Mirroring
2218	Maximum Value of K Coins From Piles	2D DP (Knapsack Variant)
2219	Maximum Sum Score of Array	Prefix Sums / Running Totals
2220	Minimum Bit Flips to Convert Number	Bitwise XOR + Brian Kernighan's Alg
2221	Find Triangular Sum of an Array	Math / Binomial Coefficients
2222	Number of Ways to Select Buildings	Prefix/Suffix Counts
2223	Sum of Scores of Built Strings	Z-Algorithm
2224	Minimum Number of Operations to Convert Time	Greedy Subtraction
2225	Find Players With Zero or One Losses	Hash Map / Frequency Array
2226	Maximum Candies Allocated to K Children	Binary Search on Answer
2227	Encrypt and Decrypt Strings	Reverse Encryption Mapping (Trie/Hash)
2228	Users With Two Purchases Within Seven Days	Window Function / LAG (SQL)
2229	Check if an Array Is Consecutive	Math Bounds + Hash Set
2230	The Users That Are Eligible for Discount	Targeted Filter + Distinct (SQL)
2231	Largest Number After Digit Swaps by Parity	Sorting by Parity / Max Heap
2232	Minimize Result by Adding Parentheses to Expression	Sliding Boundary Pointers
2233	Maximum Product After K Increments	Min Heap (Priority Queue)
2234	Maximum Total Beauty of the Gardens	Sorting + Prefix Sum + Two Pointers/Binary Search
2235	Add Two Integers	Direct Math Operations / ALU
2236	Root Equals Sum of Children	Direct Comparison / O(1) Access
2237	Count Positions on Street With Required Brightness	Difference Array (Sweep Line)
2238	Number of Times a Driver Was a Passenger	Left Join + Group By (SQL)
2239	Find Closest Number to Zero	Linear One-Pass Update
2240	Number of Ways to Buy Pens and Pencils	Single Loop Math
2241	Design an ATM Machine	Greedy Hash Map / Frequency Array
2242	Maximum Score of a Node Sequence	Edge-Centric + Top 3 Neighbors
2243	Calculate Digit Sum of a String	Iterative StringBuilder / Two-Pointer
2244	Minimum Rounds to Complete All Tasks	Greedy Math / Frequency Map
2245	Maximum Trailing Zeros in a Cornered Path	2D Prefix Sums (Factors of 2 and 5)
2246	Longest Path With Different Adjacent Characters	Tree DP (Post-order Traversal)

2247	Maximum Cost of Trip With K Highways	Dynamic Programming with Bitmask
2248	Intersection of Multiple Arrays	Frequency Map / Counting Sort
2249	Count Lattice Points Inside Circles	Global Bounding Box Scan
2250	Count Number of Rectangles Containing Each Point	Sorting + Binary Search (By Height)
2251	Number of Flowers in Full Bloom	Coordinate Compression / Line Sweep
2252	Dynamic Pivoting of a Table	Prepared Statements / Dynamic SQL
2253	Dynamic Unpivoting of a Table	Dynamic SQL / Information Schema
2254	Design Video Sharing Platform	Hash Map + Min-Heap (for ID reuse)
2255	Count Prefixes of a Given String	Trie (Prefix Tree) / Hash Map
2256	Minimum Average Difference	Prefix Sum / Running Total
2257	Count Unguarded Cells in the Grid	Matrix Marking / State Grid
2258	Escape the Spreading Fire	Multi-Source BFS + Arrival Time Comparison
2259	Remove Digit From Number to Maximize Result	Greedy One-Pass Comparison
2260	Minimum Consecutive Cards to Pick Up	Sliding Window / Hash Map
2261	K Divisible Elements Subarrays	Sliding Window + Rabin-Karp / Trie
2262	Total Appeal of A String	DP / Contribution of Each Character
2263	Make Array Non-decreasing or Non-increasing	Priority Queue (Slope Trick)
2264	Largest 3-Same-Digit Number in String	Iterative Comparison
2265	Count Nodes Equal to Average of Subtree	Post-order Traversal (Bottom-Up DP)
2266	Count Number of Texts	Linear DP (Climbing Stairs Variant)
2267	Check if There Is a Valid Parentheses String Path	3D Dynamic Programming
2268	Minimum Number of Keypresses	Greedy + Sorting
2269	Find the K-Beauty of a Number	Sliding Window (String or Math)
2270	Number of Ways to Split Array	Prefix Sum / Running Total
2271	Maximum White Tiles Covered by a Carpet	Two Pointers + Prefix Sum
2272	Substring With Largest Variance	Kadane’s Algorithm Variation
2273	Find Resultant Array After Removing Anagrams	Stack / Previous-Word Pointer
2274	Maximum Consecutive Floors Without Special Floors	Sorting + Gap Calculation
2275	Largest Combination With Bitwise AND Greater Than Zero	Bit-Position Counting
2276	Count Integers in Covered Intervals	Segment Tree (Dynamic) / Churning Map
2277	Closest Node to Path in Tree	LCA + Distance Formula
2278	Percentage of Letter in String	Linear Counter + Math
2279	Maximum Bags With Full Capacity of Rocks	Greedy + Sorting
2280	Minimum Lines to Represent a Line Chart	Sorting + Cross-Multiplication
2281	Sum of Total Strength of Wizards	Monotonic Stack + Prefix Sum of Prefix Sums
2282	Number of People That Can Be Seen in a Grid	Monotonic Stack (Row/Col wise)
2283	Check if Number Has Equal Digit Count and Digit Value	Frequency Array / Hash Map
2284	Sender With Largest Word Count	Hash Map + Word Split
2285	Maximum Total Importance of Roads	Greedy Degree-Based Assignment
2286	Booking Concert Tickets in Groups	Segment Tree (Range Sum & Max)
2287	Rearrange Characters to Make Target String	Frequency Map Ratio
2288	Apply Discount to Prices	Word-by-Word Parsing
2289	Steps to Make Array Non-decreasing	Monotonic Stack (Time-to-Die)
2290	Minimum Obstacle Removal to Reach Corner	0-1 BFS (Deque)
2291	Maximum Profit From Trading Stocks	1D Dynamic Programming (Knapsack)
2292	Products With Three or More Orders in Two Consecutive Years	CTE Aggregation + Window Function (LEAD)
2293	Min Max Game	In-Place Iterative Reduction
2294	Partition Array Such That Maximum Difference Is K	Sorting + Greedy Subsequences
2295	Replace Elements in an Array	Inverted Index (Hash Map)
2296	Design a Text Editor	Two Stacks / Gap Buffer
2297	Jump Game VIII	Monotonic Stack + 1D DP
2298	Tasks Count in the Weekend	Weekday Function (SQL CASE/WEEKDAY)
2299	Strong Password Checker II	Single-Pass Flagging
2300	Successful Pairs of Spells and Potions	Sorting + Binary Search
2301	Match Substring After Replacement	Sliding Window + Hash Map of Sets
2302	Count Subarrays With Score Less Than K	Sliding Window (Two Pointers)
2303	Calculate Amount Paid in Taxes	Linear Iteration (Math/Simulation)
2304	Minimum Path Cost in a Grid	Dynamic Programming (Bottom-Up)
2305	Fair Distribution of Cookies	Backtracking with Pruning
2306	Naming a Company	Group by First Letter + Venn Diagram Intersection

2307	Check for Contradictions in Equations	Union-Find with Weights / Directed Graph DFS
2308	Arrange Table by Gender	Multi-Queue Interleaving
2309	Greatest English Letter in Upper and Lower Case	Hash Set / Bitmasking
2310	Sum of Numbers With Units Digit K	Greedy Math / Modulo Arithmetic
2311	Longest Binary Subsequence Less Than or Equal to K	Greedy (Right-to-Left)
2312	Selling Pieces of Wood	2D Dynamic Programming
2313	Minimum Flips in Binary Tree to Get Result (Premium)	Tree DP (Bottom-Up State)
2314	The First Day of the Maximum Recorded Degree in Each City	Window Function (ROW_NUMBER)
2315	Count Asterisks	Single Pass State Toggle
2316	Count Unreachable Pairs of Nodes in an Undirected Graph	Connected Components (DFS/BFS) + Combinatorics
2317	Maximum XOR After Operations	Bitwise OR (Math / Bit Manipulation Insight)
2318	Number of Distinct Roll Sequences	3D Dynamic Programming (State Machine)
2319	Check if Matrix Is X-Matrix	Single Pass Matrix Traversal
2320	Count Number of Ways to Place Houses	1D DP (Fibonacci) + Math Independence
2321	Maximum Score Of Spliced Array	Kadane's Algorithm on Difference Array
2322	Minimum Score After Removals on a Tree	DFS Subtree Precomputation + Ancestor Check
2323	Find Minimum Time to Finish All Jobs II (Premium)	Greedy Sorting Alignment
2324	Product Sales Analysis IV (SQL Premium)	Window Functions (Dense_Rank / Max over Partition)
2325	Decode the Message	Hash Map / Array Substitution Cipher
2326	Spiral Matrix IV	Direction Array + Matrix Simulation
2327	Number of People Aware of a Secret	1D Dynamic Programming / Sliding Window
2328	Number of Increasing Paths in a Grid	DFS + Memoization (or Sorting + DP)
2329	Product Sales Analysis V (Premium SQL)	GROUP BY + Aggregation (SUM)
2330	Valid Palindrome IV (Premium)	Two Pointers (Inward Convergence)
2331	Evaluate Boolean Binary Tree	Post-Order Traversal (DFS)
2332	The Latest Time to Catch a Bus	Sorting + Two Pointers + Backtracking
2333	Minimum Sum of Squared Difference	Bucket Sort / Frequency Map + Greedy Flattening
2334	Subarray With Elements Greater Than Varying Threshold	Monotonic Stack (Next Smaller Element)
2335	Minimum Amount of Time to Fill Cups	Greedy Math / Max Priority Queue
2336	Smallest Number in Infinite Set	Min-Heap + Hash Set (or Min_Tracker)
2337	Move Pieces to Obtain a String	Two Pointers (Relative Order & Constraints)
2338	Count the Number of Ideal Arrays	Math (Combinatorics + Stars and Bars)
2339	All the Matches of the League	CROSS JOIN / SELF JOIN
2340	Minimum Adjacent Swaps to Make a Valid Array (Premium)	Greedy (Extreme Finding)
2341	Maximum Number of Pairs in Array	Frequency Map (Hash Table)
2342	Max Sum of a Pair With Equal Sum of Digits	Hash Map of Max Values
2343	Query Kth Smallest Trimmed Number	Radix Sort Logic / Precomputed Sorted States
2344	Minimum Deletions to Make Array Divisible	GCD + Sorting
2345	Finding the Number of Visible Mountains (Premium)	Interval Sorting + Monotonic Scan
2346	Compute the Rank as a Percentage (SQL Premium)	Window Functions (RANK() / PERCENT_RANK())
2347	Best Poker Hand	Frequency Counting (Hash Map/Array)
2348	Number of Zero-Filled Subarrays	Linear Strike Counting (Math)
2349	Design a Number Container System	Two Hash Maps + Min-Heap (Sorted Set)
2350	Shortest Impossible Sequence of Rolls	Greedy Set Partitioning
2351	First Letter to Appear Twice	Hash Set / Boolean Array (Bitmask)
2352	Equal Row and Column Pairs	Hash Map (Stringified Rows)
2353	Design a Food Rating System	Two Hash Maps + Priority Queues (Sorted Sets)
2354	Number of Excellent Pairs	Bit Count Frequency + Inclusion-Exclusion
2355	Maximum Number of Books You Can Take (Premium)	Monotonic Stack + Dynamic Programming
2356	Number of Unique Subjects Taught by Each Teacher (SQL Premium)	GROUP BY + COUNT(DISTINCT)
2357	Make Array Zero by Subtracting Equal Amounts	Unique Positive Integers (Hash Set)
2358	Maximum Number of Groups Entering a Competition	Greedy Math (Arithmetic Progression)
2359	Find Closest Node to Given Two Nodes	Two Independent BFS/DFS Passes
2360	Longest Cycle in a Graph	DFS with Timestamping (Kahn’s or Visited Array)
2361	Minimum Costs Using the Train Line (Premium)	1D Dynamic Programming (Two States)
2362	Generate the Invoice (SQL Premium)	CTE + Window Functions (RANK)
2363	Merge Similar Items	Treemap (Sorted Dictionary)
2364	Count Number of Bad Pairs	Complement Counting (Total - Good)
2365	Task Scheduler II	Greedy Time Calculation (Hash Map)
2366	Minimum Replacements to Sort the Array	Greedy (Right-to-Left) + Math Division
2367	Number of Arithmetic Triplets	Hash Set / Boolean Lookahead

2368	Reachable Nodes With Restrictions	Hash Set + Single BFS/DFS
2369	Check if There is a Valid Partition For The Array	1D Dynamic Programming (Boolean State)
2370	Longest Ideal Subsequence	1D DP (Character Bucket State)
2371	Minimize Maximum Value in a Grid (Premium)	Sorting + Greedy Row/Col Max Tracking
2372	Calculate the Influence Score (SQL Premium)	JOIN + SUM Aggregation
2373	Largest Local Values in a Matrix	Sliding Window Maximum
2374	Node With Highest Edge Score	In-Degree Accumulator (Array)
2375	Construct Smallest Number From DI String	Stack-Based Segment Reversal
2376	Count Special Integers	Digit Dynamic Programming (Bitmask)
2377	Sort the Olympic Table (SQL Premium)	ORDER BY Multiple Columns
2378	Choose Edges to Maximize Score in a Tree (Premium)	Tree Dynamic Programming (House Robber Style)
2379	Minimum Recolors to Get K Consecutive Black Blocks	Sliding Window (Constant Space)
2380	Time Needed to Rearrange a Binary String	Dynamic Programming / Mathematical Greedy
2381	Shifting Letters II	Difference Array (Prefix Sum of Shifts)
2382	Maximum Segment Sum After Removals	Reverse Processing + Union Find (DSU)
2383	Minimum Hours of Training to Win a Competition	Single Pass Greedy Training
2384	Largest Palindromic Number	Frequency Map + Greedy Construction
2385	Amount of Time for Binary Tree to Be Infected	Adjacency List + BFS (Graph Conversion)
2386	Find the K-Sum of an Array	Max Heap + Absolute Value Sorting
2387	Median of a Row-Wise Sorted Matrix (Premium)	Binary Search on Range + Upper Bound
2388	Change Null Values in a Table to the Previous Value (SQL Premium)	Window Functions with IGNORE NULLS
2389	Longest Subsequence With Limited Sum	Sorting + Prefix Sum + Binary Search
2390	Removing Stars From a String	Stack (LIFO) / Two Pointers
2391	Minimum Amount of Time to Collect Garbage	Prefix Sums + Last Occurrence Tracking
2392	Build a Matrix With Conditions	Topological Sort (Kahn's Algorithm)
2393	Count Strictly Increasing Subarrays (Premium)	Dynamic Programming / Linear Counting
2394	Employees With Deductions (SQL Premium)	CEIL + SUM + LEFT JOIN
2395	Find Subarrays With Equal Sum	Hash Set (Single Pass)
2396	Strictly Palindromic Number	Mathematical Insight (Constant Return)
2397	Maximum Rows Covered by Columns	Bitmasking + Gosper's Hack
2398	Maximum Number of Robots Within Budget	Sliding Window + Monotonic Deque
2399	Check Distances Between Same Letters	Single Pass First-Occurrence Map
2400	Number of Ways to Reach a Position After Exactly k Steps	Combinatorics (Pascal's Triangle / Math)
2401	Longest Nice Subarray	Sliding Window / Two Pointers with Bitmask Running OR
2402	Meeting Rooms III	Two Min-Heaps (Priority Queues): One for unused rooms, one for engaged rooms (endTime, roomIdx).
2403	Minimum Time to Kill All Monsters	Bitmask DP
2404	Most Frequent Even Element	Hash Map / Dictionary Counting
2405	Optimal Partition of String	Greedy + Bitmask (or HashSet)
2406	Divide Intervals Into Minimum Number of Groups	Line Sweep Algorithm (Prefix Sum over time) OR Min-Heap
2407	Longest Increasing Subsequence II	Segment Tree / Fenwick Tree (Range Maximum Query)
2408	Design SQL	Hash Map of Lists (Indexed Access)
2409	Count Days Spent Together	Math / Line Intersection (Convert MMDD to Day of Year Integer)
2410	Maximum Matching of Players With Trainers	Sorting + Two Pointers (Greedy)
2411	Smallest Subarrays With Maximum Bitwise OR	Reverse Traversal + 32-Bit Position Tracking
2412	Minimum Money Required Before Transactions	Greedy / Math Accumulator
2413	Smallest Even Multiple	Math (Modulo Arithmetic / Bitwise)
2414	Length of the Longest Alphabetical Continuous Substring	Single Pass / State Tracking (Current Streak)
2415	Reverse Odd Levels of Binary Tree	DFS/BFS with Symmetrical Dual-Pointers
2416	Sum of Prefix Scores of Strings	Trie (Prefix Tree) with Frequency Counters
2417	Closest Fair Integer	Digit DP / Constructive Greedy
2418	Sort the People	Hash Map Binding / Array of Objects + Sort
2419	Longest Subarray With Maximum Bitwise AND	Linear Scan (Max Property Observation)
2420	Find All Good Indices	Prefix/Suffix Dynamic Programming (Precomputation)
2421	Number of Good Paths	Disjoint Set Union (DSU) + Sorting by Value
2422	Merge Operations to Make Array Palindrome	Two Pointers (Greedy)
2423	Remove Letter To Equalize Frequency	Frequency of Frequencies (Count Map)
2424	Longest Uploaded Prefix	Persistent Pointer (Amortized Analysis)
2425	Bitwise XOR of All Pairings	XOR Parity Property (Even/Odd Counts)
2426	Number of Pairs Satisfying Inequality	Merge Sort (Divide & Conquer) or Fenwick Tree after transforming to (nums1[i]-nums2[i])
2427	Number of Common Factors	GCD + Factor Counting

2428	Maximum Sum of an Hourglass	Sliding Window (Optimised Summing)
2429	Minimize XOR	Greedy Bit Construction
2430	Maximum Deletions on a String	DP + Longest Common Prefix (LCP) with Rolling Hash or Suffix Array
2431	Maximize Total Tastiness of Purchased Fruits	3D Dynamic Programming (Knapsack Variant)
2432	The Employee That Worked on the Longest Task	Single Pass Tracking (Max/Min Greedy)
2433	Find The Original Array of Prefix XOR	XOR Inverse Property
2434	Using a Robot to Print the Lexicographically Smallest String	Greedy + Stack + Suffix Minimums
2435	Paths in Matrix Whose Sum Is Divisible by K	3D Dynamic Programming (Grid + Modulo State)
2436	Minimum Split Into Subarrays With GCD Greater Than One	Greedy / Running GCD
2437	Number of Valid Clock Times	Rule-Based Combinatorics (Independent Slots)
2438	Range Product Queries of Powers	Precomputation (Prefix Products)
2439	Minimize Maximum Value in Array	Prefix Sum Average (Greedy Ceiling)
2440	Create Components With Same Value	Divisor Search + Post-order DFS
2441	Largest Positive Integer That Exists With Its Negative	Hash Set / Two Pointers (after Sort)
2442	Count Number of Distinct Integers After Reverse Operations	Digit Manipulation + Hash Set
2443	Sum of Number and Its Reverse	Math (Direct Search is already efficient given N <= 10^5)
2444	Count Subarrays With Fixed Bounds	Three-Pointer Sliding Window (Tracking Indices)
2445	Number of Nodes With Value One	Prefix Sum on Trees (Parity)
2446	Determine if Two Events Have Conflict	String Comparison / Line Overlap
2447	Number of Subarrays With GCD Equal to K	Optimal Brute (Observation: GCD only decreases)
2448	Minimum Cost to Make Array Equal	Weighted Median / Binary Search on Answer
2449	Minimum Number of Operations to Make Arrays Similar	Sorting + Parity Segregation
2450	Number of Distinct Binary Strings After Operations	Mathematical Observation (Powers of 2)
2451	Odd String Difference	Hash Map Counting (Frequency of Arrays)
2452	Words Within Two Edits of Dictionary	Optimized Linear Scan (Early Exit)
2453	Destroy Sequential Targets	Hash Map (Remainder Frequency)
2454	Next Greater Element IV	Two Monotonic Stacks + Priority Queue
2455	Average Value of Even Numbers Divisible by Three	Single Pass (Running Sum and Count)
2456	Most Popular Video Creator	Hash Map (Aggregate & Track Max)
2457	Beautiful Number / Minimum Addition to Make Integer Beautiful	Greedy Zero-ing (Place Value Increment)
2458	Height of Binary Tree After Subtree Removal Queries	DFS + Precomputation (Level-wise Max Depths)
2459	Sort Array by Moving Items to Empty Space	Graph Cycle Decomposition
2460	Apply Operations to an Array	In-Place Two-Pointer Swap
2461	Maximum Sum of Distinct Subarrays With Length K	Sliding Window + Frequency Map (or Set)
2462	Total Cost to Hire K Workers	Two Min-Heaps (Priority Queues)
2463	Minimum Total Distance Traveled	DP (Sorted Robots & Factories)
2464	Minimum Subarrays in a Valid Split	1D Dynamic Programming + GCD Check
2465	Number of Distinct Averages	Sorting + Two Pointers
2466	Count Ways To Build Good Strings	1D Dynamic Programming (Climbing Stairs Variant)
2467	Most Profitable Path in a Tree	Two-Pass DFS (Time Map + Profit DFS)
2468	Split Message Based on Limit	Math (Suffix Length Precalculation)
2469	Convert the Temperature	Arithmetic Computation
2470	Number of Subarrays With LCM Equal to K	Optimized Brute Force with LCM Early Exit
2471	Minimum Number of Operations to Sort a Binary Tree by Level	BFS + Cycle Decomposition (Minimum Swaps)
2472	Maximum Number of Non-overlapping Palindrome Substrings	DP / Center Expansion + Greedy
2473	Minimum Cost to Buy Apples	Multi-Source Dijkstra / Graph Reversal
2474	Customers With Strictly Increasing Purchases (SQL)	Window Functions (LAG / Difference)
2475	Number of Unequal Triplets in Array	Frequency Counting + Combinatorics
2476	Closest Nodes Queries in a Binary Search Tree	In-order Traversal + Binary Search (Sorted Array)
2477	Minimum Fuel Cost to Report to the Capital	DFS + Subtree Size Math
2478	Number of Beautiful Partitions	2D Dynamic Programming with Prefix Sum Optimization
2479	Maximum XOR of Two Non-Overlapping Subtrees	DFS + Trie (Bitwise)
2480	Form a Chemical Bond (SQL)	CROSS JOIN
2481	Minimum Cuts to Divide a Circle	Parity-Based Math (Even/Odd Geometry)
2482	Difference Between Ones and Zeros in Row and Column	Precomputation (Row/Col Sum Arrays)
2483	Minimum Penalty for a Shop	Prefix Sums / Single Pass Logic
2484	Count Palindromic Subsequences	Prefix/Suffix DP (Length 5)
2485	Find the Pivot Integer	Math (Perfect Square Property)
2486	Append Characters to String to Make Subsequence	Two-Pointer Greedy Matching
2487	Remove Nodes From Linked List	Monotonic Stack or Reverse Recursion

2488	Count Subarrays With Median K	Prefix Sums + Frequency Map (Balance Tracking)
2489	Number of Substrings With Fixed Ratio	Prefix Sum + Hash Map (Math Restructuring)
2490	Circular Sentence	Single Pass (Space Detection)
2491	Divide Players Into Teams of Equal Skill	Sorting + Two Pointers
2492	Minimum Score of a Path Between Two Cities	Graph Connectivity (BFS/DFS or Union-Find)
2493	Divide Nodes Into the Maximum Number of Groups	Bipartite Check + Component BFS Max Depth
2494	Merge Overlapping Events in the Same Hall	Sorting + Linear Interval Merge
2495	Number of Subarrays Having Even Product	Complementary Counting (Total - Odd)
2496	Maximum Value of a String in an Array	Linear Scan (Manual Digit Check)
2497	Maximum Star Sum of a Graph	Greedy (Sort Neighbors per Node)
2498	Frog Jump II	Greedy (Skipping Alternate Stones)
2499	Minimum Total Cost to Make Arrays Unequal	Greedy / Majority Element
2500	Delete Greatest Value in Each Row	Sorting Rows
2501	Longest Square Streak in an Array	Hash Set + Greedy Growth
2502	Design Memory Allocator	1D Array tracking with consecutive free-block counting and slice assignment.
2503	Maximum Number of Points From Grid Queries	Offline Queries (Sort Queries) + Min-Heap (Priority Queue) + BFS to prevent re-processing nodes.
2504	Concatenate the Name and the Profession (SQL)	SQL String Functions (CONCAT , LEFT) running directly on the database engine.
2505	Bitwise OR of All Subsequence Sums	Bit contribution counting array tracking 1s, coupled with right-shift carry propagation logic.
2506	Count Pairs Of Similar Strings	Bitmasking for character presence + Hash Map frequency counting.
2507	Smallest Value After Replacing With Sum of Prime Factors	Simulation + Prime Factorization
2508	Add Edges to Make Degrees of All Nodes Even	Graph Theory (Degree counting). Odd-degree nodes must total 0, 2, or 4. Check specific adjacency rules.
2509	Cycle Length Queries in a Tree	Lowest Common Ancestor (LCA) via integer division (node / 2) in a Complete Binary Tree.
2510	Check if There is a Path With Equal Number of 0's And 1's	Dynamic Programming (DP) tracking the minimum and maximum possible sum of `1s - 0s` at each cell.
2511	Maximum Enemy Forts That Can Be Captured	Two Pointers / Single-Pass Array Traversal checking alternating non-zero boundary states.
2512	Reward Top K Students	Hash Set for O(1) word lookup + Custom Comparator Sorting (or Min-Heap for top K selection).
2513	Minimize the Maximum of Two Arrays	Binary Search on Answer Space + Number Theory (Inclusion-Exclusion Principle using LCM).
2514	Count Anagrams	Combinatorics (Permutations with Repetition) + Modular Multiplicative Inverse.
2515	Shortest Distance to Target String in a Circular Array	Linear Array Scan with Modulo Arithmetic (comparing absolute distance vs circular wrap-around distance).
2516	Take K of Each Character From Left and Right	Inverse Sliding Window (Two Pointers) seeking the maximum middle subarray to *leave behind*.
2517	Maximum Tastiness of Candy Basket	Sort the array + Binary Search on the Answer Space + Greedy Checking.
2518	Number of Great Partitions	0/1 Knapsack Dynamic Programming (Inverse Problem). Find subsets summing to strictly less than K, and subtract from total sub
2519	Count the Number of K-Big Indices	Two Passes (Prefix and Suffix) using Max-Heaps of size K.
2520	Count the Digits That Divide a Number	Mathematical Modulo-10 Digit Extraction.
2521	Distinct Prime Factors of Product of Array	On-the-fly Trial Division + Hash Set deduplication.
2522	Partition String Into Substrings With Values at Most K	Greedy Algorithm / String Parsing with continuous accumulation.
2523	Closest Prime Numbers in Range	Sieve of Eratosthenes (Precomputation) + Linear Array Scan.
2524	Maximum Frequency Score of a Subarray	Fixed Sliding Window + Persistent Hash Map (or Array Cache) for O(1) incremental score updates.
2525	Categorize Box According to Criteria	Pre-evaluated Boolean Variables mapping to strict logic combinations.
2526	Find Consecutive Integers from a Data Stream	State Machine / Single Counter tracking contiguous valid states.
2527	Find Xor-Beauty of Array	Mathematical Bitwise Cancellation (Symmetry). The expression simplifies to just XORing all elements in the array.
2528	Maximize the Minimum Powered City	Binary Search on the Answer Space + Greedy Sliding Window.
2529	Maximum Count of Positive Integer and Negative Integer	Two separate Binary Searches (Upper Bound for negatives, Lower Bound for positives) leveraging the sorted nature of the array.
2530	Maximal Score After Applying K Operations	Max-Heap (Priority Queue) to maintain the largest element dynamically.
2531	Make Number of Distinct Characters Equal	26x26 Frequency Map Simulation.
2532	Time to Cross a Bridge	Event-Driven Simulation using 4 Priority Queues (Heaps).
2533	Number of Good Binary Strings	Linear Dynamic Programming (DP) - Staircase variation.
2534	Time Taken to Cross the Door	Two Queues (Enter and Exit) with State Tracking.
2535	Difference Between Element Sum and Digit Sum of an Array	Single-Pass Accumulation with Modulo Digit Extraction.
2536	Increment Submatrices by One	2D Difference Array (Prefix Sums).
2537	Count the Number of Good Subarrays	Sliding Window with Frequency Map.
2538	Difference Between Maximum and Minimum Price Sum	Tree Dynamic Programming (Rerooting or Diameters).
2539	Count the Number of Good Subsequences	Combinatorics + Frequency Counting.
2540	Minimum Common Value	Two Pointers (Sorted Array Property).
2541	Minimum Operations to Make Array Equal II	Greedy Difference Tracking + Parity Check.
2542	Maximum Subsequence Score	Sorting + Min-Heap (Sliding Window of Top K).
2543	Check if Point Is Reachable	Number Theory (GCD + Power of Two).
2544	Alternating Digit Sum	Math-based digit extraction (Modulo/Division).
2545	Sort the Students by Their Kth Score	Custom Comparator / Lambda Sorting.
2546	Apply Bitwise Operations to Make Strings Equal	Logical Invariance (Existence of '1').
2547	Minimum Cost to Split an Array	1D Dynamic Programming with Frequency Map optimization.
2548	Maximum Price to Fill a Bag (Premium)	Greedy Algorithm (Price-per-Weight Ratio).

2549	Count Distinct Numbers on Board	Mathematical Observation (Cascade Effect).
2550	Count Collisions of Monkeys on a Polygon	Complementary Counting + Binary Exponentiation.
2551	Put Marbles in Bags	Sorting Pair Sums (Boundary Logic).
2552	Count Increasing Quadruplets	2D Prefix Sums / Precomputation of "Smaller" and "Greater" counts.
2553	Separate the Digits in an Array	Iterative Digit Extraction (Modulo & Stack).
2554	Maximum Number of Integers to Choose From a Range I	Greedy Selection + Hash Set Lookup.
2555	Maximize Win From Two Segments	Sliding Window + Dynamic Programming (Prefix Max).
2556	Disconnect Path in a Binary Matrix by at Most One Flip	Double DFS / Max Flow Min Cut (Path Neutralization).
2557	Maximum Number of Integers to Choose From a Range II (Premium)	Sorting + Binary Search on Answer Space.
2558	Take Gifts From the Richest Pile	Max-Heap (Priority Queue).
2559	Count Vowel Strings in Ranges	Prefix Sum Array.
2560	House Robber IV	Binary Search on Answer Space + Greedy Counting.
2561	Rearranging Fruits	Frequency Mapping + Greedy Pairing + Min-Cost Observation.
2562	Find the Array Concatenation Value	Two-Pointer Iteration + Mathematical Concatenation.
2563	Count the Number of Fair Pairs	Sorting + Two Pointers (or Binary Search).
2564	Substring XOR Queries	Precomputation of All Substrings (up to 30 bits) + Hash Map.
2565	Subsequence With the Minimum Score	Two-Pass Prefix/Suffix Matching + Sliding Window.
2566	Maximum Difference by Remapping a Digit	Greedy First-Digit Replacement.
2567	Minimum Score by Changing Two Elements	Boundary Analysis (Sorting).
2568	Minimum Impossible OR	Power of Two Check (Bit Manipulation).
2569	Handling Sum Queries After Update	Segment Tree with Lazy Propagation.
2570	Merge Two 2D Arrays by Summing Values	Two-Pointer Merging (Sorted Property).
2571	Minimum Operations to Reduce an Integer to 0	Greedy Bit Manipulation (Low-bit Analysis).
2572	Count the Number of Square-Free Subsets	Bitmask Dynamic Programming (Primes as Bits).
2573	Find the String with LCP	Union-Find / Greedy Coloring + LCP Verification.
2574	Left and Right Sum Differences	Prefix and Suffix Running Totals.
2575	Find the Divisibility Array of a String	Modular Arithmetic (Iterative Remainder).
2576	Find the Maximum Number of Marked Indices	Sorting + Greedy Two-Pointer (Split Search).
2577	Minimum Time to Visit a Cell In a Grid	Dijkstra's Algorithm + Parity Waiting Logic.
2578	Split With Minimum Sum	Sorting + Greedy Alternating Distribution.
2579	Count Total Number of Colored Cells	Mathematical Progression (Sum of Arithmetic Series).
2580	Count Ways to Group Overlapping Ranges	Sorting + Merge Intervals + Modular Exponentiation.
2581	Count Number of Possible Roots	Tree DP with Rerooting (Change of Root).
2582	Pass the Pillow	Modular Arithmetic (Cycles).
2583	Kth Largest Sum in a Binary Tree	Level Order Traversal (BFS) + Sorting/Min-Heap.
2584	Split the Array to Make Coprime Products	Prime Factorization + Range Merging / Interval Coverage.
2585	Number of Ways to Earn Points	Bounded Knapsack Dynamic Programming.
2586	Count the Number of Vowel Strings in Range	Direct Linear Scan with Set Lookup.
2587	Rearrange Array to Maximize Prefix Score	Greedy Sorting (Descending Order).
2588	Count the Number of Beautiful Subarrays	Prefix XOR + Hash Map (Frequency Counting).
2589	Minimum Time to Complete All Tasks	Greedy Interval Scheduling (End-Point Prioritization).
2590	Design a Todo List (Premium)	Hash Map (User to Task-List) + Sorting/Filtering.
2591	Distribute Money to Maximum Children	Greedy Case-Based Logic.
2592	Maximize Greatness of an Array	Sorting + Greedy Two-Pointer.
2593	Find Score of an Array After Marking All Elements	Sorting with Index Tracking + Boolean Marking.
2594	Minimum Time to Repair Cars	Binary Search on Answer Space (Time).
2595	Number of Even and Odd Bits	Bitwise Shift and Masking.
2596	Check Knight Tour Configuration	Coordinate Mapping (Direct Indexing).
2597	The Number of Beautiful Subsets	Recursive Backtracking with Frequency Map (or House Robber DP).
2598	Smallest Missing Non-negative Integer After Operations	Modulo Frequency Mapping (Remainders).
2599	Make the Prefix Sum Non-negative (Premium)	Greedy with Min-Heap (Priority Queue).
2600	K Items With the Maximum Sum	Greedy Case-Based Logic.
2601	Prime Subtraction Operation	Sieve of Eratosthenes + Binary Search (Greedy)
2602	Minimum Operations to Make All Array Elements Equal	Sorting + Prefix Sums + Binary Search
2603	Collect Coins in a Tree	Topological Sorting (Peeling leaves like an onion)
2604	Minimum Time to Eat All Grains	Binary Search on Answer + Greedy
2605	Form Smallest Number From Two Digit Arrays	Hash Set / Bitmask + Math.min
2606	Find the Substring With Maximum Cost	Kadane's Algorithm
2607	Make K-Subarray Sums Equal	Graph Cycles + Median Math
2608	Shortest Cycle in a Graph	BFS from every node (or Edge Removal)
2609	Find the Longest Balanced Substring of a Binary String	Two Variables / Counting Consecutive Blocks

2610	Convert an Array Into a 2D Array With Conditions	Frequency Map / Hash Table
2611	Mice and Cheese	Greedy Algorithm + Sorting
2612	Minimum Reverse Operations	BFS + Ordered Sets (Parity sets)
2613	Beautiful Pairs	Divide and Conquer / Sweep Line
2614	Prime In Diagonal	Matrix Traversal + O(√N) Prime Check
2615	Sum of Distances	Hash Map + Prefix Sum
2616	Minimize the Maximum Difference of Pairs	Sort + Binary Search on Answer + Greedy
2617	Minimum Number of Visited Cells in a Grid	BFS + Priority Queue / Monotonic Stack
2618	Check if Object Instance of Class (JS)	JavaScript Prototype Chain Traversal
2619	Array Prototype Last (JS)	Array Length Property (`this.length`)
2620	Counter (JS)	Lexical Scope / Closures
2621	Sleep (JS)	Promises + setTimeout
2622	Cache With Time Limit (JS)	Hash Map + clearTimeout
2623	Memoize (JS)	Closure + Hash Map (Caching)
2624	Snail Traversal (JS)	Math-based column parity (Odd/Even columns)
2625	Flatten Deeply Nested Array (JS)	Recursive DFS with Depth Tracking
2626	Array Reduce Transformation (JS)	Iterative Accumulator Update
2627	Debounce (JS)	Closure + clearTimeout / setTimeout
2628	JSON Deep Equal (JS)	Recursive DFS Type Checking
2629	Function Composition (JS)	Right-to-Left Array Reduce
2630	Memoize II (JS)	Nested Maps / Prefix Tree (Trie)
2631	Group By (JS)	Hash Map (Object Literal) Accumulator
2632	Curry (JS)	Recursive Closure / Function Arity
2633	Convert Object to JSON String (JS)	Recursive DFS Serialization
2634	Filter Elements from Array (JS)	Iterative New Array Push
2635	Apply Transform Over Each Element (JS)	Iterative Mapping (New Array)
2636	Promise Pool	Recursive Chain / Worker Pool
2637	Promise Time Limit	Promise.race() with Timer
2638	Count the Number of K-Free Subsets	Sorting + DP on Chains
2639	Find the Width of Columns of a Grid	Column-Wise Linear Scan
2640	Find the Score of All Prefixes of an Array	Running Max + Prefix Sum (Single Pass)
2641	Cousins in Binary Tree II	Two-Pass BFS (Level Sums)
2642	Design Graph Shortest Path Calculator	Dijkstra's Algorithm (with Priority Queue)
2643	Row With Maximum Ones	Linear Row Scan + Tracker
2644	Find the Maximum Divisibility Score	Frequency Map + Divisibility Check
2645	Minimum Additions to Make Valid String	Greedy / Math-based Cycle Counting
2646	Minimize the Total Price of the Trips	Tree DP + Frequency Map (BFS/DFS)
2647	Color the Triangle Red	Constructive Pattern / Mathematical Observation
2648	Generate Fibonacci Sequence (JS)	Generators (`yield`)
2649	Nested Array Generator (JS)	Recursive Yield* (Delegation)
2650	Design Cancellable Function	JavaScript Generators & Promises
2651	Calculate Delayed Arrival Time	Modulo Arithmetic (%)
2652	Sum Multiples	Inclusion-Exclusion Principle (Math)
2653	Sliding Subarray Beauty	Sliding Window + Frequency Array (Counting Sort)
2654	Minimum Number of Operations to Make All Array Elements Equal to 1	Iterative GCD + Greedy Expansion
2655	Find Maximal Uncovered Ranges	Sorting + Merging Intervals
2656	Maximum Sum With Exactly K Elements	Greedy + Arithmetic Progression
2657	Find the Prefix Common Array of Two Arrays	Frequency Map / Bitset (Single Pass)
2658	Maximum Number of Fish in a Grid	DFS / BFS + Grid Traversal (Visited tracking)
2659	Make Array Empty	Sorting + Fenwick Tree (Binary Indexed Tree)
2660	Determine the Winner of a Bowling Game	Linear Scan with State Tracker
2661	First Completely Painted Row or Column	Hash Map + Frequency Counters
2662	Minimum Cost of a Path With Special Roads	Dijkstra's on a Coordinate Graph
2663	Lexicographically Smallest Beautiful String	Greedy String Mutation (Right-to-Left)
2664	The Knight's Tour	Backtracking (Warnsdorff’s Rule optional)
2665	Counter II (JS)	Closures / Factory Pattern
2666	Allow One Function Call	Closure / Higher-Order Function
2667	Create Hello World Function	Higher-Order Function (Factory)
2668	Find Latest Salaries (SQL)	Window Functions (`RANK()` or `MAX() OVER`)
2669	Count Artist Occurrences On Spotify (SQL)	GROUP BY + COUNT

2670	Find the Distinct Difference Array	Two-Pass Hash Set (Prefix/Suffix)
2671	Frequency Tracker	Dual Hash Maps (Count of Counts)
2672	Number of Adjacent Elements With the Same Color	Local Incremental Update
2673	Make Costs of Paths Equal in a Binary Tree	Bottom-Up Greedy (Post-order Traversal)
2674	Split a Circular Linked List	Fast and Slow Pointers (Tortoise and Hare)
2675	Array of Objects to Matrix	Recursive DFS Flattening + Sorted Key Set
2676	Throttle	Closure + setTimeout + Stored Arguments
2677	Chunk Array	Iterative Slicing (Step = Size)
2678	Number of Senior Citizens	Substring Slicing + Direct Integer Comparison
2679	Sum in a Matrix	Row Sorting + Column Maximums
2680	Maximum OR After Operations	Prefix/Suffix OR + Greedy Power of 2
2681	Power of Heroes	Sorting + Dynamic Programming (Prefix Sum)
2682	Find the Losers of the Circular Game	Boolean Array + Step Increment
2683	Neighboring Bitwise XOR	XOR Sum Property
2684	Maximum Number of Moves in a Grid	DP / BFS (Visited Grid)
2685	Count the Number of Complete Components	DFS + Degree Counting
2686	Immediate Food Delivery III (SQL)	Group By + CASE WHEN + ROUND
2687	Bikes Last Time Used	Window Function (MAX) / GROUP BY
2688	Find Active Users	GROUP BY + COUNT(DISTINCT)
2689	Extract Kth Character From The Rope Tree	Recursive Length-Based Search
2690	Infinite Method Objects (JS)	JavaScript Proxy (`get` trap)
2691	Immutability Helper (JS)	Proxy + Structural Sharing
2692	Make Object Immutable	Proxy Object / Recursive Object.freeze
2693	Call Function with Custom Context (JS)	Function.prototype.apply / Symbol Key
2694	Event Emitter (JS)	Hash Map of Sets (Callback Registry)
2695	Array Wrapper (JS)	valueOf() and toString() Overriding
2696	Minimum String Length After Removing Substrings	Stack-Based Character Matching
2697	Lexicographically Smallest Palindrome	Two-Pointer Greedy Replacement
2698	Find the Punishment Number of an Integer	Recursion with Backtracking
2699	Modify Graph Edge Weights	Dual-Pass Dijkstra (Target Distance Matching)
2700	Differences Between Two Objects	Recursive DFS (Object Traversal)
2701	Consecutive Transactions with Increasing Amounts (SQL)	Row-by-Row Memory Copy (Intermediate tables filling up Cartesian space).
2702	Minimum Operations to Make Numbers Non-positive	Array modification tracking (snapshotting the entire array state at each simulated step).
2703	Return Length of Arguments Passed	Call Stack & Arguments Object frame allocation.
2704	To Be Or Not To Be (JS)	Heap memory allocation map (Drawing separate object allocations in memory for every closure context created).
2705	Compact Object	Call Stack & Redundant Object Allocation (Draw stacked boxes for deep recursion + identical clone structures in heap).
2706	Buy Two Chocolates	Loop pointers footprint (Draw i and j pointers constantly jumping back and forth across an array).
2707	Extra Characters in a String	Deep Call Stack Grid (Showing massive stack frames for O(N) depth and O(2^N) repeated states).
2708	Maximum Strength of a Group	Exponential State Tracking (Visualizing memory holding all subsets simultaneously or deep recursion tree).
2709	Greatest Common Divisor Traversal	Massive NxN Grid Matrix (Highlighting the memory waste of storing every direct edge).
2710	Remove Trailing Zeros From a String	Heap Allocation Overload (Visualize multiple intermediate string objects piling up in memory due to immutable string slicing).
2711	Difference of Number of Distinct Values on Diagonals	Disposable HashSets (Draw hundreds of temporary Sets being created and garbage-collected for every individual cell).
2712	Minimum Cost to Make All Characters Equal	Deep Recursion Stack (Visualizing maximum stack depth O(N) but an exploding number of string copies across states).
2713	Maximum Strictly Increasing Cells in a Matrix	Unbounded Call Stack (Deep recursion for every valid path permutation).
2714	Find Shortest Path with K Hops	Brute Force Path Memory (Storing every conceivable route configuration simultaneously).
2715	Timeout Cancellation (JS)	Floating Closure Memory (The callback and its lexical environment stay in memory until the timer naturally finishes).
2716	Minimize String Length	Massive String Reallocations (Immutable strings mean creating a brand new string in memory for every deletion).
2717	Semi-Ordered Permutation	In-place array mutations (Unnecessary write operations to memory).
2718	Sum of Matrix After Queries	O(N^2) 2D Array Allocation (Massive memory waste for large N).
2719	Count of Integers	Trillions of temporary string allocations for digit extraction.
2720	Popularity Percentage	UNION CTE + Window/Aggregate
2721	Execute Asynchronous Functions in Parallel (JS)	Blocking Execution Context (One task occupies the stack while others wait in the queue).
2722	Join Two Arrays by ID	Temporary Object Accumulation (Creating massive amounts of intermediate merged objects).
2723	Add Two Promises	Serialized Memory Wait (The closure for the addition waits twice).
2724	Sort By	Repeated Function Execution Stack (Multiple redundant calls to the selector function).
2725	Interval Cancellation (JS)	Accumulating Closure Scopes (Each timeout creates a new closure, potentially keeping old variables in memory longer).
2726	Calculator with Method Chaining	Multiple Statement Contexts (Each call is a separate statement in the execution stack).
2727	Is Object Empty	Intermediate Array Allocation (Visualize a potentially massive array of strings being created in memory just for a boolean check).
2728	Count Houses in a Circular Street	Unbounded Integer Accumulator (Memory usage of the counter variable growing as it fails to terminate).
2729	Check if The Number is Fascinating	Temporary String Buffers (Visualizing multiple string concatenations and copies).

2730	Find the Longest Semi-Repetitive Substring	String Slice Memory (Draw many temporary substring copies being created in the heap).
2731	Movement of Robots	Temporal State Snapshots (Visualizing the position of every robot at every tick of time).
2732	Find a Good Subset of Matrix	Combinatorial Row Storage (Visualizing all row combinations in memory).
2733	Neither Minimum nor Maximum	Sort Buffer (Visualize the O(N) or O(log N) extra space used by sorting).
2734	Lexicographically Smallest String After Substring Operation	Exponential String Allocation (Visualize the heap filling with modified versions of the string).
2735	Collecting Chocolates	Redundant Array Rotation Buffers (Storing the state of the array for every single shift).
2736	Maximum Sum Queries	Query/Pair Stack (Drawing the search range for every individual query).
2737	Find the Closest Marked Node	Distance Array + Priority Queue (Drawing the standard Dijkstra memory footprint).
2738	Count Occurrences in Text (SQL)	Buffer Scan (Visualizing the database engine reading the text column repeatedly for each word).
2739	Total Distance Traveled	Integer State Mutation (Visualizing the variables main and additional changing over time).
2740	Find the Value of the Partition	Partition Memory Sets (Visualizing two massive separate heaps for every single partition attempt).
2741	Special Permutations	Stack Frame Explosion (Visualizing a call stack N-levels deep repeated billions of times).
2742	Painting the Walls	Redundant Decision Paths (Visualizing same wall counts being re-calculated in different branches).
2743	Count Substrings Without Repeating Character	Heap Fragment Memory (Visualizing thousands of temporary string objects created during slicing).
2744	Find Maximum Number of String Pairs	Temporary Reversed String Buffers (Creating a reversed copy for every comparison).
2745	Construct the Longest New String	Recursive Call Stack (Visualizing depth-first search for the longest possible valid string path).
2746	Decremental String Concatenation	Massive String Heap (Visualizing huge strings being copied and stored at every level of recursion).
2747	Count Zero Request Servers	Linear Search Pointers (Drawing the search range for every individual query).
2748	Number of Beautiful Pairs	Repeated Digit Extraction (Drawing the logic to pull digits from numbers inside the nested loops).
2749	Minimum Operations to Make the Integer Zero	Queue Memory (Visualizing the BFS queue filling with millions of possible intermediate numbers).
2750	Ways to Split Array Into Good Subarrays	Recursive Call Stack (Visualizing the depth-first search through all valid and invalid partition combinations).
2751	Robot Collisions	State Array Snapshots (Drawing the entire array state for every microscopic time increment).
2752	Customers with Maximum Number of Transactions on Consecutive Days (SQL)	Temporary Join Tables (Visualizing the SQL engine creating massive intermediate tables for each join).
2753	Count Houses in a Circular Street II	Unbounded Step Counter.
2754	Bind Function to Context (JS)	Temporary Property Allocation (Mutation of the target object).
2755	Deep Merge of Two Objects (JS)	Deep Call Stack (Visualize memory frames for every nested level of the JSON structure).
2756	Query Batching (JS)	Network Socket Overload (Visualizing multiple open connections consuming system file descriptors).
2757	Generate Circular Array Values	Iterative Pointer Mutation (Visualizing the CPU cycle overhead for millions of small increments).
2758	Next Day (JS)	Multiple Boolean Flags (Visualizing all the manual state variables needed for date math).
2759	Convert JSON String to Object	Recursive Descent Parser / Stack
2760	Longest Even Odd Subarray With Threshold	Intermediate Subarray Slices (Visualizing the heap filling with transient array segments).
2761	Prime Pairs With Target Sum	Sieve of Eratosthenes + Two Pointers
2762	Continuous Subarrays	Redundant Min/Max Scans (Drawing a search arrow within every window).
2763	Sum of Imbalance Numbers of All Subarrays	Massive Sort Buffer Allocation (Drawing the memory used to sort N^2 subarrays).
2764	Is Array a Preorder of Some Binary Tree	Monotonic Stack (Parent Tracking)
2765	Longest Alternating Subarray	Subarray Buffer (Visualizing the heap filling with transient slices of the input array).
2766	Relocate Marbles	Linear Memory Scan (Visualizing the CPU reading the entire marbles array for every single instruction).
2767	Partition String Into Minimum Beautiful Substrings	Dynamic Programming (1D)
2768	Number of Black Blocks	O(M· N) Matrix Allocation (Massive memory waste for grids like 10^5 x 10^5).
2769	Find the Maximum Achievable Number	BFS Queue (Visualizing the search space widening with every time step).
2770	Maximum Number of Jumps to Reach the Last Index	Recursive Call Stack (Visualizing deep paths and redundant calculations of the same sub-problems).
2771	Longest Non-decreasing Subarray From Two Arrays	Exponential State Memory (Visualizing the storage of all valid subarray combinations).
2772	Apply Operations to Make All Array Elements Zero	In-place Array Mutations (Visualizing the CPU repeatedly writing to memory for every range update).
2773	Height of Special Binary Tree	Visited Node HashSet (Visualizing extra memory to store every node ID to prevent infinite loops).
2774	Array Upper Bound (JS)	Minimal Variable State (Only index i).
2775	Undefined to Null (JS)	Recursive Call Stack (Visualize memory frames for every nested level of the object).
2776	Convert Callback to Promise (JS)	Blocking Main Thread (Visualizing the CPU stuck in a loop waiting for the async task).
2777	Date Range Generator (JS)	Heap Array Allocation (Visualizing thousands of string objects being stored simultaneously).
2778	Sum of Squares of Special Elements	O(1) State Variables (Draw one sum and one i).
2779	Maximum Beauty of an Array After Applying Operation	Frequency Map over Range (Visualizing a large array tracking every possible integer value).
2780	Minimum Index of a Valid Split	Dual Frequency Maps (Visualizing two HashMaps being fully rebuilt for every possible split point).
2781	Length of the Longest Valid Substring	Redundant String Slicing (Visualizing thousands of temporary substring copies being checked against a HashSet).
2782	Number of Unique Categories	Minimal Memory (Only tracking unique category representatives).
2783	Flight Occupancy and Waitlist Analysis (SQL)	Large Intermediate Join (Visualizing the DB loading every single booking record into memory before counting).
2784	Check if Array is Good	Sort Buffer (Visualizing the O(N) or O(log N) extra space used by the sorting algorithm).
2785	Sort Vowels in a String	String Mutation Buffer (Visualizing repeated string recreations for each swap in languages with immutable strings).
2786	Visit Array Positions to Maximize Score	Recursive Call Stack (Visualizing the massive overhead of re-calculating the best path for the same index multiple times).
2787	Ways to Express an Integer as Sum of Powers	Exponential State Search (Visualizing all possible combinations of powers being summed).
2788	Split Strings by Separator	Repeated String Concatenation (Visualizing memory churn from adding one char at a time to a string).
2789	Largest Element in an Array after Merge Operations	Intermediate Array Reallocations (Visualizing the heap creating new arrays for every merge pass).
2790	Maximum Number of Groups With Increasing Lengths	Priority Queue Heap State (Visualizing the heap re-balancing for every element used in a group).

2791	Count Paths That Can Form a Palindrome in a Tree	Path Frequency Maps (Visualizing thousands of HashMaps created to check character counts for every path).
2792	Count Nodes That Are Great Ancestors	Redundant Traversal Stack (Visualizing the CPU re-visiting the same bottom nodes for every ancestor).
2793	Status of Flight Tickets (SQL)	Row-by-Row Comparison Buffer (Visualizing the SQL engine re-scanning the bookings table for every single record).
2794	Create Object from Two Arrays (JS)	Linear Search Overhead (Visualizing indexOf being called repeatedly).
2795	Parallel Execution of Promises for Individual Results Retrieval	Promise.allSettled Polyfill
2796	Repeat String (JS)	Heap Allocation Churn (Visualizing the heap filling with intermediate string fragments: "a", "aa", "aaa"...).
2797	Partial Function with Placeholders (JS)	Argument Object Cloning (Visualizing the memory used to copy the arguments list for every call).
2798	Number of Employees Who Met the Target	Sort Buffer (Visualizing the O(N) memory used to rearrange the array).
2799	Count Complete Subarrays in an Array	Massive Set Allocation (Visualizing thousands of HashSets being created and destroyed in memory).
2800	Shortest String That Contains Three Strings	String Concatenation Buffer (Visualizing the creation of 6 massive new strings in the heap).
2801	Count Stepping Numbers in Range	Trillions of Digit Arrays (Visualizing the CPU bottleneck from converting numbers to strings repeatedly).
2802	Find The K-th Lucky Number	Queue Memory (Visualizing the BFS queue growing linearly with K, potentially millions of strings).
2803	Factorial Generator (JS)	Static Array Memory (Visualizing a large block of heap memory occupied by fixed integers).
2804	Array Prototype ForEach (JS)	Execution Context Overhead (Visualizing the engine creating and destroying complex scopes for every iteration).
2805	Custom Interval	Recursive setTimeout via Closure
2806	Account Balance After Rounded Purchase	Arithmetic Snapping / Modulo Math
2807	Insert Greatest Common Divisors in Linked List	In-Place Pointer Splicing + Euclidean Algorithm
2808	Minimum Seconds to Equalize a Circular Array	Hash Map Grouping + Circular Maximum Gap Calculation
2809	Minimum Time to Make Array Sum At Most x	Sorting + 1D Dynamic Programming (Knapsack variant)
2810	Faulty Keyboard	Double Ended Queue (Deque) / Two Pointers
2811	Check if it is Possible to Split Array	Greedy / Math Observation
2812	Find the Safest Path in a Grid	Multi-source BFS + Max-Heap/Dijkstra (or Binary Search + BFS)
2813	Maximum Elegance of a K-Length Subsequence	Greedy Sorting + Stack-based Substitution
2814	Minimum Time Takes to Reach Destination Without Drowning	Multi-source BFS (Water) + Standard BFS (Player)
2815	Max Pair Sum in an Array	Hash Map / Array Grouping by Max Digit
2816	Double a Number Represented as a Linked List	Single Pass with Lookahead (In-Place)
2817	Minimum Absolute Difference Between Elements With Constraint	Balanced Binary Search Tree (TreeSet) + Sliding Pointer
2818	Apply Operations to Maximize Score	Monotonic Stack + Sieve of Eratosthenes + Greedy (Contribution Technique)
2819	Minimum Relative Loss After Buying K Items	Sorting + Prefix Sums + Binary Search (Two-Pointer Optimization)
2820	Election Results (SQL/Logic)	Common Table Expressions (CTE) + Window Functions (RANK/DENSE_RANK)
2821	Delay the Resolution of Each Promise	setTimeout + Promise Wrapper
2822	Inversion of Object	Object.entries() + Array.reduce()
2823	Deep Object Filter	Post-Order Recursive Reconstruction
2824	Count Pairs Whose Sum is Less than Target	Sorting + Two Pointers
2825	Make String a Subsequence Using Cyclic Increments	Two Pointers (Greedy Subsequence Match)
2826	Sorting Three Groups	Dynamic Programming (LIS Variation)
2827	Number of Beautiful Integers in the Range	Digit DP with Remainder State
2828	Check if a String Is an Acronym of Words	Early-Exit Linear Comparison
2829	Determine the Minimum Sum of a k-avoiding Array	Mathematical Greedy Partitioning
2830	Maximize the Profit as the Salesman	DP with Interval Grouping (Bucket Sort style)
2831	Find the Longest Equal Subarray	Hash Map of Indices + Sliding Window
2832	Maximal Range of Each Element	Monotonic Stack (Next Greater Element)
2833	Furthest Point From Origin	Greedy Absolute Displacement
2834	Find the Minimum Possible Sum of a Beautiful Array	Arithmetic Series Segmentation (Modulo Math)
2835	Minimum Operations to Form Subsequence with Target Sum	Greedy Bit Manipulation + Min-Priority Queue (or Frequency Array)
2836	Maximize Value of Function in a Ball Passing Game	Binary Lifting (Sparse Table)
2837	Total Traveled Distance (SQL)	LEFT JOIN with COALESCE / IFNULL
2838	Maximum Coins Heroes Can Collect	Sorting + Prefix Sums + Binary Search
2839	Check if Strings Can be Made Equal With Operations I	Parity-Based Character Comparison
2840	Check if Strings Can be Made Equal With Operations II	Frequency Mapping by Index Parity
2841	Maximum Sum of Almost Unique Subarray	Sliding Window + Hash Map (Frequencies)
2842	Count K-Subsequences of a String With Maximum Beauty	Greedy Frequency Sorting + Combinatorics
2843	Count Symmetric Integers	Direct Digit Extraction (Modulo/Div)
2844	Minimum Operations to Make a Special Number	Suffix Greedy Search (Targeting "00", "25", "50", "75")
2845	Count Interesting Subarrays	Prefix Sums + Hash Map (Frequency of remainders)
2846	Minimum Edge Weight Equilibrium Queries in a Tree	LCA (Lowest Common Ancestor) + Binary Lifting + Prefix Frequencies
2847	Smallest Number with Given Product of Digits	Greedy Prime Factorization (Reverse Digits)
2848	Points That Intersect With Cars	Difference Array / Prefix Sums
2849	Determine if a Cell Is Reachable at a Given Time	Chebyshev Distance (Math/Geometry)
2850	Minimum Moves to Spread Stones Over Grid	Backtracking with Pruning / BFS for State Space
2851	String Transformation	KMP (String Matching) + Matrix Exponentiation (DP)

2852	Sum of Remoteness of All Cells	Connected Components (BFS/DFS) + Global Totals
2853	Highest Salary Difference (SQL)	MAX() with Conditional Aggregation (CASE/IF)
2854	Rolling Average Steps (SQL)	Window Functions (AVG ... OVER ROWS BETWEEN 2 PRECEDING)
2855	Minimum Right Shifts to Sort the Array	Single-Pivot Circularity Check
2856	Minimum Array Length After Pair Removals	Frequency Dominance / Median Comparison
2857	Count Pairs of Points With Distance k	Hash Map + Target XOR Decomposition
2858	Minimum Edge Reversals So Every Node Is Reachable	Tree DP / Re-rooting Technique
2859	Sum of Values at Indices With K Set Bits	Built-in Bit Manipulation (popcount)
2860	Happy Students	Sorting + Greedy Boundary Check
2861	Maximum Number of Alloys	Binary Search on Answer
2862	Maximum Element-Sum of a Complete Subset of Indices	Square-Free Part Grouping
2863	Maximum Length of Semi-Decreasing Subarrays	Monotonic Stack + Two Pointers (Right-to-Left)
2864	Maximum Odd Binary Number	Greedy Bit Alignment
2865	Beautiful Towers I	Monotonic Stack + Prefix/Suffix DP
2866	Beautiful Towers II	Optimised Monotonic Stack + Dynamic Programming
2867	Count Valid Paths in a Tree	Sieve of Eratosthenes + Disjoint Set Union (DSU) or DFS
2868	The Wording Game	Greedy Game State Simulation (Two Pointers)
2869	Minimum Operations to Collect Elements	Backward Scan + Boolean Frequency Array
2870	Minimum Number of Operations to Make Array Empty	Frequency Counting + Greedy Math
2871	Split Array Into Maximum Number of Subarrays	Greedy AND-Reset Logic
2872	Maximum Number of K-Divisible Components	Post-Order Traversal (Bottom-Up Summing)
2873	Maximum Value of an Ordered Triplet I	Prefix Maximum & Suffix Maximum (or Pre-calculation)
2874	Maximum Value of an Ordered Triplet II	Dynamic Prefix Maximum & Max-Difference Tracking
2875	Minimum Size Subarray in Infinite Array	Modulo Math + Sliding Window on Doubled Array
2876	Count Visited Nodes in a Directed Graph	Functional Graph Decomposition (Cycles & Tails)
2877	Create a DataFrame from List (Pandas)	Native Pandas Constructor (Vectorized Internal Logic)
2878	Get the Size of a DataFrame	df.shape Attribute Access
2879	Display the First Three Rows	df.head(3) (Slicing)
2880	Select Data (Filtering)	Boolean Indexing: df[df['id'] == 101]
2881	Create a New Column	Vectorized Assignment: df['bonus'] = df['salary'] * 2
2882	Drop Duplicate Rows	df.drop_duplicates(subset=['id'])
2883	Drop Missing Data	df.dropna(subset=['name'])
2884	Modify Columns	df['salary'] *= 2 (Vectorized Math)
2885	Rename Columns	df.rename(columns={...})
2886	Change Data Type	df['grade'].astype(int)
2887	Fill Missing Data	df['quantity'].fillna(0)
2888	Reshape Data: Concatenate	pd.concat([df1, df2])
2889	Reshape Data: Pivot	df.pivot(index='month', columns='city', values='temperature')
2890	Reshape Data: Melt	df.melt(id_vars=['product'], ...)
2891	Method Chaining	Chained Operations: df.loc[...].sort_values(...)[['name']]
2892	Minimizing Array After Replacing Pairs With Their Product	Greedy Array Compression
2893	Calculate Orders Within Each Interval	Mathematical Binning (Floor Division)
2894	Divisible and Non-divisible Sums Difference	Arithmetic Progression (AP) Formula
2895	Minimum Processing Time	Greedy (Dual Sorting)
2896	Apply Operations to Make Two Strings Equal	1D Dynamic Programming (Memoization)
2897	Apply Operations on Array to Maximize Sum of Squares	Bit-Counting + Greedy Reconstruction
2898	Maximum Linear Stock Score	Frequency Mapping (Coordinate Transformation)
2899	Last Visited Integers	List Maintenance + Pointer Tracking
2900	Longest Unequal Adjacent Groups Subsequence I	Greedy Alternation
2901	Longest Unequal Adjacent Groups Subsequence II	Dynamic Programming (LIS Variation)
2902	Count of Sub-Multisets With Bounded Sum	DP with Frequency Map + Sliding Window Optimization
2903	Find Indices With Index and Value Difference I	Two Pointers / Tracking Min & Max limits
2904	Shortest and Lexicographically Smallest Beautiful String	Sliding Window
2905	Find Indices With Index and Value Difference II	Two Pointers + Prefix Min/Max Tracking
2906	Construct Product Matrix	Prefix & Suffix Products (Flattened 2D to 1D)
2907	Maximum Profitable Triplets With Increasing Prices I	Middle Element Fix + Left/Right Max Search
2908	Minimum Sum of Mountain Triplets I	Prefix Min & Suffix Min Arrays
2909	Minimum Sum of Mountain Triplets II	Prefix Min & Suffix Min Arrays (Identical to 2908)
2910	Minimum Number of Groups to Create a Valid Assignment	Hash Map Frequencies + Greedy Math Downward Search
2911	Minimum Changes to Make K Semi-palindromes	Precomputation + 2D Partition Dynamic Programming

2912	Number of Ways to Reach Destination in the Grid	Combinatorics / 4-State DP (Markov Chain approach)
2913	Subarrays Distinct Element Sum of Squares I	Nested Loop Sliding Window with Running Set
2914	Minimum Number of Changes to Make Binary String Beautiful	Greedy (Adjacent Pair Checking)
2915	Length of the Longest Subsequence That Sums to Target	0/1 Knapsack Dynamic Programming
2916	Subarrays Distinct Element Sum of Squares II	Segment Tree with Lazy Propagation + Sweepline
2917	Find the K-or of an Array	Bit Manipulation (Column-wise Bit Counting)
2918	Minimum Equal Sum of Two Arrays After Replacing Zeros	Greedy Sum Matching
2919	Minimum Increment Operations to Make Array Beautiful	1D Dynamic Programming
2920	Maximum Points After Collecting Coins From All Nodes	Tree DP + Memoization (State Reduction via Bit Shifts)
2921	Maximum Profitable Triplets With Increasing Prices II	Binary Indexed Tree (Fenwick Tree) / Segment Tree
2922	Market Analysis III (SQL)	SQL Window Functions (`RANK()` or `DENSE_RANK()`)
2923	Find Champion I	Graph In-Degree Counting / Row Sum Optimization
2924	Find Champion II	Directed Acyclic Graph (DAG) - Count In-Degrees
2925	Maximum Score After Applying Operations on a Tree	Tree Dynamic Programming (Min Cost to Satisfy)
2926	Maximum Balanced Subsequence Sum	LIS-style DP + Coordinate Compression + Fenwick Tree
2927	Distribute Candies Among Children II	Combinatorics (Inclusion-Exclusion Principle)
2928	Distribute Candies Among Children I	Iterative Brute Force (Optimized Range)
2929	Distribute Candies Among Children III	Inclusion-Exclusion (Math)
2930	Number of Strings Which Can Be Rearranged to Contain Substring	Inclusion-Exclusion Principle with Modular Exponentiation
2931	Maximum Spending After Buying Items	Greedy (Min-Heap / Sorting)
2932	Maximum Strong Pair XOR I	Brute Force (Small Constraints)
2933	High-Access Employees	Sorting + Sliding Window (Fixed Size 3)
2934	Minimum Operations to Maximize Last Elements	Greedy (Two Scenarios)
2935	Maximum Strong Pair XOR II	Sliding Window + Binary Trie
2936	Number of Equal Numbers Blocks	Binary Search / Jump Search
2937	Make Three Strings Equal	Greedy Prefix Matching
2938	Separate Black and White Balls	Two Pointers / Greedy (Counting 1s)
2939	Maximum Xor Product	Bit Manipulation (Greedy Bit-by-Bit)
2940	Find Building Where Alice and Bob Can Meet	Offline Queries + Monotonic Stack / Priority Queue
2941	Maximum GCD-Sum of a Subarray	Sparse Table / GCD Property (Unique Prefix GCDs)
2942	Find Words Containing Character	Simple Linear Scan
2943	Maximize Area of Square Hole in Grid	Longest Consecutive Sequence (Greedy)
2944	Minimum Number of Coins for Fruits	Dynamic Programming + Monotonic Queue (Sliding Window)
2945	Find Maximum Non-decreasing Array Length	Greedy DP + Prefix Sums + Monotonic Queue
2946	Matrix Similarity After Cyclic Shifts	Modulo Arithmetic Indexing
2947	Count Beautiful Substrings I	Prefix Sums + Hash Map (Frequency Tracking)
2948	Make Lexicographically Smallest Array by Swapping Elements	Sorting + Grouping (Disjoint Connectivity)
2949	Count Beautiful Substrings II	Math Optimization + Prefix Map (Advanced)
2950	Number of Divisible Substrings	Prefix Sums (Brute Force is expected for N=1000)
2951	Find the Peaks	Linear Array Scanning
2952	Minimum Number of Coins to be Added	Greedy Range Extension (Patching)
2953	Count Complete Substrings	Sliding Window + Segment Splitting
2954	Count the Number of Infection Sequences	Combinatorics + Modular Inverse
2955	Number of Same-End Substrings	Prefix Sums + Combinatorics (nC2)
2956	Find Common Elements Between Two Arrays	Hash Set Lookups
2957	Remove Adjacent Almost-Equal Characters	Greedy (Pairs Comparison)
2958	Length of Longest Subarray With at Most K Frequency	Sliding Window (Two Pointers)
2959	Number of Possible Sets of Closing Branches	Bitmask DP + Floyd-Warshall
2960	Count Tested Devices After Test Operations	Greedy (Running Offset)
2961	Double Modular Exponentiation	Binary Exponentiation (Modular Pow)
2962	Count Subarrays Where Max Element Appears at Least K Times	Sliding Window (Two Pointers)
2963	Count the Number of Good Partitions	Merge Intervals + Combinatorics
2964	Number of Divisible Triplets	Prefix Sum Hash Map / Two-Sum Pattern
2965	Find Missing and Repeated Values	Frequency Array / Math (Sum & Squares)
2966	Divide Array Into Arrays With Max Difference	Greedy (Sorting + Linear Grouping)
2967	Minimum Cost to Make Array Equalindromic	Median + Palindrome Generation
2968	Apply Operations to Maximize Frequency Score	Sorting + Sliding Window + Prefix Sums
2969	Minimum Number of Coins for Fruits II	DP + Monotonic Queue (Deque)
2970	Count the Number of Incremovable Subarrays I	Two Pointers (Prefix/Suffix Scan)
2971	Find Polygon With the Largest Perimeter	Greedy (Sorting + Prefix Sums)
2972	Count the Number of Incremovable Subarrays II	Two Pointers (Optimized Prefix/Suffix)

2973	Find Number of Coins to Place in Tree Nodes	Tree DP (Post-order Traversal)
2974	Minimum Number Game	Sorting + Adjacent Swaps
2975	Maximum Square Area by Removing Fences	Hash Set Intersection (Difference Checking)
2976	Minimum Cost to Convert String I	Floyd-Warshall (All-Pairs Shortest Path)
2977	Minimum Cost to Convert String II	Trie + Floyd-Warshall + DP
2978	Symmetric Coordinates	Self-Join with ROW_NUMBER()
2979	Most Expensive Item That Cannot Be Bought	Frobenius Coin Problem (Math)
2980	Check if Bitwise OR Has Trailing Zeros	Greedy (Count Even Numbers)
2981	Find Longest Special Substring That Occurs Thrice I	Sliding Window + Frequency Map
2982	Find Longest Special Substring That Occurs Thrice II	Top-3 Length Tracking per Character
2983	Palindrome Rearrangement Queries	Prefix Sums + Parity Symmetry Check
2984	Find Peak Calling Hours for Each City	Window Function (RANK)
2985	Calculate Compressed Mean	Weighted Average Formula
2986	Find Third Transaction	Window Function (`RANK()` / `ROW_NUMBER()`)
2987	Find Expensive Cities	Aggregate Comparison (HAVING Clause)
2988	Manager of the Largest Department	Window Function (`DENSE_RANK()`)
2989	Class Performance	Aggregate Math (`MAX` - `MIN`)
2990	Loan Types	Grouping + `COUNT(DISTINCT)` with `HAVING`
2991	Top Three Wineries	Window Function (`RANK()`) + Pivoting
2992	Number of Self-Divisible Permutations	Backtracking with State Pruning (Bitmask)
2993	Friday Purchases I	Date Filtering + Grouping
2994	Friday Purchases II	Recursive CTE (Date Generation) + Left Join
2995	Viewers Turned Streamers	Window Function (`FIRST_VALUE`) / `RANK()`
2996	Smallest Missing Integer Greater Than Sequential Prefix Sum	Simulation + Hash Set
2997	Minimum Number of Operations to Make Array XOR Equal to K	Bitwise XOR Properties + Hamming Distance
2998	Minimum Number of Operations to Make X and Y Equal	Breadth-First Search (BFS) / Shortest Path
2999	Count the Number of Powerful Integers	Digit Dynamic Programming (Digit DP)
3000	Maximum Area of Longest Diagonal Rectangle	Single Pass Comparison (Pythagorean Theorem)
3001	Minimum Moves to Capture The Queen	Math / Geometry (Line of Sight checks)
3002	Maximum Size of a Set After Removals	Greedy Selection + Hash Sets
3003	Maximize the Number of Partitions After Operations	Dynamic Programming with Bitmasking (State Compression)
3004	Maximum Subtree of the Same Color	Bottom-Up Post-Order DFS (Tree DP)
3005	Count Elements With Maximum Frequency	Single/Double Pass Hash Map (Frequency Array)
3006	Find Beautiful Indices in the Given Array I	KMP String Matching + Binary Search / Two Pointers
3007	Maximum Number That Sum of Prices Is Less Than or Equal to K	Binary Search on Answer + Bit Digit DP (Math)
3008	Find Beautiful Indices in the Given Array II	KMP Prefix Function (LPS) + Monotonic Two Pointers
3009	Maximum Number of Intersections on the Chart	Line Sweep Algorithm / Difference Array
3010	Divide an Array Into Subarrays With Minimum Cost I	Greedy Selection (Math/Sorting)
3011	Find if Array Can Be Sorted	Array Grouping by Set Bits + Min/Max Tracking
3012	Minimize Length of Array Using Operations	Math / Number Theory (Global Minimum modulo reduction)
3013	Divide an Array Into Subarrays With Minimum Cost II	Sliding Window + Two Heaps (or Balanced BST)
3014	Minimum Number of Pushes to Type Word I	Greedy Chunking (Math)
3015	Count the Number of Houses at a Certain Distance I	Floyd-Warshall Algorithm (All-Pairs Shortest Path)
3016	Minimum Number of Pushes to Type Word II	Greedy Assignment + Frequency Sorting
3017	Count the Number of Houses at a Certain Distance II	Prefix Sums / Difference Array (Line Sweep)
3018	Maximum Number of Removal Queries That Can Be Processed I	2D Dynamic Programming (Interval DP)
3019	Number of Changing Keys	Single-Pass Constant Space Scan
3020	Find the Maximum Number of Elements in Subset	Hash Map Frequencies + Mathematical Chaining
3021	Alice and Bob Playing Flower Game	O(1) Combinatorics (Parity Math)
3022	Minimize OR of Remaining Elements Using Operations	Greedy Bitwise Masking (MSB to LSB)
3023	Find Pattern in Infinite Stream I	Knuth-Morris-Pratt (KMP) Automaton
3024	Type of Triangle	Sorting + Comparison Math
3025	Find the Number of Ways to Place People I	Coordinate Sorting + Monotonic Optimization
3026	Maximum Good Subarray Sum	Prefix Sum + Hash Map (Minimum Prefix Tracking)
3027	Find the Number of Ways to Place People II	Sorting + Single Pass Optimization
3028	Ant on the Boundary	Prefix Sum Simulation
3029	Minimum Time to Revert Word to Initial State I	KMP Prefix Function (LPS Array)
3030	Find the Grid of Region Average	Simulation with Condition Validation
3031	Minimum Time to Revert Word to Initial State II	Z-Algorithm / KMP Prefix Function
3032	Count Numbers With Unique Digits II	Precomputation / Bitmasking
3033	Modify the Matrix	Column-Wise Max Pre-calculation

3034	Number of Subarrays That Match a Pattern I	Pattern Conversion + KMP Algorithm
3035	Maximum Palindromes After Operations	Greedy Counting + Sorting lengths
3036	Number of Subarrays That Match a Pattern II	KMP / Z-Algorithm on Trend Array
3037	Find Pattern in Infinite Stream II	KMP with Stream State
3038	Maximum Number of Operations With the Same Score I	Greedy Simulation
3039	Apply Operations to Make String Empty	Frequency Counting + Last Occurrence Mapping
3040	Maximum Number of Operations With the Same Score II	Interval DP (Top-Down Memoization)
3041	Maximize Consecutive Elements in an Array After Modifications	Sorting + Dynamic Programming
3042	Count Prefix and Suffix Pairs I	Brute Force (Small Constraints) / Trie (Optimal)
3043	Find the Length of the Longest Common Prefix	Prefix Hash Set / Trie
3044	Most Frequent Prime	Directional Ray Casting + Primality Sieve
3045	Count Prefix and Suffix Pairs II	Trie with Pairwise Character Keys
3046	Split the Array	Pigeonhole Principle (Frequency Counting)
3047	Find the Largest Area of Square Inside Two Rectangles	Coordinate Min-Max Comparison
3048	Earliest Second to Mark Indices I	Binary Search on Answer + Greedy Verification
3049	Earliest Second to Mark Indices II	Binary Search + Greedy Priority Queue
3050	Pizza Toppings Cost Analysis (SQL)	Join Filtering + Order By Clause
3051	Find Candidates for Data Scientist Position (SQL)	Group By + Having Count Clause
3052	Maximize Items in Partitions (SQL)	Window Functions + Floor Math
3053	Classifying Triangles by Lengths (SQL)	Case When Logic (Geometric Constraints)
3054	Binary Tree Nodes (SQL)	Case When + Correlated Subquery (or In/Not In)
3055	Top Percentile Fraud	Window Function (PERCENT_RANK)
3056	Snaps Analysis (SQL)	Join + Group By with Filtered Aggregation
3057	Employees Project Allocation (SQL)	Window Function (AVG over Partition)
3058	Friends With No Common Friends (SQL)	CTEs + Not In / Except (Set Difference)
3059	Find All Unique Email Domains (SQL)	SUBSTRING_INDEX / Regex + Group By
3060	User Activities Queries (SQL)	Common Table Expressions (CTE) + Aggregation
3061	Calculate Trapping Rain Water (SQL)	Window Functions (Max Over Partition)
3062	Winner of the Linked List Game	Two-Pointer Jump (Step of 2)
3063	Linked List Frequency	Hash Map + Result Linked List Construction
3064	Guess the Number Using Bitwise Questions I	Interactive Bit Manipulation
3065	Minimum Operations to Exceed Threshold Value I	Single-Pass Filtering
3066	Minimum Operations to Exceed Threshold Value II	Min-Priority Queue (Heap)
3067	Count Pairs of Connectable Servers in a Weighted Tree	DFS + Combinatorics (Product of Subtree Counts)
3068	Find the Maximum Sum of Node Values	Greedy + Delta Sorting (or Simple DP)
3069	Distribute Elements Into Two Arrays I	Simple Greedy Simulation
3070	Count Submatrices with Top-Left Element and Sum Less Than k	2D Prefix Sum (In-Place)
3071	Minimum Operations to Write the Letter Y on a Grid	Frequency Counting (Y vs. Non-Y)
3072	Distribute Elements Into Two Arrays II	Fenwick Tree (BIT) / Segment Tree + Coordinate Compression
3073	Maximum Increasing Triplet Value	Right Max Array + Sorted Set
3074	Apple Redistribution into Boxes	Sorting + Greedy Accumulation
3075	Maximize Happiness of Selected Children	Greedy + Decay Offset
3076	Shortest Uncommon Substring in an Array	Hash Map Frequency + Sorting
3077	Maximum Strength of K Disjoint Subarrays	Dynamic Programming (Pick/Skip State)
3078	Match Alphanumerical Pattern in Matrix I	2D Sliding Window / Submatrix Hashing
3079	Find the Sum of Encrypted Integers	Mathematical Digit Decomposition
3080	Mark Elements on Array by Performing Queries	Min-Heap + Boolean Visited Array
3081	Replace Question Marks in String to Minimize Its Value	Greedy Frequency Counting + Min-Heap
3082	Find the Sum of the Power of All Subsequences	3D Dynamic Programming (Knapsack Variant)
3083	Existence of a Substring in a String and Its Reverse	2D Bitset / Hash Set of Pairs
3084	Count Substrings Starting and Ending with Given Character	Combinatorics (Handshake Lemma)
3085	Minimum Deletions to Make String K-Frequency Special	Frequency Sorting + Sliding Threshold
3086	Minimum Moves to Pick K Ones	Sliding Window + Median Distance + Prefix Sums
3087	Find Trending Hashtags (SQL)	Regex Extraction + Group By + Order By
3088	Make String Anti-palindrome	Sorting + Greedy Swapping
3089	Find Bursty Behavior (SQL)	Window Functions (COUNT with Range)
3090	Maximum Length Substring With Two Occurrences	Sliding Window (Two Pointers)
3091	Apply Operations to Make Sum Greater Than or Equal to k	Math / Square Root Optimization
3092	Most Frequent IDs	Hash Map + Max-Priority Queue (Lazy Removal)
3093	Longest Common Suffix Queries	Reverse Trie (Suffix Trie) with Precomputed Min
3094	Guess the Number Using Bitwise Queries II	Bit-by-Bit Reconstruction

3095	Shortest Subarray With OR at Least K I	Two Pointers / Sliding Window
3096	Minimum Levels to Gain More Points	Prefix Sum / Running Total
3097	Shortest Subarray With OR at Least K II	Sliding Window + Bit Count Array
3098	Find the Sum of Subsequences Powers	5D Dynamic Programming + Sorting
3099	Harshad Number	Digit Extraction (Modulo Arithmetic)
3100	Water Bottles II	Greedy Simulation (While Loop)
3101	Count Alternating Subarrays	One-Pass DP / Sliding Window (Arithmetic Progressions)
3102	Minimize Manhattan Distances	Coordinate Rotation (Chebyshev Transformation)
3103	Find Trending Hashtags II (SQL)	REGEXP_SUBSTR + Window Aggregation
3104	Find Longest Self-Contained Substring	Interval Merging / Prefix Sums
3105	Longest Strictly Inc. or Dec. Subarray	Greedy / One-Pass State Machine
3106	Lexicographically Smallest String After Ops	Greedy Left-to-Right / Math
3107	Min Operations to Make Median Equal to K	Sorting + Greedy Target Matching
3108	Min Cost Walk in Weighted Graph	Disjoint Set Union (DSU) / Connected Components
3109	Find the Index of Permutation	Combinatorics + Fenwick Tree (BIT)
3110	Score of a String	Linear Simulation / Iteration
3111	Minimum Rectangles to Cover Points	1D Greedy / Sorting
3112	Minimum Time to Visit Disappearing Nodes	Modified Dijkstra's Algorithm (Min-Heap)
3113	Find the Number of Subarrays Where Boundary Elements Are Maximum	Monotonic Stack
3114	Latest Time You Can Obtain After Replacing Characters	Greedy Constraint Matching
3115	Maximum Prime Difference	Two Pointers (Inward Search)
3116	Kth Smallest Amount With Single Denomination Combination	Binary Search + Inclusion-Exclusion (PIE)
3117	Minimum Sum of Values by Dividing Array	DP + Memoization + Bitwise AND
3118	Friday Purchase III (SQL)	Recursive CTE + LEFT JOIN
3119	Maximum Number of Potholes That Can Be Fixed	Greedy + Counting / Bucket Sort
3120	Count the Number of Special Characters I	Hash Set / Boolean Arrays
3121	Count the Number of Special Characters II	Hash Table / Array Mapping (Single Pass)
3122	Min Operations to Satisfy Conditions	Dynamic Programming (Column by Column)
3123	Find Edges in Shortest Paths	Dijkstra's (Bi-directional) + Edge Relaxation
3124	Find Longest Calls (SQL)	JOIN + Window Function (RANK)
3125	Max Number That Makes Result of Bitwise AND Zero	Bit Manipulation / Math
3126	Server Utilization Time (SQL)	Window Function (LEAD) + CTE
3127	Make a Square with the Same Color	Sliding Window (Fixed 2x2) / Counting
3128	Right Triangles	Precomputation / Combinatorics
3129	Find All Possible Stable Binary Arrays I	DP + Memoization (State: zeros, ones, last_bit)
3130	Find All Possible Stable Binary Arrays II	DP + Prefix Sum Optimization
3131	Find the Integer Added to Array I	Min-Element Comparison / Math
3132	Find the Integer Added to Array II	Sorting + Small Constant Search
3133	Minimum Array End	Bit Manipulation (Zero-Bit Filling)
3134	Median of Uniqueness Array	Binary Search on Answer + Sliding Window
3135	Equalize Strings by Adding or Removing Characters at Ends	Longest Common Substring (LCS Variant)
3136	Valid Word	Single-Pass Flagging
3137	Minimum Number of Operations to Make Word K-Periodic	Hash Map / Chunk Frequency
3138	Minimum Length of Anagram Concatenation	Divisor Iteration + Freq Map Comparison
3139	Minimum Cost to Make Array Elements Equal	Greedy Math + Case Analysis
3140	Consecutive Available Seats II	Greedy Sliding Window
3141	Maximum Hamming Distances	State Tracking (Parity / Flip Count)
3142	Check if Grid Satisfies Conditions	Matrix Traversal (Local Constraint)
3143	Maximum Points Inside Square	Sorting / Binary Search on Answer
3144	Min Substring Partition of Equal Character Frequency	Dynamic Programming + Freq Counting
3145	Find Products of Elements of Big Array	Binary Search + Bit Counting Math
3146	Permutation Difference between Two Strings	Hash Map / Frequency Indexing
3147	Taking Maximum Energy From the Mystic Dungeon	Dynamic Programming (Suffix Sum with Jump)
3148	Maximum Difference Score in a Grid	2D DP (Prefix Minimum)
3149	Find the Minimum Cost Array Permutation	Bitmask DP (TSP Variant)
3150	Invalid Tweets II (SQL)	String Predicate Filtering
3151	Special Array I	Single-Pass Parity Check
3152	Special Array II	Prefix Sum (Parity Violations)
3153	Sum of Digit Differences of All Pairs	Contribution Technique (Position-wise Counting)
3154	Number of Ways to Reach the K-th Stair	DP + Memoization / Combinatorics
3155	Maximum Number of Upgradable Servers	Binary Search on Answer / Math

3156	Employee Task Duration and Concurrent Tasks	CTE + Interval Merging (Sweep Line SQL)
3157	Find the Level of Tree with Minimum Sum	BFS (Level Order Traversal)
3158	Find the XOR Sum of All Numbers Appears Twice	Hash Set / Frequency Counting
3159	Find Occurrences of an Element in an Array	Pre-processing (Index Collection)
3160	Find the Number of Distinct Colors Among the Balls	Dual Hash Maps (Frequency Tracking)
3161	Block Placement Queries	Segment Tree + Balanced BST (Sorted Set)
3162	Find the Number of Good Pairs I	Simple Iteration (Small Constraints)
3163	String Compression III	Two-Pointer / Greedy Counting
3164	Find the Number of Good Pairs II	Frequency Map + Multiples / Divisors
3165	Max Sum of Subsequence With Non-adjacent Elements	Segment Tree (DP State Merging)
3166	Calculate Parking Fees and Duration	CTE / Window Functions
3167	Better Compression of String	Hash Table + Alphabetical Sorting
3168	Min Number of Chairs in a Waiting Room	Greedy / Simulation (Peak Tracking)
3169	Count Days Without Meetings	Interval Merging
3170	Lexicographically Minimum String After Removing Stars	Priority Queue / Array of Stacks
3171	Find Subarray With Bitwise OR Closest to K	Shrinking Bitwise Set
3172	Second Day Verification	Window Function (DENSE_RANK)
3173	Bitwise OR of Adjacent Elements	Single-Pass Array Mapping
3174	Clear Digits	Stack (LIFO)
3175	Find The First Player to win K Games in a Row	Greedy / One-Pass Simulation
3176	Max Length of a Good Subsequence I	Dynamic Programming (O(N^2 * k))
3177	Max Length of a Good Subsequence II	DP + Global Max Tracking (O(Nk))
3178	Find the Child Who Has the Ball After K Seconds	Math (Modulo Arithmetic)
3179	Find the N-th Value After K Seconds	Combinatorics (Stars and Bars)
3180	Maximum Total Reward Using Operations I	DP / Bitset Optimization
3181	Maximum Total Reward Using Operations II	DP + Bitset (Bit-Level Parallelism)
3182	Find Top Scoring Students	JOIN + Aggregation (HAVING Count Match)
3183	The Number of Ways to Make the Sum	Unbounded Knapsack + Case Addition
3184	Count Pairs That Form a Complete Day I	Frequency Array / Hashing (Modulo 24)
3185	Count Pairs That Form a Complete Day II	Frequency Array / Hashing (Long Int)
3186	Maximum Total Damage With Spell Casts	DP + Freq Map + Two Pointers
3187	Peaks in Array	Segment Tree / Fenwick Tree
3188	Find Top Scoring Students II (SQL)	Window Functions + CTE Aggregation
3189	Minimum Moves to Get a Peaceful Board	Greedy / Independent Sorting
3190	Find Minimum Operations to Make All Elements Divisible by Three	Modulo Simulation
3191	Minimum Operations to Make Binary Array Elements Equal to One I	Greedy / Fixed Window Flip
3192	Minimum Operations to Make Binary Array Elements Equal to One II	Greedy / State-Flip Parity
3193	Find the Number of Inversions in Permutation	Dynamic Programming (Prefix Sum Opt)
3194	Minimum Average of Smallest and Largest Elements	Sorting + Two Pointers
3195	Find the Minimum Area to Cover All 1s I	Min-Max Coordinate Tracking
3196	Maximize Total Cost of Alternating Subarrays	1D Dynamic Programming (2 States)
3197	Find the Minimum Area to Cover All 1s II	Grid Partitioning + Bounding Box
3198	Find Cities in Each State (SQL)	String Aggregation (GROUP_CONCAT)
3199	Count Triplets with Even XOR Sum	Bitwise Parity Counting
3200	Maximum Height of a Triangle	Greedy Simulation (Two Cases)
3201	Find the Maximum Length of Valid Subsequence I	State Machine / 1D Dynamic Programming
3202	Find the Maximum Length of Valid Subsequence II	2D Dynamic Programming (Tracking mod pairs)
3203	Find Minimum Diameter After Merging Two Trees	Graph Tree Diameter (2-Pass BFS/DFS)
3204	Bitwise User Permissions Analysis (SQL)	Bitwise AND (&) Operator in SELECT/WHERE
3205	Maximum Array Hopping Score I	1D Dynamic Programming / Greedy
3206	Alternating Groups I	Modulo Sliding Window (Size 3)
3207	Maximum Points After Enemy Battles	Greedy + Sorting (Two-Pointer logic)
3208	Alternating Groups II	Variable Sliding Window / State Counting
3209	Number of Subarrays With AND Value of K	LogTrick / Hash Map (Monotonic Bitwise AND)
3210	Find the Encrypted String	String Slicing / Cyclic Shift
3211	Generate Binary Strings Without Adjacent Zeros	Constrained Backtracking / DFS
3212	Count Submatrices With Equal Frequency of X and Y	2D Prefix Sum Matrix
3213	Construct String with Minimum Cost	Aho-Corasick / Trie + 1D Dynamic Programming
3214	Year on Year Growth Rate (SQL)	Window Functions (LAG / LEAD)
3215	Count Triplets with Even XOR Set Bits II	Bitwise Parity Math / Combinatorics

3216	Lexicographically Smallest String After a Swap	Greedy Linear Scan
3217	Delete Nodes From Linked List Present in Array	HashSet + Dummy Node 1-Pass
3218	Minimum Cost for Cutting Cake I	Greedy Sorting (Largest Cost First)
3219	Minimum Cost for Cutting Cake II	Greedy Sorting / Two Pointers
3220	Odd and Even Transactions (SQL)	Conditional Aggregation (SUM + IF/CASE)
3221	Maximum Array Hopping Score II	Monotonic Stack / Greedy Suffix Max
3222	Find the Winning Player in Coin Game	Mathematical Parity / Bottleneck Check
3223	Minimum Length of String After Operations	Frequency Counting Map / Parity Math
3224	Minimum Array Changes to Make Differences Equal	Difference Array / Sweep Line Algorithm
3225	Maximum Score From Grid Operations	Dynamic Programming (Column State Transitions)
3226	Number of Bit Changes to Make Two Integers Equal	Bit Manipulation (XOR, AND)
3227	Vowels Game in a String	Mathematical Invariant / Game Theory
3228	Maximum Number of Operations to Move Ones to the End	Greedy Counting / Cumulative Sum
3229	Minimum Operations to Make Array Equal to Target	Difference Array / Greedy Peaks
3230	Customer Purchasing Behaviors Analysis (SQL)	Window Functions / CTE (Common Table Expressions)
3231	Minimum Number of Increasing Subsequence to Be Removed	Patience Sorting / Segment Tree (LIS variant)
3232	Find if Digit Game Can Be Won	Greedy Summation / Math
3233	Find the Count of Numbers Which Are Not Special	Number Theory / Sieve of Eratosthenes
3234	Count Substrings With Dominant Ones	Enumeration by Zero-Count (O(N√N))
3235	Check if Rectangle Corner Is Reachable	Graph connectivity (Union Find / BFS on Circles)
3236	CEO Subordinate Hierarchy (SQL)	Recursive CTE (WITH RECURSIVE)
3237	Alt and Tab Simulation	Reverse Traversal + HashSet
3238	Find the Number of Winning Players	2D Frequency Matrix (Counting)
3239	Minimum Number of Flips to Make Binary Grid Palindromic I	Independent Row/Column Counting
3240	Minimum Number of Flips to Make Binary Grid Palindromic II	Symmetry Grouping (2x2 Blocks + Centroids)
3241	Time Taken to Mark All Nodes	Tree DP (Rerooting / 2-Pass DFS)
3242	Design Neighbor Sum Service	Coordinate Mapping (Precomputation)
3243	Shortest Distance After Road Addition Queries I	Dynamic BFS / Incremental DP
3244	Shortest Distance After Road Addition Queries II	Greedy Range Removal (Set/Segment Tree)
3245	Alternating Groups III	Sliding Window + Deficit Counter
3246	Premier League Table Ranking (SQL)	Window Function (RANK)
3247	Number of Subsequences with Odd Sum	Combinatorics / DP (Parity Switch)
3248	Snake in Matrix	Coordinate Math (i, j) Simulation
3249	Count the Number of Good Nodes	Post-Order DFS (Bottom-Up Size Tally)
3250	Find the Count of Monotonic Pairs I	2D Dynamic Programming (State: Index, Last Value)
3251	Find the Count of Monotonic Pairs II	DP + Prefix Sum Optimization
3252	Premier League Table Ranking II (SQL)	Window Functions (RANK, COUNT, CEIL)
3253	Construct String with Minimum Cost (Easy)	Iterative Dynamic Programming (Substring Check)
3254	Find the Power of K-Size Subarrays I	Linear Count Tracker
3255	Find the Power of K-Size Subarrays II	Amortized Linear Scan / Precomputation
3256	Maximum Value Sum by Placing Three Rooks I	Top-3 Row Candidates + Backtracking
3257	Maximum Value Sum by Placing Three Rooks II	2D Prefix/Suffix Max + Greedy
3258	Count Substrings That Satisfy K-Constraint I	Sliding Window (Two Pointers)
3259	Maximum Energy Boost From Two Drinks	2D Dynamic Programming (State Switching)
3260	Find the Largest Palindrome Divisible by K	Greedy + Remainder DP
3261	Count Substrings That Satisfy K-Constraint II	Sliding Window + Prefix Sums
3262	Find Overlapping Shifts	Sieve of Eratosthenes + Math
3263	Convert Doubly Linked List to Array II	Linear Traversal (Bidirectional)
3264	Final Array State After K Multiplication Operations I	Min-Priority Queue (Heap)
3265	Count Almost Equal Pairs I	Hash Map + Digit Perturbation
3266	Final Array State After K Multiplication Operations II	Simulation + Modular Exponentiation
3267	Count Almost Equal Pairs II	Optimized BFS / DFS on Digit States
3268	Find Overlapping Shifts II	Sorting + Sweeping Max-Y
3269	Constructing Two Increasing Arrays	2D Dynamic Programming (dp[i][j])
3270	Find the Key of the Numbers	Arithmetic Digit Extraction
3271	Hash Divided String	Single-Pass Index Traversal (Step K)
3272	Find the Count of Good Integers	Palindrome Generation + Combinatorial Counting
3273	Minimum Amount of Damage Dealt to Bob	Greedy Sorting (Smith's Rule)
3274	Check if Two Chessboard Squares Have the Same Color	Parity Summation
3275	K-th Nearest Obstacle Queries	Fixed-Size Max-Heap
3276	Select Cells in Grid with Maximum Score	Bitmask DP + Sorting Values

3277	Maximum XOR Score Subarray Queries	2D DP (Subarray XOR Triangle + Range Max)
3278	Find Candidates for Data Scientist Position II	GROUP BY + HAVING COUNT()
3279	Maximum Total Area Occupied by Pistons	Linear Scan (Min/Max Tracking)
3280	Convert Date to Binary	Standard Library formatting (bin/format)
3281	Maximize Score of Numbers in Ranges	Binary Search on Answer + Greedy
3282	Reach End of Array With Maximum Score	Greedy (Monotonic Suffix Max)
3283	Maximum Number of Moves to Kill All Pawns	BFS Precomputation + Bitmask DP
3284	Sum of Consecutive Subarrays	Dynamic Programming (State: Value & Direction)
3285	Find Indices of Stable Mountains	Single-Pass Linear Scan
3286	Find a Safe Walk Through a Grid	0-1 BFS or Dijkstra
3287	Find the Maximum Sequence Value	DP with Bitset (Prefix/Suffix ORs)
3288	Length of the Longest Increasing Path Through Two Points	Sort + Binary Search LIS
3289	The Two Sneaky Numbers of Digitville	XOR Math or Frequency Array
3290	Maximum Multiplication Score	1D Rolling DP (4 Stages)
3291	Minimum Number of Valid Strings to Form Target I	Trie-Based Dynamic Programming
3292	Minimum Number of Valid Strings to Form Target II	Aho-Corasick + Greedy Jump / Segment Tree
3293	Calculate Product Final Cost (SQL)	LEFT JOIN + COALESCE (Conditional Logic)
3294	Convert Doubly Linked List to Array II	Two-Pass Linear Traversal (Head-to-Tail)
3295	Report Spam Message	HashSet Lookup
3296	Minimum Number of Seconds to Make Mountain Height Zero	Binary Search on Answer / Priority Queue
3297	Count Substrings That Can Be Rearranged to Contain a String I	Sliding Window (Two-Pointers)
3298	Count Substrings That Can Be Rearranged to Contain a String II	Optimized Sliding Window (Variable Frequency)
3299	Sum of Consecutive Subsequences	Dynamic Programming (DP with Hash Table)
3300	Minimum Element After Replacement With Digit Sum	Arithmetic Digit Summation
3301	Maximize the Total Height of Unique Towers	Greedy (Sort + Boundary Cap)
3302	Find the Lexicographically Smallest Valid Sequence	Greedy + Suffix Matching Precomputation
3303	Find the Occurrence of First Almost Equal Substring	Z-Algorithm (Prefix & Suffix Matching)
3304	Find the K-th Character in String Game I	Bit Manipulation (Popcount) / Math
3305	Count of Substrings Containing Every Vowel and K Consonants I	Sliding Window (atLeast(k) - atLeast(k+1))
3306	Count of Substrings Containing Every Vowel and K Consonants II	Sliding Window: exact(k) = atLeast(k) - atLeast(k+1)
3307	Find the K-th Character in String Game II	Recursion / Divide and Conquer
3308	Find Top Performing Driver (SQL)	SQL JOINs + Window Functions (RANK)
3309	Maximum Possible Number by Binary Concatenation	Custom Greedy Sorting
3310	Remove Methods From Project	Graph Traversal (DFS/BFS) + Reachability
3311	Construct 2D Grid Matching Graph Layout	Graph Degree Analysis + BFS
3312	Sorted GCD Pair Queries	Divisor Counting + Inclusion-Exclusion
3313	Find the Last Marked Nodes in Tree	Tree Diameter Endpoints (2/3 DFS Passes)
3314	Construct the Minimum Bitwise Array I	Bit Manipulation (Flipping Trailing Ones)
3315	Construct the Minimum Bitwise Array II	Bitwise Shift & Subtract
3316	Find Maximum Removals From Source String	2D Dynamic Programming (Source Idx vs Pattern Idx)
3317	Find the Number of Possible Ways for an Event	Math / Combinatorics (Stirling Numbers of the Second Kind)
3318	Find X-Sum of All K-Long Subarrays I	Sliding Window + Ordered Set / Hash Map
3319	K-th Largest Perfect Subtree Size in Binary Tree	Bottom-Up DFS (Post-order Traversal)
3320	Count The Number of Winning Sequences	3D Dynamic Programming (Index, Last Move, Score Diff)
3321	Find X-Sum of All K-Long Subarrays II	Sliding Window + Two Balanced BSTs (Heaps)
3322	Premier League Table Ranking III (SQL)	SQL Window Functions (RANK / DENSE_RANK)
3323	Minimize Connected Groups by Inserting Interval	Merge Intervals + Binary Search
3324	Find the Sequence of Strings Appeared on the Screen	Prefix Expansion / Direct String Construction
3325	Count Substrings With K-Frequency Characters I	Sliding Window (Monotonic Property)
3326	Minimum Division Operations to Make Array Non Decreasing	Greedy Traversal (Right-to-Left) + Smallest Prime Factor (SPF)
3327	Check if DFS Strings Are Palindromes	DFS Tree Flattening + Manacher's Algorithm
3328	Find Cities in Each State II (SQL)	SQL GROUP BY + String Aggregation + HAVING
3329	Count Substrings With K-Frequency Characters II	Sliding Window (Expanding Right, Shrinking Left)
3330	Find the Original Typed String I	Linear Scan / Adjacent Identical Pairs Math
3331	Find Subtree Sizes After Changes	DFS + Ancestor Hash Map
3332	Maximum Points Tourist Can Earn	2D Dynamic Programming
3333	Find the Original Typed String II	DP + Sliding Window (Prefix Sums)
3334	Find the Maximum Factor Score of Array	Prefix & Suffix Arrays
3335	Total Characters in String After Transformations I	Frequency Array / Matrix Exponentiation
3336	Find the Number of Subsequences With Equal GCD	3D Dynamic Programming

3337	Total Characters in String After Transformations II	Matrix Exponentiation
3338	Second Highest Salary II (SQL)	Window Functions (DENSE_RANK)
3339	Find the Number of K-Even Arrays	DP (Index, Remaining K, Last Parity)
3340	Check Balanced String	One-Pass Parity Tally
3341	Find Minimum Time to Reach Last Room I	Dijkstra's Algorithm
3342	Find Minimum Time to Reach Last Room II	Modified Dijkstra (State-Space Expansion)
3343	Count Number of Balanced Permutations	DP + Combinatorics (Digits, Count, Sum)
3344	Maximum Sized Array	Greedy / Set Intersection
3345	Smallest Divisible Digit Product I	Brute Force with Early Exit (Small Constraints)
3346	Maximum Frequency of an Element After Performing Operations I	Sliding Window / Sweep Line
3347	Maximum Frequency of an Element After Performing Operations II	Coordinate Compression / Sweep Line Map
3348	Smallest Divisible Digit Product II	Greedy + Backtracking + Prime Factorization
3349	Adjacent Increasing Subarrays Detection I	One-Pass Precomputation
3350	Adjacent Increasing Subarrays Detection II	Two Pointers / Maximum Overlap
3351	Sum of Good Subsequences	DP with Frequency & Sum Mapping
3352	Count K-Reducible Numbers Less Than N	Digit DP + Precomputed Sieve
3353	Minimum Total Operations	Difference Array / Greedy
3354	Make Array Elements Equal to Zero	Prefix Sum Balance
3355	Zero Array Transformation I	Difference Array (Prefix Sum of Queries)
3356	Zero Array Transformation II	Binary Search on Answer + Difference Array
3357	Minimize the Maximum Adjacent Element Difference	Binary Search on Answer
3358	Books with NULL Ratings	SQL IS NULL
3359	Find Sorted Submatrices With Maximum Element at Most K	CASE WHEN / Aggregate Filtering
3360	Stone Removal Game	Math / Simulation
3361	Shift Distance Between Two Strings	Prefix Sums (Cost Optimization)
3362	Zero Array Transformation III	Greedy + Max-Heap + Difference Array
3363	Find the Maximum Number of Fruits Collected	Dynamic Programming (State Independence)
3364	Minimum Positive Sum Subarray	Sliding Window + Prefix Sums
3365	Rearrange K Subsets to Form Target String	Hash Map (Frequency Counting)
3366	Minimum Array Sum	3D Dynamic Programming
3367	Maximize Sum of Weights after Edge Removals	Tree DP with Greedy Sorting
3368	First Letter Capitalization I (SQL)	SQL String Functions (INITCAP / Regex)
3369	Design an Array Statistics Tracker	Two Heaps (Median) + Hash Map & Treap/SortedSet (Mode)
3370	Smallest Number With All Set Bits	Bit Manipulation (Power of 2 - 1)
3371	Identify the Largest Outlier in an Array	Frequency Hash Map + Sum Equation
3372	Maximize the Number of Target Nodes After Connecting Trees I	Independent Precomputed BFS
3373	Maximize the Number of Target Nodes After Connecting Trees II	Bipartite Coloring (Parity Counting)
3374	First Letter Capitalization II (SQL)	SQL REGEXP_REPLACE / Nested String Functions
3375	Minimum Operations to Make Array Values Equal to K	Set-Based Unique Value Counting
3376	Minimum Time to Break Locks I	Backtracking / Greedy (Small N)
3377	Digit Operations to Make Two Integers Equal	Dijkstra’s Algorithm + Prime Sieve
3378	Count Connected Components in LCM Graph	DSU + Multiples (Sieve-like)
3379	Transformed Array	Modulo Arithmetic (Circular Indexing)
3380	Maximum Area Rectangle With Point Constraints I	Diagonal Iteration + Hash Set
3381	Maximum Subarray Sum With Length Divisible by K	Prefix Sum + Hash Map (Modulo Tracking)
3382	Maximum Area Rectangle With Point Constraints II	Sweep Line + Segment Tree
3383	Minimum Runes to Add to Cast Spell	Graph Reachability / SCC (Tarjan's)
3384	Team Dominance (SQL)	Window Functions (SUM OVER) + CASE
3385	Minimum Time to Break Locks II	Bitmask DP
3386	Button with Longest Push Time	Single-Pass Linear Scan
3387	Maximize Amount After Two Days of Conversions	BFS / Dijkstra (Day-by-Day)
3388	Count Beautiful Splits in an Array	LCP Array (Longest Common Prefix) / Z-Algorithm
3389	Minimum Operations to Make Character Frequencies Equal	Dynamic Programming (Frequency Transitions)
3390	Longest Team Pass Streak	3D Dynamic Programming
3391	Design a 3D Binary Matrix with Efficient Layer Tracking	Greedy / Math (Reverse Operations)
3392	Count Subarrays of Length Three With a Condition	Linear Scan / Fixed Sliding Window
3393	Count Paths with the Given XOR Value	3D Dynamic Programming
3394	Check if Grid can be Cut into Sections	Dimension Reduction + Interval Merging
3395	Subsequences with a Unique Middle Mode I	Combinatorics + Prefix/Suffix Maps
3396	Minimum Number of Operations to Make Elements in Array Distinct	Hash Set (Backward Scan)
3397	Maximum Number of Distinct Elements After Operations	Sorting + Greedy Allocation

3398	Smallest Substring With Identical Characters I	Binary Search on Answer
3399	Smallest Substring With Identical Characters II	Binary Search on Answer + Edge Cases
3400	Maximum Number of Matching Indices After Right Shifts	KMP / String Matching (or O(N^2) if constraints allow)
3401	Find Circular Gift Exchange Chains	Recursive CTE (SQL) / Graph Cycle Detection with Visited Set.
3402	Minimum Operations to Make Columns Strictly Increasing	Greedy Prefix Maximum Tracking (O(N x M)).
3403	Find the Lexicographically Largest String From the Box I	Two Pointers / Suffix Analysis / Greedy Char Selection.
3404	Count Special Subsequences	Hash Map with Math Ratio (GCD) and Middle-element Fixing (O(N^2)).
3405	Count the Number of Arrays with K Matching Adjacent Elements	Combinatorics (nCr) + Fast Power Math. (O(N))
3406	Find the Lexicographically Largest String From the Box II	Largest Suffix Algorithm (Two Pointers) + Substring Truncation. (O(N))
3407	Substring Matching Pattern	Greedy String Matching (Split by wildcard, search Prefix then Suffix). (O(N))
3408	Design Task Manager	Hash Map + Max-Heap/SortedList with Lazy Deletion. (O(log N))
3409	Longest Subsequence With Decreasing Adjacent Difference	2D Dynamic Programming with Suffix Max Optimization (State: dp[num][diff]). (O(N * M))
3410	Maximize Subarray Sum After Removing All Occurrences of One Element	Segment Tree (Point Updates) tracking Max Subarray Sum or DP Prefix mapping. (O(N log N))
3411	Maximum Subarray With Equal Products	Incremental Accumulation + Upper Bound Pruning (O(N^2)).
3412	Find Mirror Score of a String	Array of 26 Stacks / Deques mapping available indices. (O(N))
3413	Maximum Coins From K Consecutive Bags	Sort Intervals + Sliding Window snapped to Interval Boundaries (Two passes: Left/Right). (O(N log N))
3414	Maximum Score of Non-overlapping Intervals	Sort by End Time + Binary Search + 2D Dynamic Programming (dp[i][quota]). (O(N log N))
3415	Find Products with Three Consecutive Digits	Sliding Window / Regex Pattern Matching. O(N)
3416	Subsequences with a Unique Middle Mode II	Advanced Combinatorics + Inclusion-Exclusion
3417	Zigzag Grid Traversal With Skip	Row-based Parity Processing + Array Slicing. O(M · N)
3418	Maximum Amount of Money Robot Can Earn	3D Dynamic Programming / Grid DP (State: row, col, neutralizations_left). O(M · N · K)
3419	Minimize the Maximum Edge Weight of Graph	Binary Search on Answer + Reversed Graph Traversal. O(E log W)
3420	Count Non-Decreasing Subarrays After K Operations	Monotonic Stack + Two Pointers
3421	Find Students Who Improved	Group By / Sort by ID & Date + Linear Scan. O(N log N)
3422	Minimum Operations to Make Subarray Elements Equal	Sliding Window + Two Heaps (Min/Max) to maintain dynamic median. O(N log K)
3423	Maximum Difference Between Adjacent Elements in a Circular Array	Linear loop checking `nums[i]-nums[i-1]` + one explicit edge check for `nums[N-1]-nums[0]`. O(N)
3424	Minimum Cost to Make Arrays Identical	Sort Both Arrays + Sequential Pairing. O(N log N)
3425	Longest Special Path	DFS with Sliding Window on Tree Paths + Hash Map. O(N)
3426	Manhattan Distances of All Arrangements of Pieces	Math (Contribution to Sum) + Combinatorics
3427	Sum of Variable Length Subarrays	Prefix Sum Array. O(N)
3428	Maximum and Minimum Sums of at Most K Subsequences	Sorting + Contribution Technique + Combinatorics. O(N log N)
3429	Paint House IV	Dynamic Programming (State: index, last_color_left, last_color_right). O(N)
3430	Maximum and Minimum Sums of at Most Size K Subarrays	Monotonic Stack + Contribution of Each Element + Math (Counting valid subarrays). O(N)
3431	Minimum Unlocked Indices to Sort Nums	Finding Bound Indices (Last 1, First 3) + Linear locked-state count. O(N)
3432	Count Partitions with Even Sum Difference	Parity Logic (Difference is even iff Total Sum is even). O(N)
3433	Count Mentions Per User	Sorting Events + Lazy Counter for "ALL" mentions. O(M log M + N)
3434	Maximum Frequency After Subarray Operation	Modified Kadane's Algorithm for each distinct value T. O(50 · N)
3435	Frequencies of Shortest Supersequences	DP on Strings / Bitmasking
3436	Find Valid Emails	Regular Expression (Regex) Parsing. O(N)
3437	Permute Adjacent Array Elements	Greedy Swapping / Parity-Based Reordering. O(N)
3438	Find Valid Pair of Adjacent Digits in String	Precomputed Frequency Map + Single Pass. O(N)
3439	Reschedule Meetings for Maximum Free Time I	Sliding Window over Gap Arrays. O(N)
3440	Reschedule Meetings for Max Free Time II	Top-3 Largest Gaps + Prefix/Suffix Max Gaps. O(N)
3441	Minimum Cost Good Caption	DP with Lexicographical Greedy Path. O(26 x N)
3442	Max Diff Between Even and Odd Frequencies I	Extreme Frequency Identification. O(N)
3443	Max Manhattan Distance After K Changes	Greedy Quadrant Invariance. O(N)
3444	Min Increments for Target Multiples	Bitmask DP over Target Array + LCM. O(N x 2^|target|)
3445	Max Difference Between Even and Odd Frequency II	Prefix Parity State + Sliding Window on Pair Combinations (20 pairs). O(20 · N)
3446	Sort Matrix by Diagonals	Diagonal Indexing (i - j) + HashMap/Bucket Sort. O(N²)
3447	Assign Elements to Groups with Constraints	Sieve of Eratosthenes / Harmonic Series Mapping. O(M log M + N)
3448	Count Substrings Divisible By Last Digit	DP with Remainder State for each Divisor (1-9). O(10 · 10 · N)
3449	Maximize the Minimum Game Score	Binary Search on Answer + Greedy Traversal. O(N log(Max_Score))
3450	Maximum Students on a Single Bench	Hash Map of Hash Sets. O(N)
3451	Find Invalid IP Addresses	Regex (Regular Expressions) or SQL Pattern Matching. O(N)
3452	Sum of Good Numbers	Single Pass Bound Check. O(N)
3453	Separate Squares I	Binary Search on Answer (Y-coordinate). O(N log(Range/Epsilon))
3454	Separate Squares II	Sweep Line + Segment Tree (with Lazy Propagation/Coverage Count). O(N log N)
3455	Shortest Matching Substring	KMP (LPS Array) + Three-Pointer Greedy Slide. O(N + M)
3456	Find Special Substring of Length K	Linear Scan with Consecutive Counter. O(N)
3457	Eat Pizzas!	Greedy Selection on Sorted Array. O(N log N)
3458	Select K Disjoint Special Substrings	Interval Consolidation + Greedy Activity Selection. O(N)

3459	Length of Longest V-Shaped Diagonal Segment	DFS + Memoization / DP (State: r, c, dir, turn_taken, next_val). O(N · M)
3460	Longest Common Prefix After at Most One Removal	Two Pointers with "Skip" Flag. O(N)
3461	Check If Digits Are Equal in String After Operations I	Simulation (due to small N=100). O(N^2)
3462	Maximum Sum With at Most K Elements	Row Sorting + Global Priority Queue. O(N · M log M + NM log K)
3463	Check If Digits Are Equal in String After Operations II	Combinatorics (nCr mod10) using Lucas Theorem & CRT. O(N)
3464	Maximize the Distance Between Points on a Square	Perimeter Unwrapping + Binary Search on Answer + Greedy. O(N log N + N log side)
3465	Find Products with Valid Serial Numbers	Regular Expression (Regex) Matching. O(N)
3466	Maximum Coin Collection	Dynamic Programming (State: mile, lane, switchesLeft). O(N)
3467	Transform Array by Parity	Counting & Filling (Stable Parity). O(N)
3468	Find the Number of Copy Arrays	Range Narrowing (Iterative Intersection). O(N)
3469	Find Minimum Cost to Remove Array Elements	DP with State (lastRemainingIndex, nextIndex). O(N^2)
3470	Permutations IV	Combinatorial Counting + Digit-by-Digit Construction. O(n)
3471	Find the Largest Almost Missing Integer	Case-Based Boundary Analysis. O(n)
3472	Longest Palindromic Subsequence After at Most K Operations	3D Dynamic Programming (State: i, j, kRemain). O(n^2 · k)
3473	Sum of K Subarrays With Length at Least M	DP with Prefix Sum Optimization + "Ongoing" State. O(n · k)
3474	Lexicographically Smallest Generated String	Greedy Template Filling + Contradiction Check (KMP/Suffix). O(n · m)
3475	DNA Pattern Recognition	SQL `CASE` statements with `LIKE` and `REGEXP` for vectorized pattern matching. O(N)
3476	Maximize Profit from Task Assignment	Greedy Assignment + Hash Map of Max-Heaps + Special Worker Greedy choice. O((N+M) log M)
3477	Fruits Into Baskets II	Simulation with Leftmost Greedy Search. O(N^2) is acceptable given N=100.
3478	Choose K Elements With Maximum Sum	Sorting + Sliding Window with Min-Heap of size K. O(N log N + N log K)
3479	Fruits Into Baskets III	Segment Tree (Max-Range) + Binary Search for First Index. O(N log N)
3480	Maximize Subarrays After Removing One Conflicting Pair	Two-Pointer / Boundary Tracking (Min and Second Min). O(N + M)
3481	Apply Substitutions	DFS with Memoization / Dependency Graph Resolve. O(Final Length)
3482	Analyze Organization Hierarchy	Recursive CTE (SQL) / DFS with Aggregate Accumulation. O(N)
3483	Unique 3-Digit Even Numbers	Fixed-Space Iteration (100..998) + Frequency Map Tally. O(N + 450)
3484	Design Spreadsheet	Hash Map (Sparse Storage) + Direct Formula Parsing. O(1) set / O(L) get.
3485	Longest Common Prefix of K Strings After Removal	Trie + Dynamic Programming
3486	Longest Special Path II	DFS + Sliding Window on Path + Hash Map for Last Depth. O(N)
3487	Maximum Unique Subarray Sum After Deletion	Greedy (Sum of Unique Positive Numbers). O(N)
3488	Closest Equal Element Queries	Bi-directional Linear Scan with Last-Seen Map. O(N + Q)
3489	Zero Array Transformation IV	Difference Array / Sweep Line
3490	Count Beautiful Numbers	Digit Dynamic Programming (State: index, tight, productMod, sum). O(digits · sum · prod)
3491	Phone Number Prefix	Sorting + Adjacent Comparison OR Trie Insertion. O(N log N · L)
3492	Maximum Containers on a Ship	Direct Math (Division & Min). O(1)
3493	Properties Graph	Set-Based Adjacency List + DFS/BFS for Components. O(N^2 · M) (Optimized constant factor).
3494	Find the Minimum Amount of Time to Brew Potions	Prefix Sums of Skill + Gap Analysis Formula. O(N · M)
3495	Minimum Operations to Make Array Elements Zero	Math + Prefix Sums of "Division Counts". O(Q log(Range)) or O(Q · 1).
3496	Maximize Score After Pair Deletions	Greedy (Complementary Minimum Tracking). O(N)
3497	Analyze Subscription Conversion	SQL Common Table Expression (CTE) with Filtering and Grouping. O(N)
3498	Reverse Degree of a String	Single-Pass Simulation with ASCII Math. O(N)
3499	Maximize Active Section with Trade I	Greedy (Adjacent Zero-Segment Merging). O(N)
3500	Minimum Cost to Divide Array Into Subarrays	DP with Convex Hull Trick (CHT) or prefix sum optimization. O(N^2)
3501	Maximize Active Section with Trade II (Hard)	Line Sweep Algorithm + DP with Segment Tree.
3502	Minimum Cost to Reach Every Position (Easy)	Greedy (Rightward free, Leftward paid).
3503	Longest Palindrome After Substring Concatenation I (Med)	String Hashing + Center Expansion.
3504	Longest Palindrome After Substring Concatenation II (Hard)	Manacher's Algorithm + Suffix Automaton / Hash.
3505	Minimum Operations to Make Elements Within K Subarrays Equal (Hard)	Sliding Window + Running Medians (Two Heaps).
3506	Find Time Required to Eliminate Bacterial Strains (Hard)	Multi-source BFS + Priority Queue (Dijkstra Variant).
3507	Minimum Pair Removal to Sort Array I (Easy)	Dynamic Programming (LIS Variant).
3508	Implement Router (Med)	Trie (Prefix Tree) with Wildcard/Parameter Support.
3509	Max Product of Subsequences With an Alternating Sum Equal to K (Hard)	3D Dynamic Programming (Knapsack with Sign State).
3510	Minimum Pair Removal to Sort Array II (Hard)	Segment Tree or Fenwick Tree (BIT) for O(N log N) LIS.
3511	Make a Positive Array (Med)	Greedy Accumulation / Prefix Sum tracking.
3512	Minimum Operations to Make Array Sum Divisible by K (Easy)	Modular Arithmetic & Hash Map (Prefix Modulos).
3513	Number of Unique XOR Triplets I (Med)	Bitwise Prefix XOR & Frequency Counting.
3514	Number of Unique XOR Triplets II (Med)	Bit-by-Bit Counting / Mathematical Combinatorics.
3515	Shortest Path in a Weighted Tree (Hard)	Lowest Common Ancestor (LCA) with Binary Lifting.
3516	Find Closest Person (Easy)	Binary Search (if sorted) or Single Pass Min-Tracker.
3517	Smallest Palindromic Rearrangement I (Med)	Character Frequency Counting + Greedy Symmetric Placement.
3518	Smallest Palindromic Rearrangement II (Hard)	Two-Pointer Greedy Swapping (Bubble Swaps for target positions).
3519	Count Numbers with Non-Decreasing Digits (Hard)	Digit DP (Dynamic Programming) / Combinatorics.

3520	Minimum Threshold for Inversion Pairs Count (Med)	Binary Search on Answer + Fenwick Tree (BIT) / Merge Sort for Inversions.
3521	Find Product Recommendation Pairs (Med)	Graph Traversal (BFS) / Collaborative Filtering with Inverted Index.
3522	Calculate Score After Performing Instructions (Med)	Prefix Sums on 2D Coordinates / Hash Map caching.
3523	Make Array Non-decreasing (Med)	Greedy Single-Pass / Max-Tracker.
3524	Find X Value of Array I (Med)	Bitwise Evaluation / Prefix Counts.
3525	Find X Value of Array II (Hard)	Binary Search on Answer + Sliding Window / Segment Tree.
3526	Range XOR Queries with Subarray Reversals (Hard)	Implicit Treap (Randomized Binary Search Tree) / Splay Tree with Lazy Propagation.
3527	Find the Most Common Response (Med)	Hash Map Counting + Single-Pass Max Tracking.
3528	Unit Conversion I (Med)	Floyd-Warshall (Precomputation) or Weighted Disjoint Set Union (DSU).
3529	Count Cells in Overlapping Horizontal and Vertical Substrings (Med)	Line Sweep / Prefix Sums / 2D Difference Arrays.
3530	Maximum Profit from Valid Topological Order in DAG (Hard)	Dynamic Programming with Bitmask (State Compression).
3531	Count Covered Buildings (Med)	Difference Array / Line Sweep.
3532	Path Existence Queries in a Graph I (Med)	Disjoint Set Union (DSU) / Connected Components.
3533	Concatenated Divisibility (Hard)	Modular Arithmetic + Hash Map (Digit Length Bins).
3534	Path Existence Queries in a Graph II (Hard)	Offline Queries + DSU with Sorting.
3535	Unit Conversion II (Med)	Graph Precomputation (Anchored Weights) / Component Ratios.
3536	Maximum Product of Two Digits (Easy)	Single Pass Max/Min Tracker (Greedy).
3537	Fill a Special Grid (Med)	Constructive Algorithm / Greedy Pattern Filling.
3538	Merge Operations for Minimum Travel Time (Hard)	Disjoint Set Union (DSU) / Kruskal's Variant with DP.
3539	Find Sum of Array Product of Magical Sequences (Hard)	Dynamic Programming with Mathematical Contribution (Prefix Arrays).
3540	Minimum Time to Visit All Houses (Med)	Sorting / 1D Greedy Sweep (Assuming 1D coordinate plane).
3541	Find Most Frequent Vowel and Consonant (Easy)	Single Pass with Hash Map/Frequency Array.
3542	Min Ops to Convert All Elements to Zero (Med)	Greedy Selection + Difference Array / Sorting.
3543	Maximum Weighted K-Edge Path (Med)	Dynamic Programming (Bellman-Ford variant).
3544	Subtree Inversion Sum (Hard)	DFS + Fenwick Tree (BIT) / Merge Sort Tree.
3545	Min Deletions for At Most K Distinct Characters (Easy)	Sliding Window (Two Pointers).
3546	Equal Sum Grid Partition I (Med)	2D Prefix Sums (Inclusion-Exclusion).
3547	Maximum Sum of Edge Values in a Graph (Hard)	Greedy (Kruskal's) or Tree DP (if constrained).
3548	Equal Sum Grid Partition II (Hard)	2D Prefix Sums + Hash Map / Binning.
3549	Multiply Two Polynomials (Hard)	Fast Fourier Transform (FFT) or Karatsuba.
3550	Smallest Index With Digit Sum Equal to Index (Easy)	Linear Scan with Mathematical Digit Extraction.
3551	Minimum Swaps to Sort by Digit Sum (Med)	Cycle Decomposition (Permutation Graph).
3552	Grid Teleportation Traversal (Med)	Multi-Source BFS / Dijkstra with Coordinate Mapping.
3553	Min Weighted Subgraph With Required Paths II (Hard)	Dijkstra's x3 (Steiner Tree on 3 Nodes variant).
3554	Find Category Recommendation Pairs (Hard)	Inverted Index + Bitsets / Matrix Multiplication.
3555	Smallest Subarray to Sort in Every Sliding Window (Med)	Monotonic Deque / Two Pointers + Sliding Window Minimum.
3556	Sum of Largest Prime Substrings (Med)	Sieve of Eratosthenes + Rolling Hash / DP.
3557	Find Max Number of Non Intersecting Substrings (Med)	Greedy Interval Scheduling (Earliest Deadline First).
3558	Number of Ways to Assign Edge Weights I (Med)	Dynamic Programming on Trees (Tree DP).
3559	Number of Ways to Assign Edge Weights II (Hard)	DP + Generating Functions / Polynomial Interpolation or NTT.
3560	Find Minimum Log Transportation Cost (Easy)	Greedy / Single-Source Shortest Path (Dijkstra) or Prefix Min.
3561	Resulting String After Adjacent Removals (Med)	Stack-based Single Pass.
3562	Max Profit from Trading Stocks with Discounts (Hard)	3D Dynamic Programming.
3563	Lexicographically Smallest String After Adjacent Removals (Hard)	Greedy + Monotonic Stack + Frequency Counting.
3564	Seasonal Sales Analysis (Med)	Hash Map Aggregation / SQL-style GroupBy.
3565	Sequential Grid Path Cover (Med)	BFS / Dijkstra on State Space (Grid + Bitmask).
3566	Partition Array into Two Equal Product Subsets (Med)	Dynamic Programming (Knapsack on Log-Sums) / Prime Factorization.
3567	Minimum Absolute Difference in Sliding Submatrix (Med)	Sliding Window + 2D Data Structure (Segment Tree/Fenwick Tree) or Sorted Set.
3568	Minimum Moves to Clean the Classroom (Med)	BFS / Dijkstra on Grid State.
3569	Maximize Count of Distinct Primes After Split (Hard)	Sieve of Eratosthenes + Prefix Prime Frequencies.
3570	Find Books with No Available Copies (Easy)	Hash Map (Inventory Counter).
3571	Find the Shortest Superstring II (Easy)	Greedy Overlap Selection.
3572	Maximize Y-Sum by Picking a Triplet of Distinct X-Values (Med)	Sorting + Greedy / Two-Pass Max Search.
3573	Best Time to Buy and Sell Stock V (Med)	One-Pass Greedy (Peak-Valley).
3574	Maximize Subarray GCD Score (Hard)	Sparse Table + Decreasing GCD Property.
3575	Maximum Good Subtree Score (Hard)	Tree Dynamic Programming (Bottom-Up).
3576	Transform Array to All Equal Elements (Med)	Median-based Greedy.
3577	Count the Number of Computer Unlocking Permutations (Med)	Dynamic Programming with Bitmask.
3578	Count Partitions With Max-Min Difference at Most K (Med)	DP + Monotonic Deque / Sliding Window.
3579	Minimum Steps to Convert String with Operations (Hard)	Dijkstra / BFS + A* Search or Dynamic Programming.

3580	Find Consistently Improving Employees (Med)	Hash Map Grouping + Single Pass Validation.
3581	Count Odd Letters from Number (Easy)	Mathematical Digit Extraction.
3582	Generate Tag for Video Caption (Easy)	Hash Set Lookup / Single Pass Tokenization.
3583	Count Special Triplets (Med)	Prefix Sums / Frequency Mapping.
3584	Max Product of First and Last Elements of a Subsequence (Med)	Two-Pointer Greedy / Sorting.
3585	Find Weighted Median Node in Tree (Hard)	Tree Centroid Decomposition / Subtree Weight Sums.
3586	Find COVID Recovery Patients (Med)	Hash Map Grouping + Sort & Sweep.
3587	Minimum Adjacent Swaps to Alternate Parity (Med)	Greedy Positioning (Two-Target Comparison).
3588	Find Maximum Area of a Triangle (Med)	Convex Hull + Rotating Calipers.
3589	Count Prime-Gap Balanced Subarrays (Med)	Sieve + Sliding Window / Prefix Sums.
3590	Kth Smallest Path XOR Sum (Hard)	Meet-in-the-Middle + Binary Search / Trie.
3591	Check if Any Element Has Prime Frequency (Easy)	Hash Map + Sieve of Eratosthenes.
3592	Inverse Coin Change (Med)	Dynamic Programming (Generating Functions Reconstruction).
3593	Minimum Increments to Equalize Leaf Paths (Med)	Bottom-Up Tree DP (Post-order Traversal).
3594	Minimum Time to Transport All Individuals (Hard)	Binary Search on Answer + Greedy/Flow.
3595	Once Twice (Med)	Sliding Window + Frequency Map / Bitmask.
3596	Minimum Cost Path with Alternating Directions I (Med)	State-Layered Dijkstra (BFS with 2-Layer States).
3597	Partition String (Med)	Greedy Single Pass with Hash Set.
3598	Longest Common Prefix Between Adjacent Strings After Removals (Med)	Dynamic Programming / Monotonic Stack.
3599	Partition Array to Minimize XOR (Med)	Prefix XOR + Target Search.
3600	Maximize Spanning Tree Stability with Upgrades (Hard)	Prim's/Kruskal's + Greedy / DP with Knapsack.
3601	Find Drivers with Improved Fuel Efficiency (Med)	Common Table Expressions (CTEs) + GROUP BY aggregation + Self JOIN.
3602	Hexadecimal and Hexatrigesimal Conversion (Easy)	Mathematical Base Conversion (N % K, N // K) using a StringBuilder/Byte Array, followed by a single Reverse operation.
3603	Minimum Cost Path with Alternating Directions II (Med)	State-Space Dijkstra's Algorithm. State = (Node, Arriving_Direction).
3604	Minimum Time to Reach Destination in Directed Graph (Med)	Standard Dijkstra's Algorithm using an Adjacency List and a Min-Heap (Priority Queue).
3605	Minimum Stability Factor of Array (Hard)	Sliding Window with Monotonic Deque OR Segment Tree (depending on exact stability constraint).
3606	Coupon Code Validator (Easy)	Single-pass state machine or boolean flag tracking within a single iteration.
3607	Power Grid Maintenance (Med)	Kruskal's Algorithm with Disjoint Set Union (DSU) or Tarjan's Bridge Finding Algorithm.
3608	Minimum Time for K Connected Components (Med)	Sort edges by time, use Disjoint Set Union (DSU). Start with C=V components. Decrement C for every successful union. Stop whe
3609	Minimum Moves to Reach Target in Grid (Hard)	0-1 BFS (Deque) or Dijkstra's Algorithm with a 3D State Space `(row, col, direction/momentum)`.
3610	Minimum Number of Primes to Sum to Target (Med)	Sieve of Eratosthenes (precompute primes) + Unbounded Knapsack Dynamic Programming (Coin Change variant).
3611	Find Overbooked Employees (Med)	Common Table Expressions (CTEs) to group by employee_id, year, week , aggregate sums, filter > 20 , then self-join.
3612	Process String with Special Operations I (Med)	Single-pass Simulation using a Stack or a dynamically sized Array/StringBuilder.
3613	Minimize Maximum Component Cost (Med)	Binary Search on Answer + Disjoint Set Union (DSU) / Connected Components Check.
3614	Process String with Special Operations II (Hard)	Difference Array (Sweep Line) or Fenwick Tree (Binary Indexed Tree) to mathematically track shifts.
3615	Longest Palindromic Path in Graph (Hard)	Bi-directional Dynamic Programming (Meet-in-the-Middle) / Graph BFS. State = (u, v) .
3616	Number of Student Replacements (Med)	Frequency Maps / Discrepancy Counting. (Compare counts of what you have vs. what you need).
3617	Find Students with Study Spiral Pattern (Hard)	2D Prefix Sums + Binary Search for Radius, treating the "spiral rings" as concentric square boundaries.
3618	Split Array by Prime Indices (Med)	Sieve of Eratosthenes (precompute primes up to N) + Single Pass Prefix/Suffix sums.
3619	Count Islands With Total Value Divisible by K (Med)	Standard BFS/DFS Connected Components. Accumulate a `running_sum` during the traversal. Modulo K at the very end of the is
3620	Network Recovery Pathways (Hard)	Dijkstra's Algorithm with Bitmask DP (State Space Search). State = (Current_Node, Visited_Mask) .
3621	Number of Integers With Popcount-Depth Equal to K I (Hard)	Digit DP / Combinatorics. Precompute valid target "set bit counts", then build valid numbers bit by bit using combinations (nCr).
3622	Check Divisibility by Digit Sum and Product (Easy)	Early Short-Circuit Simulation. If any digit is '0', the product becomes 0 instantly (cannot divide by zero), so return false immedia
3623	Count Number of Trapezoids I (Med)	Group line segments by slope using a Hash Map. A trapezoid is formed by picking 2 segments with the same slope (bases) and en
3624	Number of Integers With Popcount-Depth Equal to K II (Hard)	Advanced Digit DP + Precomputed Combinatorics Modulo 10^9+7. Find valid popcounts up to Length L, then use nCr to place bits
3625	Count Number of Trapezoids II (Hard)	Inclusion-Exclusion Principle. (Total Trapezoids = All shapes with at least 1 pair of parallel sides MINUS Parallelograms which have
3626	Find Stores with Inventory Imbalance (Med)	Hash Map Aggregation / Group By (if SQL). Accumulate counts in a single pass, then compare against the precomputed average.
3627	Maximum Median Sum of Subsequences of Size 3 (Med)	Greedy Algorithm + Sorting. Sort the array. To maximize medians, pair the largest available, the *next* largest (the median), and
3628	Maximum Number of Subsequences After One Inserting (Med)	Prefix and Suffix Counts. Count existing '0's and '1's. Inserting '0' at the start pairs with all existing '1's. Inserting '1' at the end p
3629	Minimum Jumps to Reach End via Prime Teleportation (Med)	Sieve of Eratosthenes + BFS with a "furthest reached" pointer to strictly avoid re-scanning identical prime gaps.
3630	Partition Array for Maximum XOR and AND (Hard)	Greedy Bit-by-Bit Construction (from MSB to LSB). Check if a partition can satisfy the current bit mask using a linear scan.
3631	Sort Threats by Severity and Exploitability (Med)	Built-in O(N log N) Sort with a strictly defined Custom Comparator function returning -1, 0, or 1 based on a prioritized rule cascad
3632	Subarrays with XOR at Least K (Hard)	Prefix XOR Array combined with a Binary Trie (Prefix Tree) to count valid pairs in O(N log(Max_Value)) time.
3633	Earliest Finish Time for Land and Water Rides I (Easy)	Greedy Algorithm / Sorting. Match the longest land rides with the shortest water rides (or vice versa, depending on exact problem
3634	Minimum Removals to Balance Array (Med)	Prefix and Suffix Sums. Precalculate even/odd sums. When you "remove" an element, the suffix even/odd sums simply swap roles
3635	Earliest Finish Time for Land and Water Rides II (Med)	Binary Search on the Answer. Guess a "Finish Time" T. Write a greedy O(N) checker to see if all rides can be completed within time
3636	Threshold Majority Queries (Hard)	Segment Tree with Boyer-Moore Voting or Random Sampling + Range Binary Search (using a map of indices).
3637	Trionic Array I (Easy)	Two-pointer approach or Hash Map tracking based on the property target - (arr[i] + arr[j]) .
3638	Maximum Balanced Shipments (Med)	Dynamic Programming with a Fenwick Tree or Segment Tree to find the maximum shipments ending at index `i`.
3639	Minimum Time to Activate String (Med)	Greedy algorithm with a Queue or Monotonic Stack, focusing on the nearest available character that satisfies the activation rule.
3640	Trionic Array II (Hard)	Segment Tree or Square Root Decomposition where each "node" or "bucket" stores partial Trionic counts.

3641	Longest Semi-Repeating Subarray (Med)	Sliding Window with a single counter for "duplicate pairs".
3642	Find Books with Polarized Opinions (Med)	Hash Map Aggregation / SQL Group By with HAVING clause. Compute statistics for all books in one pass.
3643	Flip Square Submatrix Vertically (Easy)	In-place row swapping within the specified submatrix boundaries.
3644	Maximum K to Sort a Permutation (Med)	Greatest Common Divisor (GCD) of the displacements. Find the distance each element needs to move and find the GCD of all thos
3645	Maximum Total from Optimal Activation Order (Med)	Greedy Algorithm with a Max-Priority Queue. At any step, activate the item that provides the highest immediate gain/value based
3646	Next Special Palindrome Number (Hard)	Mirroring + Middle-Digit Increment. Construct the palindrome from the first half of N, then adjust based on "Special" digit constra
3647	Maximum Weight in Two Bags (Med)	Dynamic Programming (Knapsack variant) or Meet-in-the-Middle if N is around 30-40.
3648	Minimum Sensors to Cover Grid (Med)	Greedy with Set Cover or Bitmask DP (if dimensions are small). Focus on covering the "bottleneck" (uncovered cells with the fewe
3649	Number of Perfect Pairs (Med)	Algebraic Simplification + Binary Search / Fenwick Tree. Re-write the property as a range: arr[j] must be in [arr[i]/2, 2*arr[i]] .
3650	Minimum Cost Path with Edge Reversals (Med)	0-1 BFS or Dijkstra's Algorithm on a modified graph where every directed edge (u -> v) with weight 0 has a "ghost" edge (v -> u
3651	Minimum Cost Path with Teleportations (Hard)	Dijkstra + Auxiliary "Teleport Hub" nodes. Instead of V x V edges, connect all nodes to one "Hub" with specific entry/exit costs.
3652	Best Time to Buy and Sell Stock using Strategy (Med)	Dynamic Programming / State Machine. State = `(Day, HasStock)`. Use a simple greedy one-pass if unlimited trades are allowed
3653	XOR After Range Multiplication Queries I (Med)	Segment Tree or Fenwick Tree with Lazy Propagation. Store XOR values in nodes and apply multiplications as "Lazy" updates.
3654	Minimum Sum After Divisible Sum Deletions (Med)	Dynamic Programming (Remainder DP). State = `dp[i][remainder]`. Find the maximum sum we can delete that leaves a remainc
3655	XOR After Range Multiplication Queries II (Hard)	Segment Tree where each node stores a "Bit Count" array (size 32 or 64). Multiplication by a power of 2 becomes a "Bit Shift" on
3656	Determine if a Simple Graph Exists (Med)	Havel-Hakimi Algorithm. Sort the degree sequence, then repeatedly "consume" the highest degree by connecting it to the next hi
3657	Find Loyal Customers (Med)	Hash Set Aggregation. For each customer, collect `transaction_month` into a Set. Filter customers where `Set.size == 12`.
3658	GCD of Odd and Even Sums (Easy)	Euclidean Algorithm for GCD. `gcd(a, b)` where you repeatedly take the remainder until one reaches zero.
3659	Partition Array Into K-Distinct Groups (Med)	Greedy Frequency Distribution. Sort elements by frequency. Place the most frequent elements into different groups first.
3660	Jump Game IX (Med)	Monotonic Stack to find the next/previous greater/smaller elements in O(N), then Dijkstra/DP on those specific edges.
3661	Maximum Walls Destroyed by Robots (Hard)	Maximum Flow / Bipartite Matching. Model robots as one set of nodes, walls as another, and find the maximum matching/coverag
3662	Filter Characters by Frequency (Easy)	Single-pass Hash Map / Frequency Array. Count all frequencies first, then perform a single pass to filter.
3663	Find The Least Frequent Digit (Easy)	Modulo-based Frequency Array. Use `n % 10` to strip digits and an array of size 10 to store counts.
3664	Two-Letter Card Game (Med)	Greedy Sorting / Hash Map counting. Pair cards based on their first and second letters to maximize your "counter-plays".
3665	Twisted Mirror Path Count (Med)	Dynamic Programming / BFS with State. State = `(row, col, direction)`. Since the path is "twisted", we track our entering vector.
3666	Minimum Operations to Equalize Binary String (Hard)	Greedy "Flip-and-Carry" or Difference Array. Start from the left; if the current bit doesn't match the target, flip the remainder of t
3667	Sort Array By Absolute Value (Easy)	Language-native sort with a Custom Key/Comparator.
3668	Restore Finishing Order (Easy)	Hash Map (ID to Rank) or Tuple Sorting. Store (Time, ID) pairs and sort them directly.
3669	Balanced K-Factor Decomposition (Med)	Square Root Factorization + Dynamic Programming. Only check up to √N to find all factors, then use DP to find balanced subsets.
3670	Maximum Product of Two Integers With No Common Bits (Med)	Hash Map of Bitmasks. For each unique bitmask, store the maximum integer found. Then compare all pairs of bitmasks.
3671	Sum of Beautiful Subsequences (Hard)	Dynamic Programming with Counting (Combinatorics). Group elements by their remainder when divided by the "beauty" differenc
3672	Sum of Weighted Modes in Subarrays (Med)	Frequency Tracking with Contribution Counting. For each element, mathematically calculate how many subarrays it serves as the
3673	Find Zombie Sessions (Hard)	Disjoint Set Union (DSU) with Path Compression. For every session connection, union the two nodes and decrement the total clus
3674	Minimum Operations to Equalize Array (Easy)	Median-Based Greedy approach. The total cost is minimized when all elements are changed to the median of the array.
3675	Minimum Operations to Transform String (Med)	Two-Pointer Greedy Scan or Edit Distance variant. If only specific operations are allowed (like moving to front), focus on the "Long
3676	Count Bowl Subarrays (Med)	Precompute "Longest Decreasing" from left and "Longest Increasing" from right for every index.
3677	Count Binary Palindromic Numbers (Hard)	Combinatorial Construction. Generate palindromes directly by mirroring the "first half" of the binary bits.
3678	Smallest Absent Positive Greater Than Average (Easy)	Calculate average, throw array into a Hash Set, and start checking from ceil(avg) + 1 .
3679	Minimum Discards to Balance Inventory (Med)	Greedy with Frequency Maps or Dynamic Programming. Count the excess of each item type and discard only the necessary surplu
3680	Generate Schedule (Med)	Greedy Strategy: Earliest Deadline First (EDF) or Shortest Processing Time (SPT).
3681	Maximum XOR of Subsequences (Hard)	Linear Basis (Gaussian Elimination for XOR). Find a set of basis vectors that can represent any possible XOR sum of the array.
3682	Minimum Index Sum of Common Elements (Med)	Hash Map for List A (Value -> Index). Scan List B and look up values in the Map in O(1).
3683	Earliest Time to Finish One Task (Easy)	Precompute dependencies and apply a Greedy "Earliest Start" calculation.
3684	Maximize Sum of At Most K Distinct Elements (Easy)	Greedy Sorting. Sort the array descending and pick the largest values until you hit the K distinct limit.
3685	Subsequence Sum After Capping Elements (Med)	Dynamic Programming (Knapsack variant) or Combinatorial Math. Since capping is a uniform rule, use the frequency of elements
3686	Number of Stable Subsequences (Hard)	Dynamic Programming with Frequency Maps or Combinatorics. Calculate combinations of each element frequency F independently
3687	Library Late Fee Calculator (Easy)	Datetime library or Formula-based difference. `(Date2 - Date1).days * rate`.
3688	Bitwise OR of Even Numbers in an Array (Easy)	Single-pass iteration with a running bitwise OR variable and a parity check.
3689	Maximum Total Subarray Value I (Med)	Kadane's Algorithm or Sliding Window with Prefix Sums. If Value = Sum, it's strictly Kadane's.
3690	Split and Merge Array Transformation (Med)	Dynamic Programming (Interval DP). `dp[i][j]` stores whether the subarray from i to j can be transformed into the target.
3691	Maximum Total Subarray Value II (Hard)	Divide and Conquer or Segment Tree with "Monotonic Property" optimization. Focus on boundaries where the XOR value changes
3692	Majority Frequency Characters (Easy)	Single-pass Frequency Array (size 26) + Max Frequency tracking.
3693	Climbing Stairs II (Med)	Iterative Dynamic Programming (Bottom-Up). `dp[i] = dp[i-1] + dp[i-2] + dp[i-3]`.
3694	Distinct Points Reachable After Substring Removal (Med)	Prefix Sums of Coordinates. Calculate (x, y) for all prefixes and suffixes. Endpoint = Prefix[i-1] + (Total - Prefix[j]) .
3695	Maximize Alternating Sum Using Swaps (Hard)	Greedy Analysis of Parity. An alternating sum is (EvenSum - OddSum) . Swapping only helps if you move a large number from an
3696	Maximum Distance Between Unequal Words I (Easy)	Single-pass: Find the first and last occurrence of different words. The max distance is either LastIndex - FirstIndex or LastIndex -
3697	Compute Decimal Representation (Easy)	Horner's Method for Polynomial Evaluation. Process the string character by character, multiplying the current result by the base ar
3698	Split Array With Minimum Difference (Med)	Prefix Sums. Calculate all cumulative sums once. Sum of right half = TotalSum - PrefixSum[i] .
3699	Number of ZigZag Arrays I (Hard)	Dynamic Programming (DP). `dp[i][direction]` = number of zigzags ending at index i going "up" or "down".
3700	Number of ZigZag Arrays II (Hard)	DP + Segment Tree or Fenwick Tree. Use a coordinate-compressed Segment Tree to query the sum of `Up/Down` counts for all v
3701	Compute Alternating Sum (Easy)	Single-Pass Accumulator / State Machine

3702	Longest Subsequence With Non-Zero Bitwise XOR (Med)	Prefix XOR / Greedy Mathematical Observation
3703	Remove K-Balanced Substrings (Med)	Stack / Amortized Sliding Window
3704	Count No-Zero Pairs That Sum to N (Hard)	Two Pointers / Single-Pass String Check
3705	Find Golden Hour Customers (Med)	Sorting + Sliding Window (Queue)
3706	Maximum Distance Between Unequal Words in Array II (Med)	Two Pointers / Hash Map (First & Last Occurrence)
3707	Equal Score Substrings (Easy)	Sliding Window / Prefix Sum Array
3708	Longest Fibonacci Subarray (Med)	Dynamic Programming / Hash Map
3709	Design Exam Scores Tracker (Med)	Segment Tree / Balanced BST (TreeMap)
3710	Maximum Partition Factor (Hard)	Dynamic Programming with Memoization
3711	Maximum Transactions Without Negative Balance (Med)	Greedy + Priority Queue (Min-Heap of negative values)
3712	Sum of Elements With Frequency Divisible by K (Easy)	Hash Map Frequency Counting
3713	Longest Balanced Substring I (Med)	Prefix Sum + Hash Map (First Occurrence)
3714	Longest Balanced Substring II (Med)	Bitmasking + Hash Map / Advanced State Tracking
3715	Sum of Perfect Square Ancestors (Hard)	DFS with Running Sum/State
3716	Find Churn Risk Customers (Med)	Hash Map Aggregation / Sorting by User
3717	Minimum Operations to Make the Array Beautiful (Med)	Greedy Virtual Deletion (Two Pointers)
3718	Smallest Missing Multiple of K (Easy)	Hash Set + While Loop
3719	Longest Balanced Subarray I (Med)	Prefix Sum Difference + Hash Map
3720	Lexicographically Smallest Permutation Greater Than Target (Med)	Next Permutation Algorithm (Suffix Reversal)
3721	Longest Balanced Subarray II (Hard)	State Encoding (Bitmask) + Hash Map
3722	Lexicographically Smallest String After Reverse (Med)	Greedy Traversal / Peak Finder
3723	Maximize Sum of Squares of Digits (Med)	Greedy Digit Allocation
3724	Minimum Operations to Transform Array (Med)	Backwards Traversal (Target to Start)
3725	Count Ways to Choose Coprime Integers from Rows (Hard)	DP + Prime Factor Bitmasking
3726	Remove Zeros in Decimal Representation (Easy)	Mathematical Digit Extraction (Modulo & Division)
3727	Maximum Alternating Sum of Squares (Med)	Dynamic Programming (State Machine)
3728	Stable Subarrays With Equal Boundary and Interior Sum (Med)	Prefix Sums + Hash Map
3729	Count Distinct Subarrays Divisible by K in Sorted Array (Hard)	Prefix Sum Modulo K + Trie/String Hashing
3730	Maximum Calories Burnt from Jumps (Med)	1D Dynamic Programming
3731	Find Missing Elements (Easy)	Hash Set or Boolean Array Check
3732	Maximum Product of Three Elements After One Replacement (Med)	Sorting + Greedy Extremes
3733	Minimum Time to Complete All Deliveries (Med)	Binary Search on Answer + Validation (Greedy/Bipartite)
3734	Lexicographically Smallest Palindromic Permutation Greater Than Target (Hard)	Next Permutation on Half-String
3735	Lexicographically Smallest String After Reverse II (Hard)	Greedy Left-to-Right Minimization / Two Pointers
3736	Minimum Moves to Equal Array Elements III (Easy)	Sorting & Median Finding
3737	Count Subarrays With Majority Element I (Med)	Prefix Sums + Relative State Tracking (+1/-1)
3738	Longest Non-Decreasing Subarray After Replacing at Most One Element (Med)	Dynamic Programming (Left & Right Prefix/Suffix Arrays)
3739	Count Subarrays With Majority Element II (Hard)	Prefix Sum + Binary Indexed Tree (Fenwick)
3740	Minimum Distance Between Three Equal Elements I (Easy)	Hash Map with Index Lists / Sliding Window of Size 3
3741	Minimum Distance Between Three Equal Elements II (Med)	Hash Map with Size-3 Deque (Sliding Window of Indices)
3742	Maximum Path Score in a Grid (Med)	2D Dynamic Programming
3743	Maximize Cyclic Partition Score (Hard)	Double Array Concatenation + Monotonic Queue/DP
3744	Find Kth Character in Expanded String (Med)	Reverse Recursive Math / Divide and Conquer
3745	Maximize Expression of Three Elements (Easy)	Prefix Max / Suffix Max Arrays
3746	Minimum String Length After Balanced Removals (Med)	Stack / Amortized Character Matching
3747	Count Distinct Integers After Removing Zeros (Med)	Mathematical Digit Extraction + Hash Set
3748	Count Stable Subarrays (Hard)	Sliding Window + Monotonic Deques (Min/Max Tracking)
3749	Evaluate Valid Expressions (Hard)	Shunting Yard Algorithm / Two Stacks (Operands & Operators)
3750	Minimum Number of Flips to Reverse Binary String (Easy)	State Transition Counting (Adjacent Differences)
3751	Total Waviness of Numbers in Range I (Med)	Digit DP (Dynamic Programming)
3752	Lexicographically Smallest Negated Permutation that Sums to Target (Med)	Greedy Selection with Math Constraints
3753	Total Waviness of Numbers in Range II (Hard)	Advanced Digit DP with Modulo/State constraints
3754	Concatenate Non-Zero Digits and Multiply by Sum I (Easy)	Single-Pass Mathematical Extraction
3755	Find Maximum Balanced XOR Subarray Length (Med)	Prefix XOR + Hash Map (First Occurrence)
3756	Concatenate Non-Zero Digits and Multiply by Sum II (Med)	Digit DP (Combinatorial Math)
3757	Number of Effective Subsequences (Hard)	Sorting + Dynamic Programming / Two Pointers
3758	Convert Number Words to Digits (Med)	Stack-Based Parsing / Grammar state machine
3759	Count Elements With at Least K Greater Values (Med)	Sorting / Frequency Map + Suffix Sum
3760	Maximum Substrings With Distinct Start (Med)	Greedy Selection + Boolean Array/Set
3761	Minimum Absolute Distance Between Mirror Pairs (Med)	Greedy / Sorting Mirror Elements

3762	Minimum Operations to Equalize Subarrays (Hard)	DP + Median Properties / Slopes
3763	Maximum Total Sum With Threshold Constraints	Greedy / Sorting
3764	Most Common Course Pairs (Hard)	Hash Map with Tuple Keys / Bitsets
3765	Complete Prime Number (Med)	Sieve of Eratosthenes + Digit Mask
3766	Minimum Operations to Make Binary Palindrome (Med)	Two Pointers / Mirror Matching
3767	Maximize Points After Choosing K Tasks (Med)	Greedy with Priority Queue (Max-Heap)
3768	Minimum Inversion Count in Subarrays of Fixed Length (Hard)	Sliding Window + Fenwick Tree / Segment Tree
3769	Sort Integers by Binary Reflection (Easy)	Custom Comparator with Bitwise Manipulation
3770	Largest Prime from Consecutive Prime Sum (Med)	Sieve of Eratosthenes + Sliding Window / Prefix Sums
3771	Total Score of Dungeon Runs (Med)	2D Dynamic Programming (Path Counting)
3772	Maximum Subgraph Score in a Tree (Hard)	Tree Dynamic Programming (Kadane's on Trees)
3773	Maximum Number of Equal Length Runs (Med)	Hash Map + Grouping (Run-Length Encoding)
3774	Absolute Difference Between Maximum and Minimum K Elements (Easy)	Sorting + Sliding Window (Fixed Size K)
3775	Reverse Words With Same Vowel Count (Med)	Grouping by Vowel Count + Reverse Pointers
3776	Minimum Moves to Balance Circular Array (Med)	Prefix Sums on Doubled Array + Median of Differences
3777	Minimum Deletions to Make Alternating Substring (Hard)	Greedy Single Pass / State Machine
3778	Minimum Distance Excluding One Maximum Weighted Edge (Med)	Modified Dijkstra / BFS with State
3779	Minimum Number of Operations to Have Distinct Elements (Med)	Reverse Scan + Hash Set
3780	Maximum Sum of Three Numbers Divisible by Three (Med)	Greedy / Bucketing by Remainder
3781	Maximum Score After Binary Swaps (Med)	Greedy Counting / Frequency Analysis
3782	Last Remaining Integer After Alternating Deletion Operations (Hard)	Mathematical Simulation (Step & Head Tracking)
3783	Mirror Distance of an Integer (Easy)	Mathematical Remainder Reversal
3784	Minimum Deletion Cost to Make All Characters Equal (Med)	Total Cost - Max Character Frequency Cost
3785	Minimum Swaps to Avoid Forbidden Values (Hard)	Min-Cost Flow / Hungarian Algorithm
3786	Total Sum of Interaction Cost in Tree Groups (Hard)	Tree DP + Centroid Decomposition / Contribution of Edges
3787	Find Diameter Endpoints of a Tree (Med)	Double BFS / DFS
3788	Maximum Score of a Split (Med)	Prefix and Suffix Accumulators
3789	Minimum Cost to Acquire Required Items (Med)	Bitmask Dynamic Programming (Knapsack Variant)
3790	Smallest All-Ones Multiple (Med)	Pigeonhole Principle + Iterative Modulo
3791	Number of Balanced Integers in a Range (Hard)	Digit Dynamic Programming (Digit DP)
3792	Sum of Increasing Product Blocks (Med)	Linear DP + Monotonic Optimization
3793	Find Users with High Token Usage (Easy)	Hash Map Aggregation
3794	Reverse String Prefix (Easy)	Two Pointers (In-place Swap)
3795	Minimum Subarray Length With Distinct Sum At Least K (Med)	Sliding Window + Frequency Map
3796	Find Maximum Value in a Constrained Sequence (Med)	Dynamic Programming / Greedy Range Extremes
3797	Count Routes to Climb a Rectangular Grid (Hard)	2D Dynamic Programming (Layered)
3798	Largest Even Number (Easy)	Greedy Sorting + Last Digit Swap
3799	Word Squares II (Med)	Trie-Based Backtracking
3800	Minimum Cost to Make Two Binary Strings Equal (Med)	Dynamic Programming / Greedy Matching
3801	Minimum Cost to Merge Sorted Lists (Hard)	Bitmask Dynamic Programming (DP on Subsets).
3802	Number of Ways to Paint Sheets (Hard)	Matrix Exponentiation / 1D DP.
3803	Count Residue Prefixes (Easy)	Prefix Sums + Frequency Counting (HashMap).
3804	Number of Centered Subarrays (Med.)	Prefix Sum Transformation + Fenwick Tree / Sliding Window.
3805	Count Caesar Cipher Pairs (Med.)	Difference Array String Hashing + Frequency Map.
3806	Maximum Bitwise AND After Increment Operations (Hard)	Greedy Bitwise Construction (MSB to LSB).
3807	Minimum Cost to Repair Edges to Traverse a Graph (Med.)	Modified Dijkstra's Algorithm (Shortest Path on weighted states).
3808	Find Emotionally Consistent Users (Med.)	SQL Window Functions (LAG/LEAD) / Pandas `groupby().rolling()`.
3809	Best Reachable Tower (Med.)	Multi-source Breadth-First Search (BFS).
3810	Minimum Operations to Reach Target Array (Med.)	Difference Array + Greedy Peak Tracking.
3811	Number of Alternating XOR Partitions (Med.)	Prefix XOR + Dynamic Programming with HashMap.
3812	Minimum Edge Toggles on a Tree (Hard)	Tree DP (Re-rooting Technique).
3813	Vowel-Consonant Score (Easy)	Single Pass Linear Scan / Prefix Sums.
3814	Maximum Capacity Within Budget (Med.)	Binary Search on Answer.
3815	Design Auction System (Med.)	Hash Map + Max-Heap (Priority Queue) with Lazy Deletion.
3816	Lexicographically Smallest String After Deleting Duplicate Characters (Hard)	Monotonic Stack + Frequency Map + Visited Array.
3817	Good Indices in a Digit String (Med.)	Prefix and Suffix Precomputation Arrays.
3818	Minimum Prefix Removal to Make Array Strictly Increasing (Med.)	Suffix Array Traversal (Right-to-Left Scan).
3819	Rotate Non Negative Elements (Med.)	Two Pointers / Cyclic Replacement or Sub-segment Reversal Trick.
3820	Pythagorean Distance Nodes in a Tree (Med.)	Tree DFS/BFS Depth Mapping + Frequency Hashing + Mathematical Convolution/Lookup.
3821	Find Nth Smallest Integer With K One Bits (Hard)	Combinatorics (nCr) + Bit-by-Bit Construction.
3822	Design Order Management System (Med.)	Hash Map (Dictionary) for O(1) CRUD.

3823	Reverse Letters Then Special Characters in a String (Easy)	Multi-pass Two Pointers (In-Place).
3824	Minimum K to Reduce Array Within Limit (Med.)	Binary Search on Answer + Sliding Window / Greedy Check.
3825	Longest Strictly Increasing Subsequence With Non-Zero Bitwise AND (Med.)	1D DP + Bitmasking (State tracking per bit).
3826	Minimum Partition Score (Hard)	Dynamic Programming + Monotonic Queue/Segment Tree.
3827	Count Monobit Integers (Easy)	Bitwise Powers of 2 Math.
3828	Final Element After Subarray Deletions (Med.)	Two Pointers / Stack-based simulation.
3829	Design Ride Sharing System (Med.)	Spatial Hashing / Grid-based Priority Queues.
3830	Longest Alternating Subarray After Removing At Most One Element (Hard)	State Machine DP / Left-Right Precomputation.
3831	Median of a Binary Search Tree Level (Med.)	In-Order DFS Traversal with Level Grouping.
3832	Find Users with Persistent Behavior Patterns (Hard)	Gaps and Islands (SQL Window Functions) / Sequence Hashing.
3833	Count Dominant Indices (Easy)	Single-Pass Max & Second Max Tracking.
3834	Merge Adjacent Equal Elements (Med.)	Greedy Stack (Simulation).
3835	Count Subarrays With Cost Less Than or Equal to K (Med.)	Variable-Size Sliding Window (Caterpillar Method).
3836	Maximum Score Using Exactly K Pairs (Hard)	Sorting + 2D Dynamic Programming (or Max-Heap Greedy).
3837	Delayed Count of Equal Elements (Med.)	Two Pointers (Sliding Window) + Frequency Map.
3838	Weighted Word Mapping (Easy)	Direct Hash Map / ASCII Array Indexing.
3839	Number of Prefix Connected Groups (Med.)	Union-Find (Disjoint Set) with Dynamic Group Tracking.
3840	House Robber V (Med.)	1D Dynamic Programming (State Machine).
3841	Palindromic Path Queries in a Tree (Hard)	LCA (Lowest Common Ancestor) + Binary Lifting + Prefix Bitmasks.
3842	Toggle Light Bulbs (Easy)	Mathematical Property (Perfect Squares).
3843	First Element with Unique Frequency (Med.)	Two Hash Maps / Frequency-of-Frequency Tracking.
3844	Longest Almost-Palindromic Substring (Med.)	Expand Around Center + Difference Count.
3845	Maximum Subarray XOR with Bounded Range (Hard)	Prefix XOR + Binary Trie with Range Counting.
3846	Total Distance to Type a String Using One Finger (Med.)	Coordinate Geometry + Single Pass Accumulation.
3847	Find the Score Difference in a Game (Med.)	Dynamic Programming (Interval DP).
3848	Check Digitorial Permutation (Med.)	Frequency Counting + Permutation Invariance.
3849	Maximum Bitwise XOR After Rearrangement (Med.)	Greedy Bit-Construction + Trie.
3850	Count Sequences to K (Hard)	Dynamic Programming (Partition DP) / Generating Functions.
3851	Maximum Requests Without Violating the Limit (Med.)	Backtracking with Pruning / Network Flow.
3852	Smallest Pair With Different Frequencies (Easy)	Frequency Map + Sorted Unique Elements.
3853	Merge Close Characters (Med.)	Frequency Array + Greedy Consolidation.
3854	Minimum Operations to Make Array Parity Alternating (Med.)	Frequency Counting of Top 2 Elements (Even/Odd).
3855	Sum of K-Digit Numbers in a Range (Hard)	Digit DP / Arithmetic Progression Formula.
3856	Trim Trailing Vowels (Easy)	Backward Pointer Scan.
3857	Minimum Cost to Split into Ones (Med.)	Dynamic Programming (MCM - Matrix Chain Multiplication Style).
3858	Minimum Bitwise OR From Grid (Med.)	Dynamic Programming with Bitmasking / Dijkstra.
3859	Count Subarrays With K Distinct Integers (Hard)	Sliding Window (At Most K) Transformation.
3860	Unique Email Groups (Med.)	Hash Set + Pre-processing.
3861	Minimum Capacity Box (Easy)	Sorting + Greedy Accumulation.
3862	Find the Smallest Balanced Index (Med.)	Prefix Sums / Running Totals.
3863	Minimum Operations to Sort a String (Med.)	Cycle Decomposition (Permutation Theory).
3864	Minimum Cost to Partition a Binary String (Hard)	Dynamic Programming (1D).
3865	Reverse K Subarrays (Med.)	Difference Array / Lazy Propagation (Segment Tree).
3866	First Unique Even Element (Easy)	Hash Map (Frequency Counting).
3867	Sum of GCD of Formed Pairs (Med.)	Bitmask Dynamic Programming.
3868	Minimum Cost to Equalize Arrays Using Swaps (Med.)	Greedy / Cycle Counting with Costs.
3869	Count Fancy Numbers in a Range (Hard)	Digit Dynamic Programming.
3870	Count Commas in Range (Easy)	Mathematical Range Summation.
3871	Count Commas in Range II (Med.)	Mathematical Offset Calculation (f(R) - f(L-1)).
3872	Longest Arithmetic Sequence After Changing At Most One Element (Med.)	Dynamic Programming with State (Changed vs. Unchanged).
3873	Maximum Points Activated with One Addition (Hard)	Union-Find with Component Size Tracking + Adjacency Map.